

IQVIA MARKET PROGNOSIS 2025-2029

Asia/Australia - China

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IQVIA

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IQVIA MARKET PROGNOSIS

Market Overview and Forecasts

Published: September 2025

MARKET OVERVIEW AND FORECASTS

Market Synopsis

- The Chinese pharmaceutical market is forecast to grow at a CAGR of 1.3% ($\pm 1.5\%$) between 2024 and 2029, reaching RMB1,275 billion by 2029.
- At the time of publication, there remained significant uncertainty regarding the implications of the trade tariffs imposed by the US and potential counter-tariffs by other countries, as well as the US Executive Order 14297. In particular, it was unclear if the US announcement of 25 September 2025, potentially imposing a 100% tariff on imported branded or patented medicines, would override any previous agreements. These developments pose downside risks to the prognosis.

Business Environment

- Real GDP growth is forecast to decelerate to 4.7% in 2025, from 5.0% in 2024. China's economic prospects will face strong headwinds, as higher US tariffs facing Chinese products will not only hurt exports but also dampen factory investment and consumer spending. Government stimulus measures and trade front-loading will offset some of the pressure, but GDP growth is forecast to decelerate to average 4.0% annually in 2026-2029. Deflationary pressures will cause consumer prices to fall by -0.3% in 2025, from 0.2% in 2024, before rising again under the impact of public spending and monetary easing to average 0.7% per year in 2026-2029. The renminbi will appreciate against the US dollar over the forecast period, from an average RMB7.20:US\$1.00 in 2024 to RMB7.09:US\$1.00 in 2029.
- Domestic politics will be stable in 2025-2029. China's president, Xi Jinping, has cemented his power, with an ally, Li Qiang, holding the premiership, while allies also hold key roles in the Chinese Communist Party (CCP), armed forces and state bureaucracy. In 2025-2026 China will adopt a more supportive stance towards the economy. Boosting private consumption now commands a higher priority. China has expanded its consumer goods subsidies, providing more income transfers, and enhancing social security benefits. Structural reform will remain selective and gradual. Relations with the US will remain under strain, particularly as a result of the tariff actions imposed by the US administration.

Healthcare Provision

- Health policy will continue to pursue a multi-tiered, integrated healthcare system. Short to medium-term objectives include improving health service capabilities and infrastructure, especially in the primary care sector. Priorities also include further reform of public hospitals, expansion of health insurance coverage, and encouraging innovation in the pharmaceutical sector. Health policy will also seek to improve prevention and treatment of chronic and rare diseases.
- Growth in national healthcare expenditure slowed to 6.2% in 2023 (latest publicly available data), as costs associated with COVID-19 containment measures reduced and government spending on healthcare slowed to just 0.4%. Social and personal health expenditure grew by 8.7% and 7.9%, respectively, however, under the impact of delayed demand for treatment and a recovery in

screening activity in the wake of the pandemic. Total healthcare expenditure was equivalent to 7.2% of GDP in 2023.

- Financial pressure on the basic medical insurance (BMI) schemes is growing, aggravated by the slowing economy and the aging population. The government will continue to promote an integrated multi-level health insurance system, including *huiminbao* city-level supplementary insurance schemes and commercial health insurance (CHI). While many *huiminbao* schemes have struggled financially, improving regulation should help to stabilize the sector.
- The government continues to pursue reforms in the hospital sector geared towards decentralization, cost efficiency, and accessibility. Measures include the establishment of national and regional medical centers. Reforms of medical service pricing are being rolled out, and hospital payment via flat-sum payment models has been implemented in all regions, covering all eligible institutions and 95% of diseases by April 2025. The easing of restrictions on foreign investment should help drive improvements in the private hospital sector.
- Efforts to shift more demand into primary care facilities are making gradual progress. Further improvements are being targeted in the range and quality of services available in grassroots facilities, and a push is being made to establish closely integrated medical and health consortia in all counties, with the objective of establishing a tiered diagnosis and treatment system. Progress varies significantly between provinces and individual cities, however, and the government's decentralization program will have to overcome negative perceptions of expertise in primary care.
- Further expansion of volume-based procurement (VBP) will be pursued. Following the 10th round of national VBP, implemented in April 2025, 435 multisource products are included. The 11th round, launched in July 2025, covers an additional 55 active ingredients, and is expected to be implemented before the end of the year. Most VBP contracts will be taken up by local companies. Increased multi-province VBP will mean more complexity and uncertainty for bidders. National VBP for multisource biologics may be introduced in 2026.

Prescribing and Dispensing

- The government's anti-corruption campaign has led to more conservative prescribing behavior, while hospital audits and claims controls have been stepped up. VBP is exerting a growing impact on prescribing choices for hospital doctors, with opportunities for non-winners limited. Flat-sum provider payment models will also contribute to more cost-sensitive prescribing, while growing pressures on BMI finances can affect prescribing, especially towards the end of the year.
- The 2024 update of the National Reimbursement Drug List (NRDL) added 91 new drugs to the list, including a record number of innovative drugs. The presence of domestically developed drugs is growing, with 65 newly added drugs from domestic companies in 2024. The National Health Security Administration (NHSA) announced in August 2025 that 534 drugs had passed the preliminary review for NRDL listing in 2025, pending further evaluation and price negotiations in the fall, including 310 drugs to be newly listed and 224 drugs for renewal or renegotiation.
- Despite selective improvements in hospital formulary listings following NRDL updates, barriers to access persist, including delayed or non-existent formulary committee meetings. More established NRDL-listed drugs face further difficulties accessing formularies if they lose out on VBP contracts. With the number of drugs on the NRDL increasing year by year, and with more favorable listing criteria for breakthrough therapies, competition for formulary listing will remain fierce.

- Dispensing of outpatient prescriptions outside hospitals will increase gradually, supported by the dual-channel policy, which enables reimbursement of selected NRDL drugs in designated retail pharmacies. Progress with the policy has been slower than expected, however, and controls on dual-channel dispensing will be tightened. Many companies have increased investment in the retail channel in recent years, especially for products that lost out on VBP contracts. Shifts in dispensing activity will be gradual, however.

Pricing and Reimbursement

- Of the 91 drugs included in the NRDL in November 2024, to be reimbursed from January 2025, 89 were listed following a national price negotiation or bidding process, with an average price reduction of 63% – the sharpest average cuts since annual updates of the NRDL started in 2017.
- Since 2023, the introduction of product-specific value ratings and incremental cost-effectiveness ratio (ICER) thresholds for innovative new drugs has resulted in more generous envelope prices for innovative products. New rules on the renewal of existing NRDL listings, introduced in 2022 and 2023, have reduced price erosion on listed products.
- The NHTA's new value-rating system for NRDL candidates will continue to be refined over the prognosis period. Only a relatively small percentage of drugs are likely to be categorized as 'breakthrough', which requires that China is the global first-launch market, while the vast majority will be judged as 'equivalent'. Innovative medicines will be subject to overall cost ceilings, while reliance on low-priced comparator drugs will be a source of frustration to companies.
- In recognition of the limits to reimbursement of costly innovative drugs under the BMI schemes, the NHTA in August 2025 announced that 121 drugs had passed the preliminary review stage for inclusion in a new CHI innovative drug list – subject to further evaluation and price negotiations. The new list is intended to improve and standardize CHI coverage of innovative drugs that are considered of significant clinical value and patient benefit, but that are outside the financial scope of coverage by the BMI schemes.

Regulatory Environment

- Drug approvals will continue to pick up pace over the prognosis period, with regulators making further efforts to streamline and facilitate the registration of innovative medicines especially, and of foreign drugs addressing urgent unmet needs. A record 43 innovative drugs were reported to have been approved in H1 2025, a year-on-year increase of 59%, while 48 innovative drugs received marketing approval in 2024, including 17 under a priority review pathway.
- China will remain an attractive base for clinical research, as approval times for trials of innovative medicines are being shortened. Under a draft policy issued in June 2025, the investigational new drugs (IND) review and approval process will be shortened to 30 days for certain key products. While the regulatory framework for large-scale data transfers out of China has become more accommodating, companies may still struggle with gray areas around sensitive or important data, and with persistent restrictions on data exports involving human genetic material.
- The General Office of the State Council in January 2025 called for further improvements to the legal and regulatory frameworks governing drugs and medical devices, aiming to enhance the regulatory system, with more detailed measures on data exclusivity, market exclusivity, expedited

review pathways, and enhanced compliance expected to be covered in revisions to the implementing regulations of the Drug Administration Law.

- With final rules on patent term extensions (PTEs) and patent term adjustments implemented in January 2024, the intellectual property landscape for originator companies is clearer. However, originator companies' access to PTE provisions will be severely restricted unless they choose to conduct early clinical trials in China and file for approval in China before anywhere else. Draft rules for a six-year clinical trial data protection period for innovative small molecules and biologics that have not been approved in China or globally were published for consultation in March 2025.

Pharmaceutical Business Environment

- After peaking in April 2025, tariff tensions between the US and China de-escalated following an agreement on 12 May 2025, which reduced base tariffs in both countries to 10%, but imposed a 20% 'fentanyl tariff' on Chinese imports into the US. Initially in place for a 90-day period, the agreement was extended for another 90 days, to 10 November 2025. Pharmaceuticals are exempt from 'reciprocal' tariffs under the agreement, but a Section 232 investigation into pharmaceutical imports into the US could lead to the imposition of sector-specific tariffs.
- US tariffs on pharmaceutical imports from China, as well as potential Chinese counter-tariffs on pharmaceutical imports from the US, would put upward pressure on drug prices and disrupt supply chains. US imports of finished pharmaceuticals from China were worth US\$7.84 billion in 2024, while its exports to China were worth US\$9.50 billion, according to UN Comtrade data.
- Separately, US Executive Order 14297 of 12 May 2025 contains provisions for setting US drug prices by benchmarking them to the prices of the same product in other countries, effectively amounting to international reference pricing (IRP). While no details for the mechanism had been specified by mid-2025, if GDP per capita is a consideration in the selection of reference countries, Chinese drug prices are unlikely to be considered under the mechanism, given the difference in GDP per capita between the two countries.
- The government will continue to encourage drug innovation and China's share of licensing deals has been rising rapidly. With limited funding sources locally, domestic innovators will rely largely on out-licensing of early-stage assets to generate funding for sustainable R&D, while exploring alternative financing mechanisms, such as spin-off 'NewCos'.
- Policymakers will also promote the further development of the domestic generics industry through measures such as 'encouraged generics' lists and large volume uptake of local generics in VBP. Pricing pressures and more rigorous quality standards could marginalize some smaller players. Biosimilars will gain further traction as a growing number of innovative biologics face patent expiry.
- Growth in retail pharmacy numbers has slowed, with chains accounting for 57% of outlets. There are indications that the sector is reaching saturation, and it is expected to undergo a period of adjustment. Larger chains with diversified product ranges will be better placed to ride out challenges, including slower-than-expected progress with dual-channel dispensing and growing pricing pressures. Retail pharmacies are expected to begin offering more specialized, value-added professional services in the medium to longer term.

Total Market Forecasts 2025-2029

Summary of the Prognosis

Sales in Market Prognosis are reported at the ex-manufacturer price level. All values shown are in local currency.

Key Drivers and Constraints of Market Growth

Market Drivers	Market Constraints
Recent and future product launches , dominated by novel anti-cancer drugs and including the new anti-obesity treatments, putting upward pressure on average prices.	Expansion of volume-based procurement , at national and multi-province level, with sharp price cuts for a growing number of multisource products, and biologics likely to be included in 2026.
Reimbursement of innovative new drugs following price negotiations , with an additional 91 products listed on the NRDL since January 2025, and the next update expected in late 2025.	Implementation of flat-sum hospital payment models , with Diagnosis-Related Group and Diagnosis Intervention Packet models expected to lead to more cost-conscious prescribing.
Improvements in health insurance coverage and a new CHI innovative drug list , targeting cover beyond the BMI system, and improved and standardized coverage of innovative drugs.	Demographic factors , with China's population contracting by 17.40 million over the forecast period, and a steady decline in the working age population.

Source: IQVIA

Total Market Sales

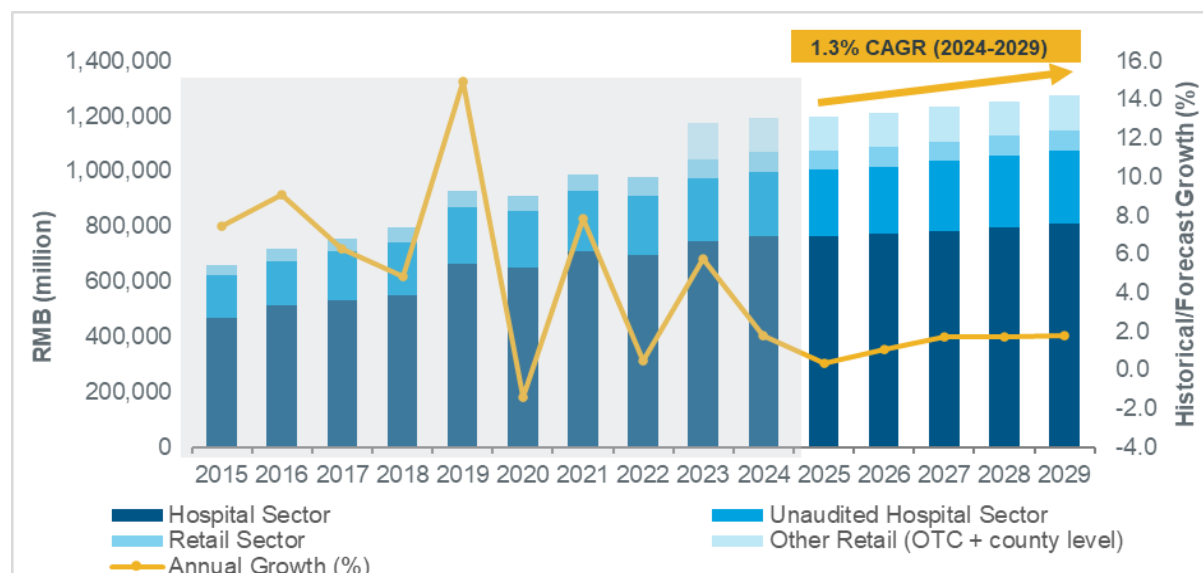


Table 1: Total Market Sales[†] at Ex-Manufacturer Level (2024-2029) RMB (million)

	2024	2025	2026	2027	2028	2029	CAGR 2024-2029 (%)
Hospital Sector (>100 beds)	765389	766201	771729	785255	798306	812198	1.2
Annual Growth (%)	2.3	0.1	0.7	1.8	1.7	1.7	
Unaudited Hospital (small hospitals, CHCs, THCs, clinics)	233986	239303	245428	251553	257679	263804	2.4
Annual Growth (%)	3.1	2.3	2.6	2.5	2.4	2.4	
Retail Sector (Rx at prefecture level)	70281	70382	70906	71644	72440	73200	0.8
Annual Growth (%)	0.7	0.1	0.7	1.0	1.1	1.0	
Other Retail (OTC + county level)	125546	123039	123391	123947	124767	125938	0.1
Annual Growth (%)	-3.1	-2.0	0.3	0.5	0.7	0.9	
Total Market	1195203	1198925	1211454	1232399	1253191	1275140	1.3
Annual Growth (%)	1.8	0.3	1.0	1.7	1.7	1.8	

[†] Forecast cut-off date: 10 September 2025

Source: IQVIA

Table 2a: Total Market Sales at Ex-Manufacturer Level (2024-2029) LC\$* (million)

	2024	2025	2026	2027	2028	2029	CAGR 2024-2029 (%)
Hospital Sector (>100 beds)	105818	105930	106694	108564	110369	112290	1.2
Unaudited Hospital (small hospitals, CHCs, THCs, clinics)	32349	33085	33931	34778	35625	36472	2.4
Retail Sector (Rx at prefecture level)	9717	9731	9803	9905	10015	10120	0.8
Other Retail (OTC + county level)	17357	17011	17059	17136	17249	17411	0.1
Total Market	165241	165756	167488	170384	173258	176293	1.3
Annual Growth (%)	1.8	0.3	1.0	1.7	1.7	1.8	

*LC\$ = local currency sales converted to US dollars using a constant exchange rate (Q2 2025)

Source: IQVIA

Table 2b: Total Market Sales at Ex-Manufacturer Level (2024-2029) US\$* (million)

	2024	2025	2026	2027	2028	2029	CAGR 2024-2029 (%)
Hospital Sector (>100 beds)	106340	106677	110141	110289	112596	114555	1.5
Unaudited Hospital (small hospitals, CHCs, THCs, clinics)	32509	33318	35027	35330	36344	37208	2.7
Retail Sector (Rx at prefecture level)	9765	9799	10120	10062	10217	10324	1.1
Other Retail (OTC + county level)	17443	17131	17610	17408	17598	17763	0.4
Total Market	166056	166925	172899	173090	176755	179851	1.6
Annual Growth (%)	0.2	0.5	3.6	0.1	2.1	1.8	
Exchange Rate RMB : US\$*	7.20	7.18	7.01	7.12	7.09	7.09	

*Exchange rate forecast sourced from the Economist Intelligence Unit (EIU) (July 2025)

Source: IQVIA

Total Audited Market Forecasts

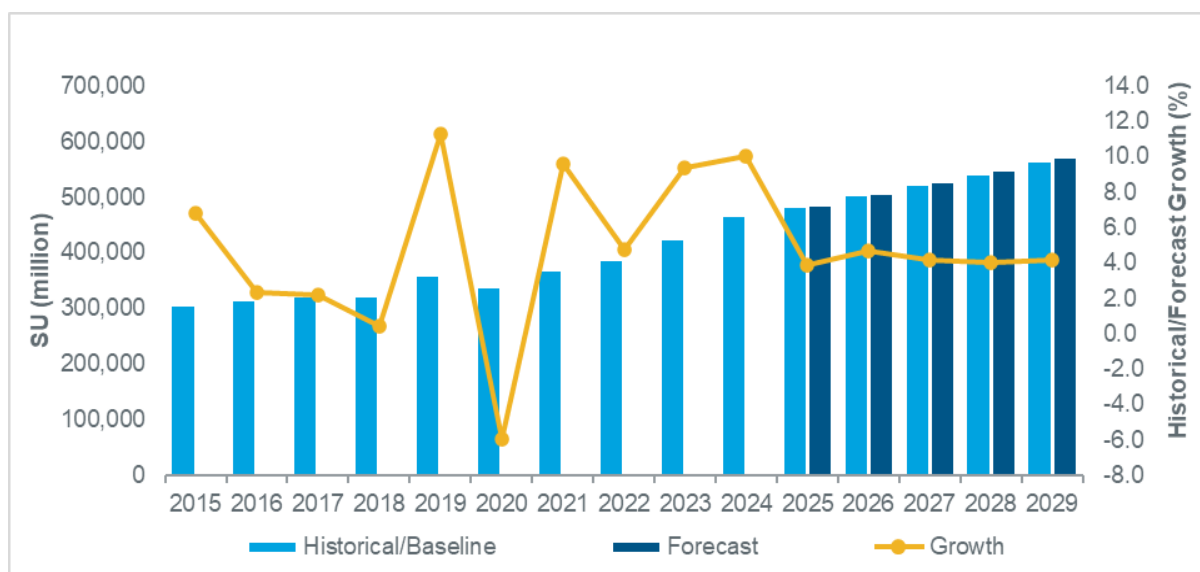
Table 3: Total Audited Market Forecasts (2024-2029)

Year	Volume (Standard Units*)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2024	466938	10.0	1.79	-7.1	835671	2.2
2025	485062	3.9	1.72	-3.6	836583	0.1
2026	507592	4.6	1.66	-3.7	842635	0.7
2027	528673	4.2	1.62	-2.4	856899	1.7
2028	550040	4.0	1.58	-2.3	870746	1.6
2029	572874	4.2	1.55	-2.4	885399	1.7

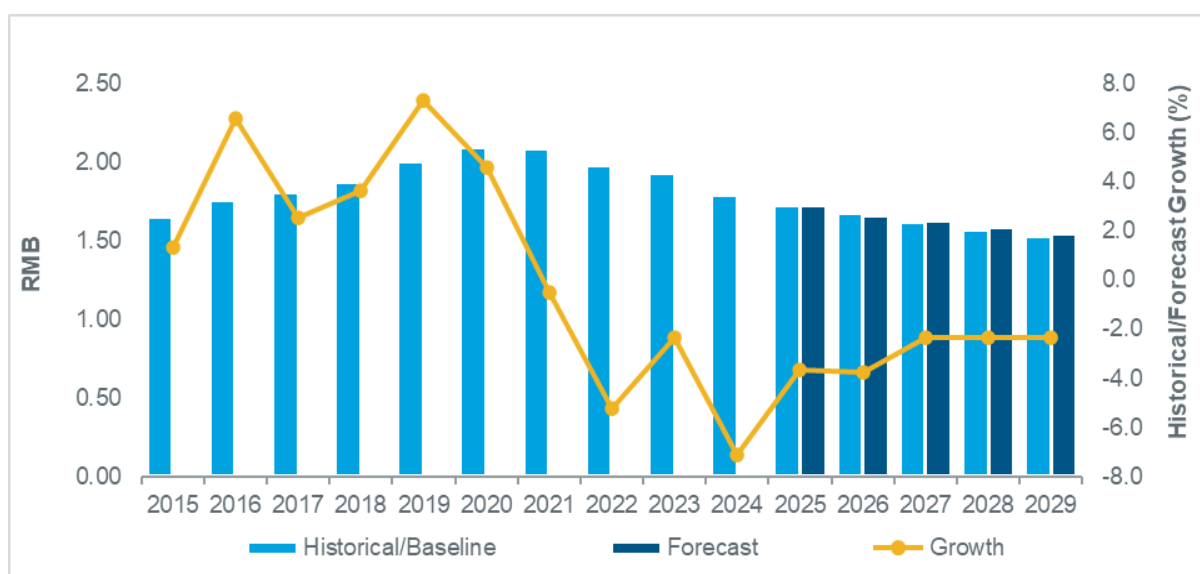
*Standard Units equate the number of milliliters of liquid preparations to the standard solid dosage of one tablet, therefore making solid and liquid preparations comparable.

Source: IQVIA

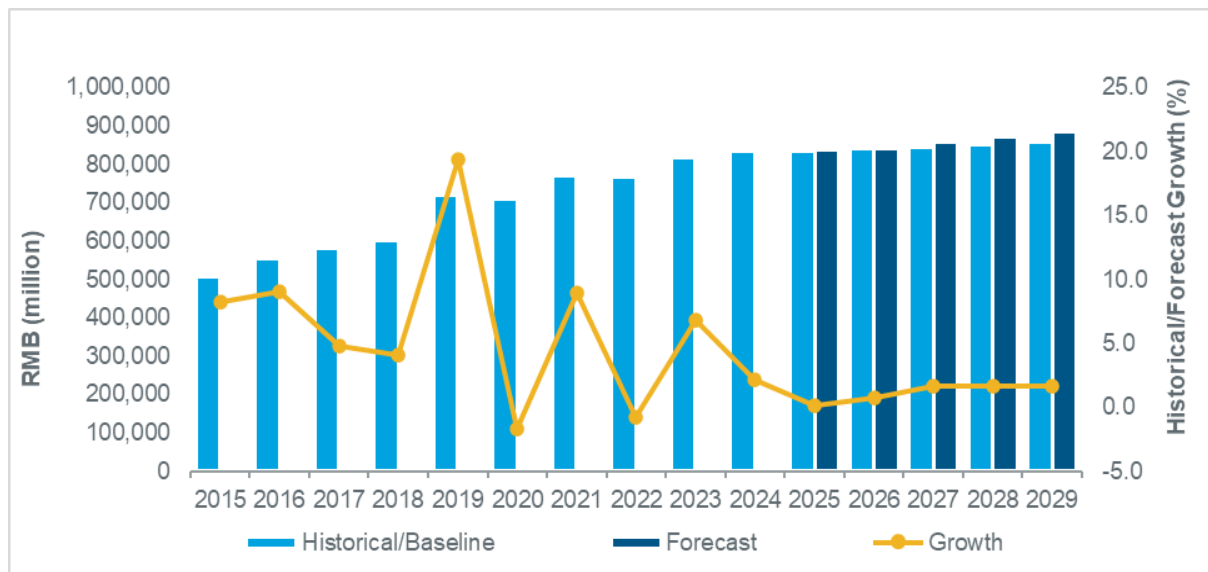
Total Audited Market Volume (Standard Units)



Total Audited Market Average Price per Standard Unit



Total Audited Market Sales



Retail Sector Overview

Retail Sector Historical Trends

Sales of prescription-only drugs through retail pharmacies have long been held back by limited reimbursement of outpatient drugs at retail pharmacies, and by the availability of drugs at lower prices in hospitals and community health centers (CHCs). Increasing control of hospital budgets and the progressive elimination of the mark-up on drugs in hospitals have begun to encourage the release of more scripts for filling in retail pharmacies in recent years. Retail pharmacies are an important sales channel for non-reimbursed products, and direct-to-patient (DTP) pharmacies have emerged as a sales channel for certain high-cost specialty products. Sales were hit by the COVID-19 outbreak in early 2020, which affected traditional Chinese medicines (TCMs) in particular. Retail volume continued to decline in 2021, in part due to competition from online pharmacies (with only online-to-offline sales captured in the audit). Growth rates are raised by a trend break in 2022, but retail pharmacy sales also saw dramatic growth in December 2022, as China's zero-COVID restrictions were lifted and COVID surged, raising volume growth for the year, while slowing price growth. Volume contracted again in 2023 and 2024, largely driven by a decline in TCMs, which account for around one-third of volume in the sector. Strong sales growth of costly anti-cancers, especially mAb antineoplastics and interleukin inhibitors, lifted the average price per standard unit in both years.

Table 4: Retail Sector (2014-2024)

Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2014*	93370	-	0.35	-	33054	-
2015	101988	9.2	0.37	5.6	38134	15.4
2016	98460	-3.5	0.42	11.9	41189	8.0
2017*	101829	3.4	0.48	14.9	48939	18.8
2018*	101935	0.1	0.52	9.0	53414	9.1
2019	98218	-3.6	0.58	10.9	57091	6.9
2020	92955	-5.4	0.62	5.9	57234	0.3
2021*	91881	-1.2	0.65	6.1	60031	4.9
2022*	103358	12.5	0.66	1.0	68216	13.6
2023	100416	-2.8	0.69	5.3	69774	2.3
2024	97145	-3.3	0.72	4.1	70281	0.7

*Growth rates increased by panel enhancements (see Forecasting Data and Methods: Data Sources).

Source: IQVIA

Retail Sector Forecasts

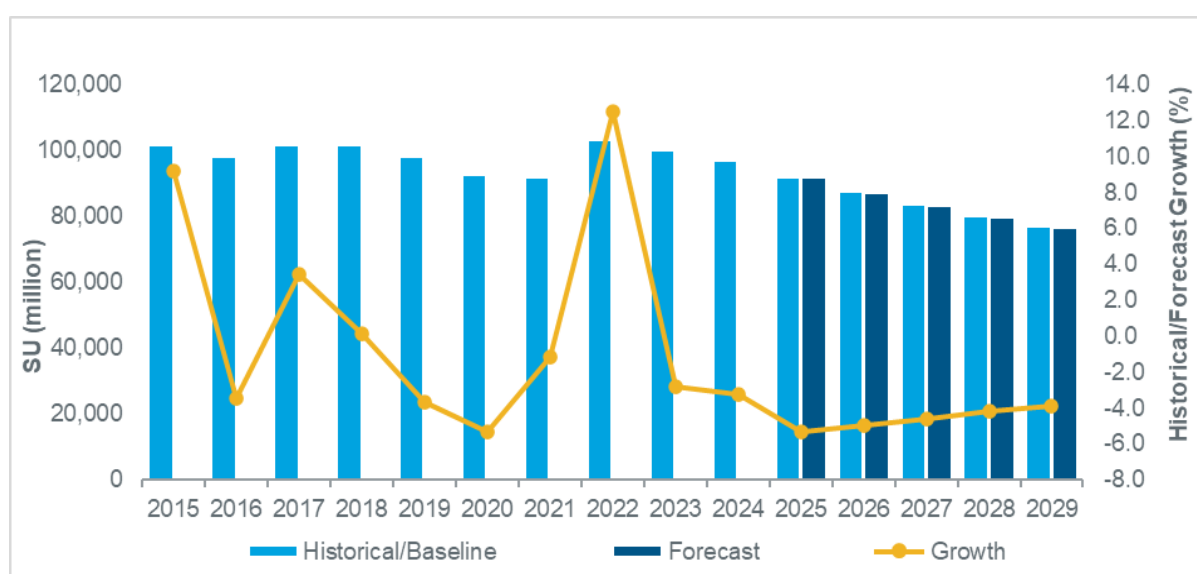
Retail volume continued to contract in year-to-date H1 2025, mainly under the impact of a sharp contraction in volume of TCMs, while therapy classes R (mainly antitussives) and J (mainly macrolides) also contributed to the decline. Volume is forecast to continue to contract throughout the forecast period. With pharmacy density reported to be nearing saturation point, the sector is expected to undergo a period of adjustment, although it is expected to remain an alternative sales channel to the dominant hospital sector, especially for products affected by volume-based procurement (VBP) in hospitals and non-reimbursed products. Improvements in health insurance coverage, as well as the rise in telemedicine, will also help retail sector volume growth, while the reform of provider payment in hospitals is expected to lead to the release of more prescriptions for dispensing in retail pharmacies. New product launches will drive a steady increase in the average price/standard unit, while generic competition will dampen price growth. The retail sector is forecast to grow at a CAGR of 0.8% over 2024-2029.

Table 5: Forecast Retail Sector (2024-2029)

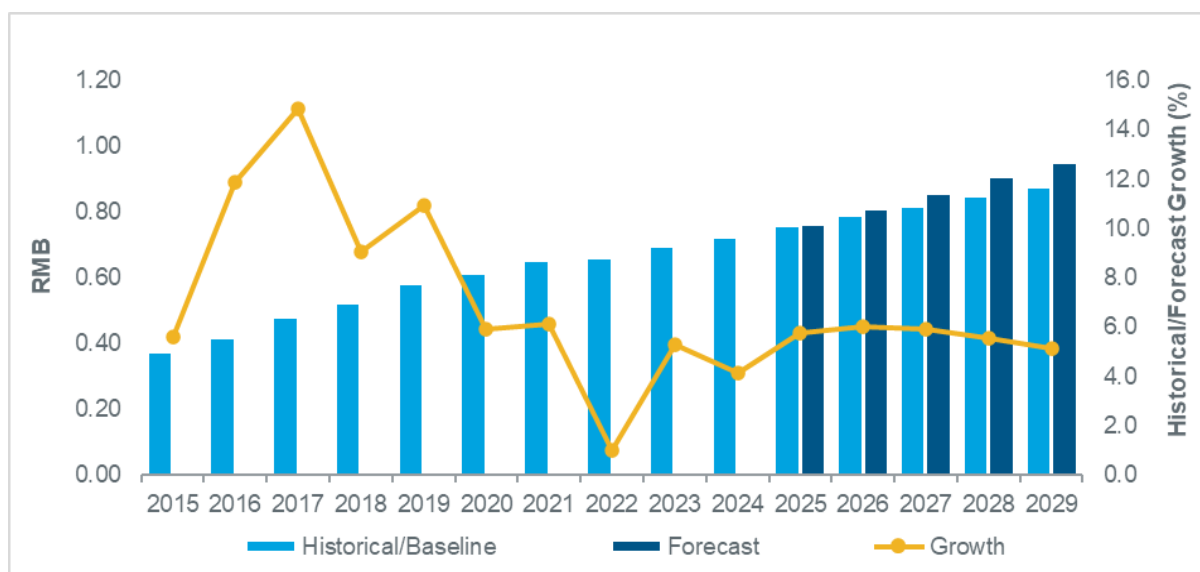
Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2024	97145	-3.3	0.72	4.1	70281	0.7
2025	91976	-5.3	0.77	5.8	70382	0.1
2026	87394	-5.0	0.81	6.0	70906	0.7
2027	83373	-4.6	0.86	5.9	71644	1.0
2028	79868	-4.2	0.91	5.5	72440	1.1
2029	76776	-3.9	0.95	5.1	73200	1.0

Source: IQVIA

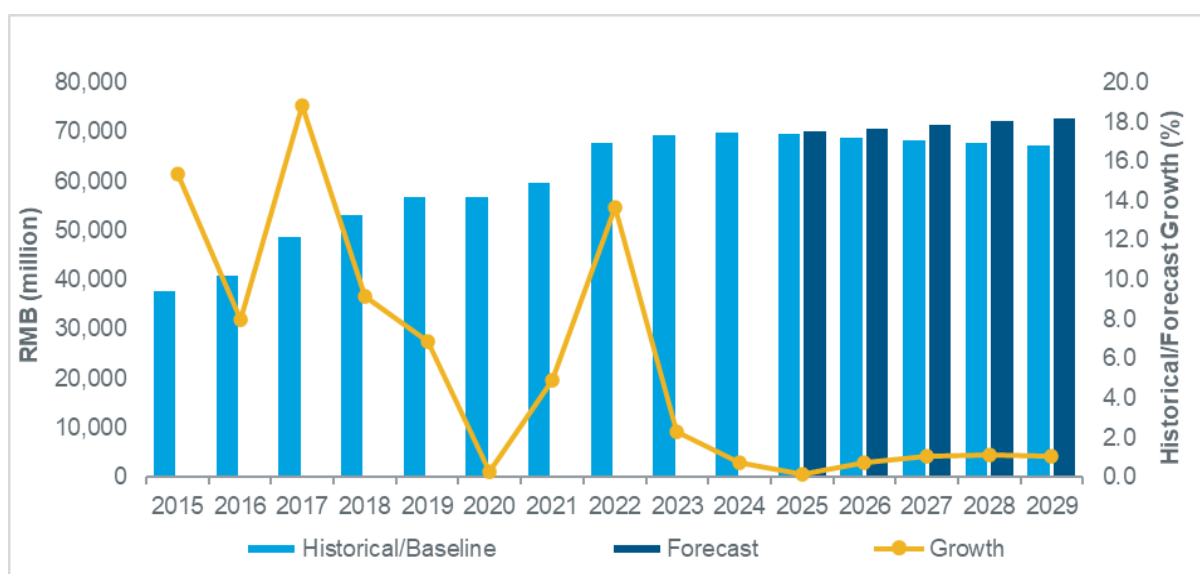
Retail Sector Volume (Standard Units)



Retail Sector Average Price per Standard Unit



Retail Sector Sales



Hospital Sector Overview

Hospital Sector Historical Trends

Growth in China's dominant hospital sector over the past decade reflects the impact of cost-containment measures and growing pricing pressures. The implementation of hospital financing reforms and tightening drug budget controls in 2017 and 2018 led to a slowdown in sales growth. Growth in 2019 is lifted by a trend break in the audit data, but also reflects improving access to innovative new drugs following their reimbursement listing, although price growth was slowed by the rollout of VBP. The COVID-19 outbreak drove a sharp contraction in volume in 2020, while price growth in 2020 reflects a trend break in the audit data. Volume rebounded strongly in 2021, but VBP drove a steep decline in the average price per standard unit. Strict localized lockdowns in 2022 and the surge in infections following the lifting of the zero-COVID restrictions in December of that year impacted regular hospital activity negatively, slowing volume growth. Combined with continuing VBP price pressures, this drove another contraction in hospital sales in 2022. As regular hospital operations resumed following COVID, volume saw a rebound in 2023, although growth in 2023 is also lifted by a data trend break. Strong volume growth continued in 2024, driven by therapy classes V (TCMs, which account for 32% of hospital volume), A (with strong growth in vitamin B12 formulations and certain antidiabetics), as well as C (cholesterol & triglyceride regulators, calcium antagonists plain, angiotensin II antagonist combinations). Solid volume growth was accompanied by a sharp contraction in the average price per standard unit, under the impact of VBP.

Table 6: Hospital Sector (2014-2024)

Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2014	194713	-	2.24	-	435707	-
2015*	205669	5.6	2.28	1.9	469107	7.7
2016	216458	5.2	2.37	3.7	512116	9.2
2017	219989	1.6	2.41	2.0	530688	3.6
2018	221189	0.5	2.48	3.0	549641	3.6
2019*	261235	18.1	2.54	2.1	662574	20.5
2020**	245135	-6.2	2.65	4.6	650611	-1.8
2021	278492	13.6	2.55	-3.8	711382	9.3
2022	284610	2.2	2.45	-4.1	697390	-2.0
2023*	323914	13.8	2.31	-5.8	748031	7.3
2024	369794	14.2	2.07	-10.4	765389	2.3

*Growth rates in 2015, 2019 and 2023 increased by audit upgrade/audit expansion.

**Trend break in 2020, with price and sales growth increased as a result of a change in the factor applied to convert data from trade level to ex-manufacturer level. No trend break in volume (see Forecasting Data and Methods: Data Sources).

Source: IQVIA

Hospital Sector Forecasts

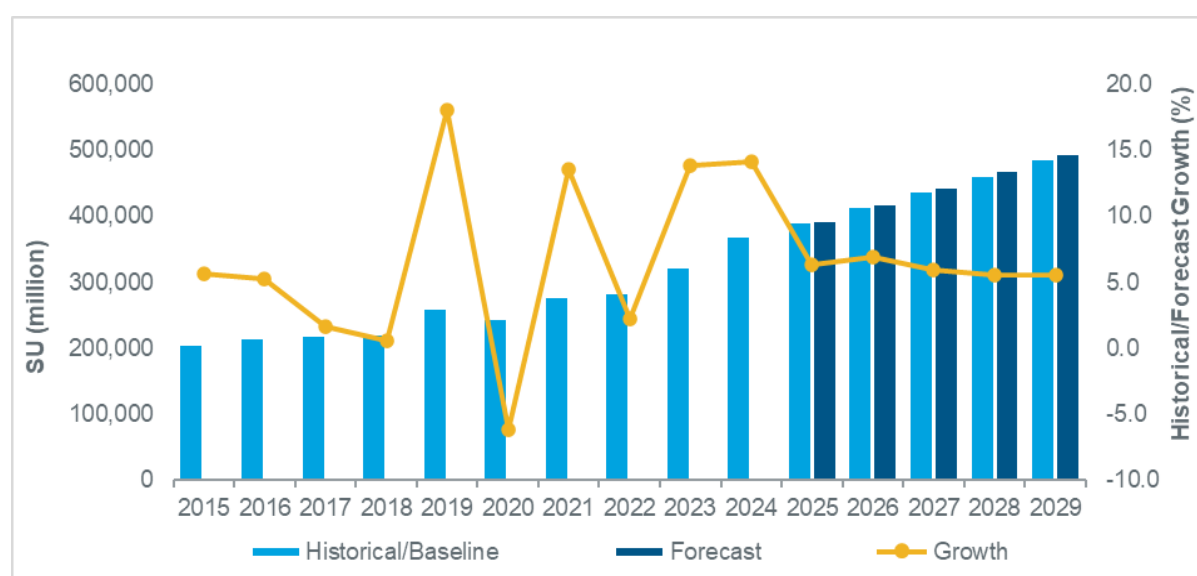
Hospital volume growth slowed sharply in H1 2025, mainly due to a deceleration in growth in TCMs. Hospital sector volume growth over the forecast period will be supported by improvements in health insurance coverage and the hospital infrastructure, while new product launches, including new anti-obesity treatments, the listing of a growing number of innovative new drugs on the NRDL, and the new commercial health insurance innovative drug list will impact the average price per standard unit positively. At the same time, the expansion of VBP will continue to put strong downward pressure on prices, while patent expiries and the onset of generic competition will also slow price growth. The implementation of flat-sum reimbursement mechanisms, as well as growing financial pressures on the BMI system, will slow both price and volume growth, and the strengthening of primary care will begin to shift more patients from major hospitals to primary care facilities, impacting volume growth negatively. The hospital sector is forecast to grow at a CAGR of 1.2% over 2024-2029.

Table 7: Forecast Hospital Sector (2024-2029)

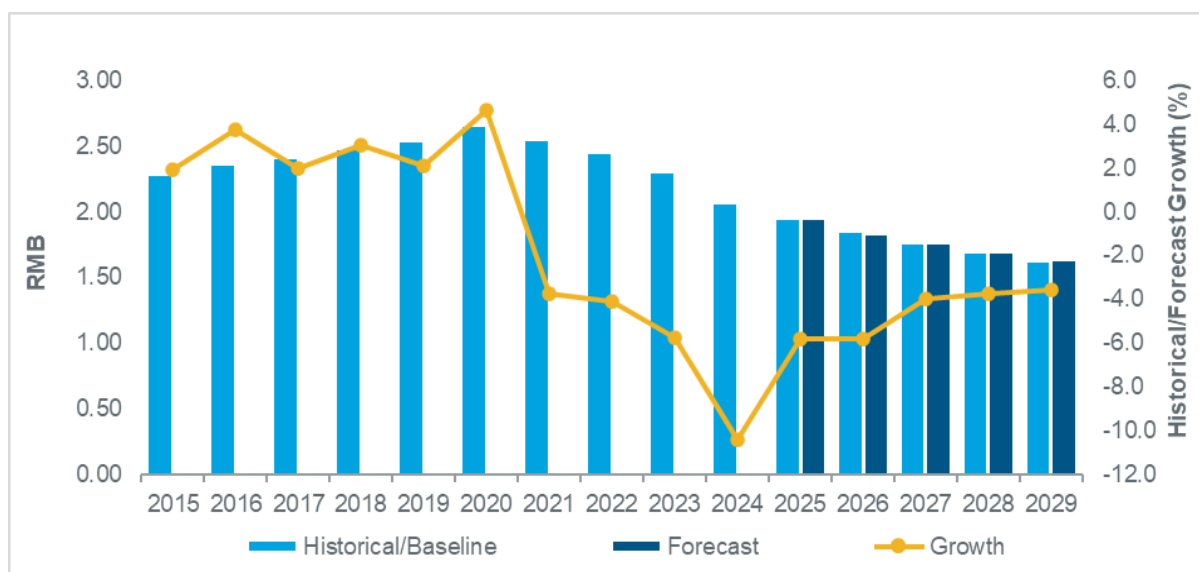
Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2024	369794	14.2	2.07	-10.4	765389	2.3
2025	393086	6.3	1.95	-5.8	766201	0.1
2026	420198	6.9	1.84	-5.8	771729	0.7
2027	445301	6.0	1.76	-4.0	785255	1.8
2028	470172	5.6	1.70	-3.7	798306	1.7
2029	496098	5.5	1.64	-3.6	812198	1.7

Source: IQVIA

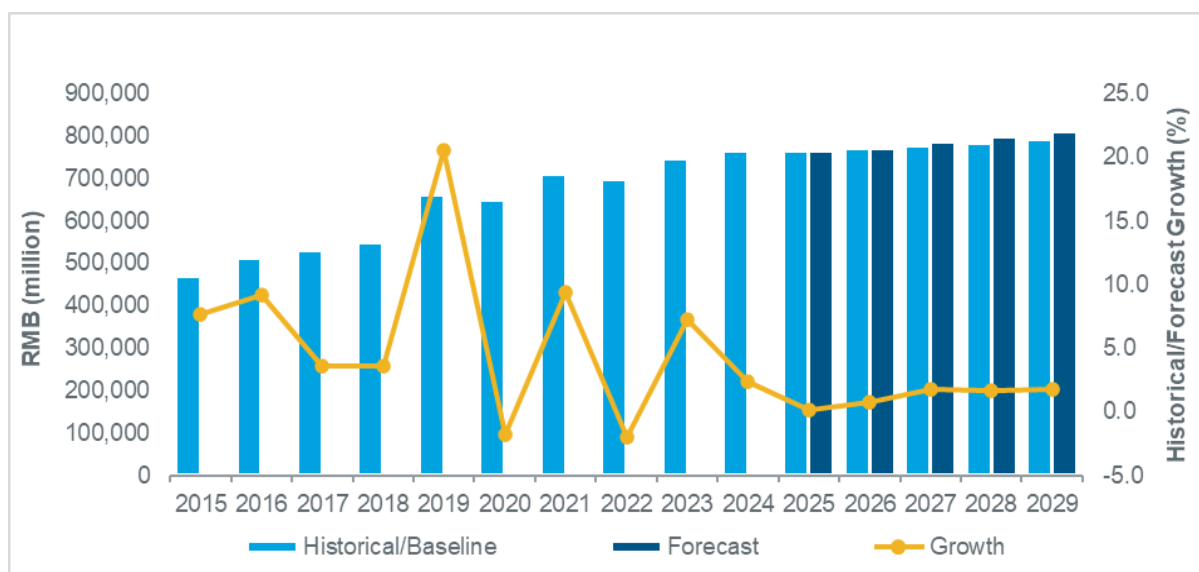
Hospital Sector Volume (Standard Units)



Hospital Sector Average Price per Standard Unit



Hospital Sector Sales



Event Analysis

Positive Events

Recent and new product launches 2025-2029: The impact of new product launches is calculated using IQVIA Analytics Link's Forecast Link data. These forecasts are based on a detailed review of future product launches in their competitive landscapes. Country-specific historical analogues are used to adjust the sales for the concomitant loss of revenues due to cannibalization of existing products. The net impact is then distributed by sector based on historical MIDAS data analysis to calculate the percentage contribution of new launches by sector. The leading new products driving sales growth over the forecast period are dominated by oncology drugs and include the new anti-obesity drugs, with both Mounjaro (tirzepatide), as well as Wegovy (semaglutide, branded as NovoCare in China) approved for the treatment of obesity in mid-2024.

Reimbursement of innovative new drugs following price negotiations: National price negotiations have paved the way for the inclusion of a growing number of innovative new drugs on the National Reimbursement Drug List (NRDL) since 2016, with price reductions averaging between 44% and 63%. The latest NRDL update, which came into effect in January 2025, added 91 products to the list, including 89 on the basis of successful price negotiations and competitive bidding procedures. The National Health Security Administration (NHSA) announced in August 2025 that 534 drugs had passed the preliminary review for NRDL listing, pending further evaluation and price negotiations in the fall, including 310 drugs to be newly listed and 224 drugs for renewal or renegotiation. While there can be challenges with hospital formulary listing for newly NRDL-listed drugs, wider usage of newly listed innovative products will lift the average price per standard unit in the hospital sector, despite the sharp price cuts, while slowing retail sector sales as dispensing shifts to hospitals.

Improvements in health insurance coverage and new CHI innovative drug list: The government is targeting the development of a comprehensive multi-level insurance system, comprising the core pillars of the current system and supplementary providers, including city-level complementary insurance (*huiminbao*) and other commercial health insurance (CHI). In a significant step towards such a system, the NHSA in August 2025 announced that 121 drugs had passed the preliminary review stage for inclusion in a new CHI innovative drug list. Drugs on the preliminary list – which are subject to further evaluation and price negotiations – include several CAR-T therapies and a gene therapy, for example. The new list is intended to improve and standardize CHI coverage of innovative drugs that are considered of significant clinical value and patient benefit, but that are outside the financial scope of coverage by the basic medical insurance (BMI) schemes.

Improvements in the hospital infrastructure: Under the 14th Five-Year Plan (2020-2025), the hospital infrastructure has been expanded and decentralized, with 13 national medical centers established across China by 2024, providing first-class diagnosis and treatment in specialized medical fields. In addition, 125 national-level regional medical centers and 114 provincial-level regional medical centers have also been set up. The number of tier 3 (tertiary) hospitals and their bed capacity continues to increase, while 92% of county-level hospitals have reached the service capacity and standards of tier 2 hospitals or above. In parallel, the number of private hospitals is growing rapidly. A pilot program launched in November 2024 allows the establishment of wholly foreign-owned hospitals in eight relatively wealthy Chinese cities, as well as in Hainan province, which are expected to provide

more diversified, specialized, and patient-centric services than the public sector, and over 150 joint-venture and wholly foreign-invested medical institutions were reportedly in operation by April 2025.

Growth in e-health and telemedicine: E-health services and their uptake will continue to grow. The user base for internet medical services in China had reached close to 420 million people by the end of 2024. The number of internet hospitals had grown to 3,340 in 2024, delivering over 100 million online diagnoses and treatments annually, and collaboration between leading institutions and lower tier facilities is expanding. One of China's leading hospitals, Peking Union Medical College Hospital, in April 2025 announced plans for an 'Internet+ Medical Alliance' with 156 medical institutions in five provinces, to help improve capabilities at grassroots facilities, for example. Commercial players will also continue to expand and broaden the capacity and capability of services, which include 24-hour access to online consultations and the delivery of prescribed drugs, but also corporate employee health management platforms. Broader use of telemedicine will expand access to care and will contribute to retail sector growth via online-to-offline drug sales through audited retail pharmacies.

Negative Events

Expansion of VBP 2025-2029: By early 2025, ten rounds of national volume-based procurement (VBP) contracts had been concluded, covering 435 multisource products. In July 2025, the Joint National Procurement Office announced that 55 drugs had been selected for the 11th round of VBP. While competition will be fierce, some requirements of the process have been fine tuned. Hospitals were allowed to report their volume requirements by brand as well as generic name, while quality requirements for bidders have been tightened, and the lowest price will no longer necessarily be the winner. National VBP will be expanded further, focusing on multisource medicines that have obtained generic quality consistency evaluation (GQCE) certification. The impact will be amplified by provincial and multi-province VBP, for the continuing procurement of drugs previously subject to national VBP contracts, as well as for additional products not previously included in national VBP rounds, and for certain biologics. National VBP for multisource biologics could be introduced in 2026.

Implementation of flat-sum hospital payment models: Reforms to hospital payment systems, including a shift to flat-sum or case-based reimbursement, are gaining pace. An updated version of the Diagnosis-Related Groups (DRG)/Diagnosis Intervention Packets (DIPs) framework, with more scientifically driven disease groupings, was introduced in July 2024. In April 2025, the NHSA reported that payment models based on either DRGs or DIPs had been adopted in all regions, covering all eligible institutions and 95% of diseases. Under the system, medical facilities are reported to be managing costs more effectively, while the co-payment burden on patients is reducing. Defined budgets under case-based payment models are expected to drive stricter adherence to prescribing guidelines and protocols in the hospital setting. The outcome will be more cost-sensitive prescribing of at least established medicines, and quite possibly of more innovative therapies too, if they cannot find a place within DRG/DIP frameworks, while shifting some dispensing into the retail pharmacy sector.

Demographic factors: China's total population will decline throughout the forecast period, at an average annual rate of -0.2% and contracting by 17.40 million between 2024 and 2029. The 0-14 age group will see numbers decline most rapidly, at an average annual rate of -3.9%, while the 'working age' 15-64 age group will contract at an average annual rate of -0.9% over the five-year period. Only the 65+ age group will see numbers grow, at an average rate of 5.8%.

Strengthening of primary care: Strengthening primary care provision is a core health policy goal. Key objectives include establishing family doctor networks and a referral system, encouraging a tiered approach to diagnosis and treatment. The National Health Commission has called for the establishment of 500 community hospitals annually in 2024 and 2025 to support community health centers, as well as further improvements in service capacity standards of community and township health centers, and village clinics. It has also called for further strengthening of staffing and training of grassroots health personnel. Separately, the government aims to establish closely integrated medical and health 'consortia' in more than 90% of China's counties by the end of 2025, and in all county-level regions by 2027. Progress with strengthening primary care has been uneven, and most patients will still prefer to seek an initial diagnosis in higher tier hospitals, but primary care facilities will play a growing role in the management of chronic conditions, and the provision of healthcare for the elderly.

Generic competition 2025-2029: The impact is calculated based on historical data analysis, taking into account the average brand erosion rate following patent expiry, the average price differential, and the average time to peak generic erosion. These factors were applied to all non-biological products expected to lose patent protection over the five-year prognosis period. Generic erosion, price differential and market share of the brand were multiplied to derive the impact of each patent expiry on each market sector. These impacts were applied over time, according to the expected date of patent expiry and time to peak generic erosion. The year-on-year impacts for each patent expiry were then totaled to generate a total sector patent impact for each of the five forecast years. The leading products expected to lose patent protection over the forecast period include Ozempic (semaglutide) and Entresto (sacubitril+valsartan).

Growing financial pressures on the BMI system: The financial situation of the BMI system is deteriorating. Urban employee BMI (UEBMI) revenues for 2024 were RMB2.4 trillion, 3.5% higher than in 2023, but UEBMI expenditure growth outstripped revenue growth, rising by 7.8% to RMB1.9 trillion. While the urban and rural resident BMI (URRBMI) saw expenditure growth slow to below revenue growth in 2024, its current balance is just 4.9% of annual expenditure. The UEBMI will come under growing pressure in the medium to long term as more employees retire, the population ages, and demand for treatment for age-related conditions escalates. The growing challenges to healthcare financing through the BMI, which are compounded by the slowing economy, were reflected in reports of reimbursement funding shortages during the second half of 2024. The deteriorating financial situation will increase pressure for cost containment over the forecast period.

Consolidation in the retail pharmacy sector: Growth in the total number of retail pharmacy outlets slowed to 2.5% in 2024, from 7.0% in 2023, to reach 683,700, with chain pharmacies accounting for 57% of total numbers. The number of chains declined by 0.9%, however, and there are indications that pharmacy density is nearing saturation, with the number of people served per pharmacy declining. Progress with dual-channel dispensing of selected NRDL drugs in designated retail pharmacies has been slower than expected, and pricing pressures on prescription medicines in retail pharmacies are increasing. The retail pharmacy sector is expected to undergo a period of adjustment, as retail pharmacies are seeking to offer more specialized, value-added services such as examination and screening services, as well as chronic disease management. The retail sector is expected to remain an alternative sales channel to the dominant hospital sector, especially for products affected by VBP in hospitals and non-reimbursed products, however.

Impact from US tariffs and Executive Order 14297: After almost constituting a trade embargo in April 2025, tariff tensions between the US and China de-escalated following the 12 May 2025 agreement, which reduced base tariffs in both countries to 10%, but imposed a 20% ‘fentanyl tariff’ on Chinese imports into the US. Initially in place for a 90-day period, the agreement was extended for another 90 days, to 10 November 2025. While pharmaceuticals are exempt from ‘reciprocal’ tariffs under the current agreement, higher US tariffs facing Chinese products — along with US policy uncertainty — will hurt Chinese exports and dampen factory investment and consumer spending. A Section 232 investigation into pharmaceutical imports into the US (including active pharmaceutical ingredients, intermediates, and finished products), launched in April 2025, could lead to the imposition of sector-specific tariffs. US tariffs on pharmaceutical imports from China, as well as potential Chinese counter-tariffs on pharmaceutical imports from the US, could put upward pressure on drug prices and disrupt supply chains. Separately, US Executive Order 14297 of 12 May 2025 contains provisions for setting US drug prices by benchmarking them to the prices of the same product in other countries, effectively amounting to international reference pricing (IRP). While no details for the mechanism had been specified by mid-2025, if GDP per capita is a consideration in the selection of reference countries, Chinese drug prices are unlikely to be considered under the mechanism, given the difference in GDP per capita between the two countries. As of mid-September 2025, the event assumes only a marginal impact on the forecast, given the limited details on the implications of tariffs and pending the outcome of the Section 232 investigation on pharmaceuticals.

Event Overview

Table 8: Events Impacting the Market (2025-2029)

Event Title	Start Month	Start Year	Probability (%) short term	Probability (%) medium term	Probability (%) long term	Time to impact months	Retail Market Impact (%) volume	Hospital Market Impact (%) volume	Retail Market Impact (%) price	Hospital Market Impact (%) price
Positive Events										
Recent and future product launches 2027	1	2027	95	95	95	0	0.0	0.0	2.6	2.0
Recent and future product launches 2026	1	2026	95	95	95	0	0.0	0.0	1.8	1.4
Recent and future product launches 2025	7	2025	95	95	95	0	0.0	0.0	1.3	1.0
Reimbursement of innovative new drugs following price negotiations	7	2025	99	99	99	54	-0.2	0.8	-0.5	1.2
Improvements in health insurance coverage and new CHI innovative drug list	7	2025	70	80	90	54	0.5	1.5	0.2	0.5

Event Title	Start Month	Start Year	Probability (%) short term	Probability (%) medium term	Probability (%) long term	Time to impact months	Retail Market Impact (%) volume	Hospital Market Impact (%) volume	Retail Market Impact (%) price	Hospital Market Impact (%) price
Recent and future product launches 2028	1	2028	95	95	95	0	0.0	0.0	2.2	1.7
Improvements in the hospital infrastructure	7	2025	99	99	99	54	0.0	1.0	0.0	0.2
Recent and future product launches 2029	1	2029	95	95	95	0	0.0	0.0	1.9	1.4
Growth in e-health and telemedicine	7	2025	90	95	99	54	0.7	0.0	0.0	0.0
Negative Events										
Expansion of VBP in 2025	12	2025	90	90	90	1	0.0	0.5	0.0	-2.0
Expansion of VBP in 2026	1	2026	90	90	90	1	0.0	0.4	0.0	-1.6
Implementation of flat-sum hospital payment models	7	2025	70	80	90	54	0.3	-0.8	0.2	-1.0
Expansion of VBP in 2027	1	2027	80	80	80	1	0.0	0.3	0.0	-1.5
Demographic factors	7	2025	90	90	90	54	-0.9	-0.9	0.0	0.0
Strengthening of primary care	7	2025	70	80	90	54	0.0	-1.0	0.0	0.0
Expansion of VBP in 2028	1	2028	70	70	70	1	0.0	0.2	0.0	-1.4
Expansion of VBP in 2029	1	2029	70	70	70	1	0.0	0.2	0.0	-1.4
Generic competition 2027	1	2027	99	99	99	0	0.0	0.0	-0.2	-0.3
Growing financial pressures on the BMI system	7	2025	70	80	90	54	0.0	-0.2	0.0	-0.2
Generic competition 2025	7	2025	99	99	99	0	0.0	0.0	-0.1	-0.1
Generic competition 2026	1	2026	99	99	99	0	0.0	0.0	-0.1	-0.1
Generic competition 2028	1	2028	99	99	99	0	0.0	0.0	-0.1	-0.2
Consolidation in the retail pharmacy sector	7	2025	50	50	50	24	-1.0	0.0	0.0	0.0
Generic competition 2029	1	2029	99	99	99	0	0.0	0.0	-0.1	-0.1
Impact from US tariffs and Executive Order 14297	7	2025	20	20	20	54	-0.1	-0.1	0.1	0.1

The events are ranked according to their cumulative sales impact over the five-year forecast period.

Therapeutic Class Forecasts

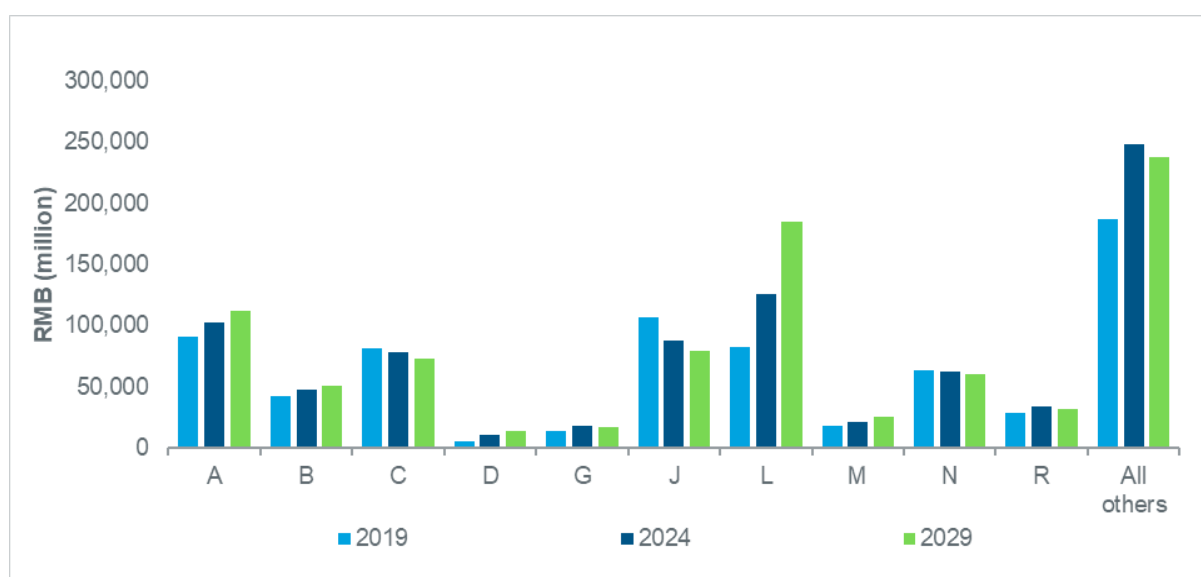
The following therapeutic class analysis focuses on ten first level categories of the Anatomical Classification System (ATC 1), and is based on audited retail and hospital sales at ex-manufacturer price level. These ten classes accounted for 70.3% of the market in 2024 and are expected to account for 73.2% in 2029.

Table 9: Therapeutic Class Forecasts* to 2029

ATC	2019		2024		2029		CAGR (%)	
	RMB (million)	Market Share (%)	RMB (million)	Market Share (%)	RMB (million)	Market Share (%)	2019-2024	2024-2029
A	91335	12.7	102500	12.3	111792	12.6	2.3	1.8
B	41838	5.8	47421	5.7	50793	5.7	2.5	1.4
C	81292	11.3	78551	9.4	73080	8.3	-0.7	-1.4
D	5451	0.8	10833	1.3	13502	1.5	14.7	4.5
G	13671	1.9	17545	2.1	17419	2.0	5.1	-0.1
J	106531	14.8	87645	10.5	79001	8.9	-3.8	-2.1
L	82837	11.5	125498	15.0	185336	20.9	8.7	8.1
M	18114	2.5	21631	2.6	25024	2.8	3.6	3.0
N	63541	8.8	62448	7.5	59942	6.8	-0.3	-0.8
R	28502	4.0	33527	4.0	31809	3.6	3.3	-1.0
All Others	186554	25.9	248072	29.7	237702	26.8	5.9	-0.9
Total Audited Market	719665	100.0	835671	100.0	885399	100.0	3.0	1.2
Unaudited Market	320164		359532		389742		2.3	1.6
Total Market	1039830		1195203		1275140		2.8	1.3

*Derived using data from IQVIA Analytics Link's Forecast Link module. For details, see Market Prognosis Sources and Methods.
Source: IQVIA

Sales by ATC



Forecasting Data and Methods

Data Sources

The forecast is based on the IQVIA **China Hospital Pharmaceutical Audit (CHPA)** and **China Retail Pharmacy Audit (CRPA)**, held on the IQVIA MIDAS database.

The forecast incorporates a projection of sales through unaudited smaller hospitals and clinics, based on the latest China National Audit (CNA) analysis conducted by IQVIA China in mid-2024, as well as segments of the retail pharmacy sector not held on MIDAS.

Hospital Sector

The CHPA covers pharmaceutical sales of ethical and over-the-counter products to hospitals with more than 100 beds nationwide. Sales to smaller hospitals, community health centers, township health centers and clinics are not covered. The CHPA accounted for approximately 64% of the total pharmaceutical market in 2024.

Enhancements to the CHPA implemented in April 2017 expanded the hospital universe, integrated new county and city audits, and improved the panel quality and design. The enhancements increased both value and volume, with back-data adjustments up to Q1 2015 inclusive, leading to a trend break.

In January 2020, the factors in use for the CHPA panel on MIDAS to convert sales from trade level prices to ex-manufacturer level prices were adjusted to reflect the now widely adopted 'two invoice' policy, which aimed to cut down distribution tiers and improve trade compliance. The adjustment increased the value of sales at ex-manufacturer level by around 10% and created a trend break in price growth and sales values between 2019 and 2020. There is no trend break in volume.

In January 2021, an upgrade involving a universe expansion, bed size class redefinition and sample quality/quantity enhancement was implemented, resulting in a significant increase in value and volume, with back-data adjustments to Q1 2019 inclusive, creating a trend break between 2018 and 2019.

In January 2025, several market measurement improvements were implemented, including an update of the hospital universe and a sample size enlargement; redefined hospital bed size classes; and quality and consistency improvements. This led to an increase in value and volume, with back-data adjustments to Q1 2023 inclusive, lifting the 2023 sales value by 2.9% and resulting in a trend break between 2022 and 2023.

Retail Sector

The CRPA covers sales of prescription-only products through retail pharmacies in cities at prefecture level and above, and accounted for around 6% of the total pharmaceutical market in 2024.

The CRPA is upgraded regularly to improve reflection of the evolving market. A universe upgrade and panel restructure were implemented in Q1 2019, resulting in a value increase of around 10%, with back-data adjustments to Q1 2017. Further improvements were implemented in Q1 2020, to reflect continued urbanization and expansion of outlets outside of the major cities and ensure continued representation of changing local market dynamics. Panel sample size increased by around 10%, while the universe size was increased by around 7%, also with back-data adjustments to January 2017.

In Q1 2021, the panel sample size was increased by a further 10%, to reflect continuing urbanization, with an increase in the universe, and back-data adjustments to January 2018. Further retail panel enhancements were introduced in Q1 2022, increasing and adjusting the retail sample size by 12% to better reflect market dynamics (with back-data adjustments to Q1 2021), and in Q1 2023, boosting the sample size by a further 10%, also with back-data adjustments to Q1 2021.

In Q1 2025, further enhancements were introduced to the CRPA, as well as the China PharmaTrend panels, including an increase in sample size and the inclusion of online-to-offline sales through retail pharmacies. The upgrade resulted in an increase of ~10% in CRPA and PharmaTrend panel sales values, with back-data adjustments to 2022, resulting in a trend break between 2021 and 2022.

Other Sectors

Unaudited hospital sector: This incorporates an estimate of the value of the unaudited hospital sector, including small hospitals with fewer than 100 beds, community health centers, township health centers and clinics, which together are estimated to account for just under 20% of the total market. The estimate is based on the latest CNA analysis conducted by IQVIA China in 2025, providing annual sales estimates up to 2024 inclusive. For the purpose of this forecast, the unaudited hospital sector has been projected on the basis of these annual estimates collected since 2012.

Other retail pharmacy segments: The prognosis also incorporates a projection of the value of segments of the retail pharmacy sector not held on MIDAS, which together are estimated to make up around 64% of the total retail pharmacy sector, and almost 11% of the total market.

These segments include:

- OTC drugs sold through retail pharmacies in prefecture-level cities and above, audited as part of the China PharmaTrend panel, which accounted for around 31% of the total retail pharmacy market in 2024.
- Sales through retail pharmacies in county-level cities, based on the new county level PharmaTrend panel, which accounted for around 33% of the total retail pharmacy sector in 2024.

For the purpose of this forecast, these segments have been projected separately, and then combined as 'other retail'.

Market Sales Share (2024)

Market Sector	Share of the Market (%)
Hospital sector > 100 beds (CHPA)	64.0
Retail sector (Rx drugs at prefecture city level)	5.9
Total audited on IQVIA MIDAS	69.9
Unaudited hospital (hospitals with < 100 beds, community health centers, township health centers, clinics)	19.6
Other retail (OTC sales at prefecture city level, county city level pharmacies)	10.5

Source: IQVIA

Rebates and Discounts

The CHPA reports the market at weighted average purchase prices, at which the panel hospitals purchase products from wholesalers, distributors and manufacturers, while the CRPA reports the market at average purchase prices at sample stores. For Market Prognosis, hospital and pharmacy purchase prices are converted into ex-manufacturer prices.

While discounting occurs on a modest scale, there are no significant off-invoice rebates or discounts causing distortion of these price levels in China for the time being.

Baseline Extrapolations

The IQVIA Market Prognosis projections are based upon ten years of volume and price quarterly sales data. Pharmaceutical volume and price data in each audited market sector are independently analyzed and projected using the Forecast Pro comprehensive forecasting and forecast management system. Forecast Pro contains a wide range of forecasting models, including exponential smoothing and ARIMA, and an Expert system that identifies the best fit for each historical data set. Adjustments are made where necessary to reflect market understanding. Econometric analysis is performed and econometric impacts are applied to the baseline where relevant. The macroeconomic variables are provided to IQVIA by a third party, the Economist Intelligence Unit (EIU).

Retail Sector Baseline Extrapolations

Table 10: Baseline Retail Sector (2024-2029)

Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2024	97145	-3.3	0.72	4.1	70281	0.7
2025	92014	-5.3	0.76	5.2	70029	-0.4
2026	87608	-4.8	0.79	3.8	69203	-1.2
2027	83704	-4.5	0.82	3.7	68536	-1.0
2028	80146	-4.3	0.85	3.6	67998	-0.8
2029	76980	-4.0	0.88	3.5	67594	-0.6

Source: IQVIA

Hospital Sector Baseline Extrapolations

Table 11: Baseline Hospital Sector (2024-2029)

Year	Volume (Standard Units)		Average Price per Standard Unit		Sales	
	SU (million)	Annual Growth (%)	RMB	Annual Growth (%)	RMB (million)	Annual Growth (%)
2024	369794	14.2	2.07	-10.4	765389	2.3
2025	392538	6.2	1.95	-5.8	765148	0.0
2026	416207	6.0	1.86	-4.8	772121	0.9
2027	439484	5.6	1.77	-4.6	777658	0.7
2028	462849	5.3	1.69	-4.4	783352	0.7
2029	487171	5.3	1.63	-4.0	791881	1.1

Source: IQVIA

Economic Indicators

Table 12: Key Macroeconomic Indicators

	2025	2026	2027	2028	2029
Real GDP 2010 RMB (billion)	105266	110003	114623	118979	123310
Annual Growth (%)	4.7	4.5	4.2	3.8	3.6
Nominal GDP RMB (billion)	140541	146134	152053	157983	163986
Annual Growth (%)	4.2	4.0	4.1	3.9	3.8
Consumer Price Index 2010 = 100	132.1	132.3	132.9	134.3	135.9
Annual Growth (%)	-0.3	0.1	0.5	1.0	1.2
Real Private Consumption 2010 RMB (billion)	39846	41719	43596	45427	47274
Annual Growth (%)	4.8	4.7	4.5	4.2	4.1
Real Government Consumption 2010 RMB (billion)	16663	17563	18476	19381	20273
Annual Growth (%)	5.2	5.4	5.2	4.9	4.6
Exchange Rate RMB:US\$	7.18	7.01	7.12	7.09	7.09
Annual Growth (%)	-0.2	-2.4	1.6	-0.4	0.0

Source: Economist Intelligence Unit, July 2025

BUSINESS ENVIRONMENT

IQVIA Market Prognosis Economic Environment and Political Environment chapters are provided by the Economist Intelligence Unit (EIU). The following are excerpts from the China five-year forecast report, published in August 2025.

Economic Environment

Economic Growth

China's economic prospects will face headwinds in 2025, although the government's stimulus measures and trade front-loading will offset some of the pressure. Tariff tensions between the US and China have de-escalated from almost reaching a trade embargo in April 2025. However, higher US tariffs facing Chinese products (with the US weighted average tariff rate on China estimated at around 40%) — along with US policy uncertainty — will not only hurt exports but also dampen factory investment and consumer spending. Some shocks will be mitigated by a greater level of government stimulus, lending support to private consumption and fixed investment. The risk of a fresh escalation of bilateral tensions will be high, but China's deployment of tariff and non-tariff measures will deter the US president, Donald Trump, from more extreme trade measures. The EIU expects real GDP growth to decelerate mildly to 4.7% in 2025, from 5.0% a year earlier.

Private consumption will be resilient, outpacing headline GDP growth, as the government tries to stimulate consumer spending with subsidies, income transfers and other non-financial incentives. The performance of fixed investment will be mixed; manufacturers grappling with excess industrial capacity and trade uncertainties will slash their capital expenditure (or divert it overseas) and property developers will continue to deleverage, but the government will expedite infrastructure spending, particularly on strategic projects that help to bolster China's resilience against natural disasters and geopolitical conflicts, such as the Yarlung Zangbo Medog dam project, a major (but geopolitically controversial) infrastructure project planned during the 14th five-year plan era. Domestic demand will therefore help to offset some of the loss of external demand.

Growth will slow further in 2026-2029. The US-China rivalry will drag on, against the backdrop of heightened US protectionism more generally; this will continue to hit China through weaker global demand. Property-related investment will continue to decline before finally stabilizing at a low level in 2027-2028, while the manufacturing sector will record a better performance as global supply chains settle into a new reality and China's own concerns over industrial overcapacity ease. Improvement in the local fiscal environment will allow for better provision of public services. China's years-long investment in advanced manufacturing and technologies will pay off in some areas, providing upside risks to our growth forecasts. This also suggests, however, that “legacy industries” (spanning traditional sectors like steel, glass and cement, but also some downstream industries like electronics, textiles and machinery) will probably face disruptive consolidation.

The EIU remains concerned about the slack in the labor market. The private sector has been cautious about hiring, and the room for the public sector — including government bodies, public institutions (such as schools and hospitals) and state-owned enterprises (SOEs) - to recruit is limited; it is insufficient to absorb the unemployed and a record high number of new graduates. The ability of the

construction sector to hire labor has been weakened by the sustained contraction of the property sector. Meanwhile, China's giant manufacturing sector has become more capital- and technology-intensive, relying less on labor input. The emergence of artificial intelligence (AI) technology has also reduced labor demand. Elevated underlying unemployment (China's official statistics underestimate the true picture of unemployment) will put a dent in consumer demand and pose risks to social stability.

Demographic issues will be a major constraint on long-term growth, especially as the working-age population shrinks more quickly by the late 2020s. The productivity outlook is mixed; while efforts at industrial upgrade will deliver gains, they will be offset in part by unambitious market reforms.

Inflation

China faces significant challenges around reflating its economy. A combination of persistent demand-side weakness, structural overcapacity and damaging sectoral price wars (most notably in the automotive sector) have kept deflationary pressures fierce, to the detriment of corporate profit margins and consumer willingness to spend (given the compounding effects of deflationary expectations). On average, the EIU expects consumer prices to fall by 0.3% in 2025, before returning to marginal positive growth of 0.1% in 2026.

Aware of the deflationary pressures gripping the country, policymakers will attempt to reflate the economy via public spending and monetary easing, which will modestly lift prices. However, the deflationary impact of the trade war — through weaker global demand, intensified industrial overcapacity and slips in the oil price — will be a dominant factor in shaping the near-term outlook for inflation.

Similar dynamics will drive producer prices lower in 2025, when the EIU expects another year of contraction (-3.6%). The central government has sought to ease industrial overcapacity by facilitating production cuts in several sectors, but the effects will be mixed. Local governments are motivated to protect their local industries (even loss-makers), especially in sectors with high employment implications, as well as strategic ones such as renewable energy and electric vehicles. This will maintain overcapacity strains. An escalation of tariffs between China and the US and its impact on commodity prices will put further pressure on ex-factory prices as US importers ask for price cuts from Chinese manufacturers to absorb tariff costs.

Inflation will pick up later in the forecast period but will remain significantly below the government's new 2% target, averaging just 0.7% in 2026-2029. The imbalance between supply and demand will persist. Limited improvements in social welfare will constrain inflation through tepid private consumption. The long-term outlook is mixed; a mismatch between job market supply and demand will emerge over time — albeit beyond the forecast period — and put a floor under wage growth, while population ageing (and resulting labor shortages) will give rise to wage-led inflation. However, AI diffusion will provide a counter-factor by subduing labor demand.

Population

China's total population will decline throughout the forecast period, at an average annual rate of -0.2%, contracting by some 17.40 million between 2024 and 2029. The 0-14 age group will decline most rapidly, at an average annual rate of -3.9%, while the 'working age' 15-64 age group will contract

at an average annual rate of -0.9% over the five-year period. Only the 65+ age group will see numbers grow, at an average rate of 5.8%. Numbers in this cohort will reach 271.20 million or 19.3% of the total by 2029, when the age dependency ratio will be 28.8%, compared with 20.7% in 2024.

Official registered average unemployment stood at 5.1% in 2024 and is forecast to rise to 5.4% in 2025, before slowing again gradually to stand at 5.0% in 2029.

Fiscal Policy

Bolder fiscal expansion will characterize much of the forecast period, as the authorities look to support growth and meet increasing spending obligations such as those related to social security. A general government budget deficit equivalent to 4% of GDP — a record high — was arranged for 2025. The actually realized budget deficit tends to be wider; the EIU expects it to reach 5.9% of GDP in 2025, wider than the average of 4.7% of GDP in 2020-2024. Additional funding will be earmarked for public-sector wage increases, pension rises, stronger social security spending and cash transfers to disadvantaged sections of the population.

Outside the headline deficit, the government will increase the issuance of special sovereign bonds and local-government special bonds, which contribute to the broader deficit. The former will support consumer subsidies, equipment upgrade, strategic investments and recapitalization of large banks. The latter will be earmarked for replacing off-the-books local debt (and therefore alleviating pressure on troubled local public finances), acquiring unsold homes and undeveloped land, facilitating urban renovation and repaying arrears owed to companies.

The EIU does not expect massive local government debt defaults in the forecast period, as the central government has stepped up its intervention. A growing amount of debt borrowed by local government financing vehicles has been repaid, taken onto the books of the central or provincial governments or refinanced at lower rates and longer terms, thereby mitigating the debt-servicing pressure on those entities. Over time, this will help local governments to alleviate deleveraging pressure, function better and implement state policies. Still, their fiscal stress will last for a couple of years as deflation in the economy hurts tax collection and the property downturn continues to suppress land sales. Local governments will face ongoing difficulties in increasing welfare spending without central support.

Fiscal reform will gain momentum over 2025-2029, with the aim of making the public finances more sustainable, especially at the local level. Authorities could allocate a larger share of tax revenue directly to lower levels of government and/or transfer some responsibilities from local government to the center. Attempts to increase fiscal revenue are likely, including by reinforcing tax administration (China has stepped up tax inspection on overseas income of Chinese nationals), exploring new forms of taxation and overhauling consumption tax. Taxation on property ownership is outside the EIU's core forecast.

Monetary Policy

China intends to support the economy through monetary easing. The People's Bank of China (PBC, the central bank) will double down on its efforts to drive up inflation, having stated "a reasonable rise in prices" as an important objective; this reflects concern over entrenched deflation. The EIU expects a multi-pronged approach to monetary easing. Reductions in the policy interest rate — the seven-day reverse repo rate — will continue, with considerations over the stability of the renminbi exchange rate

becoming less of a constraint in the context of a weaker US dollar. After a cut of 10 basis points to the policy rate in early May, the EIU expects the rate to fall by another 20 basis points, to 1.2%, by end-2025.

The PBC will also pursue other moves, such as lowering deposit rates, cutting banks' reserve requirement ratios and offering concessional interest rates for its structural monetary policy tools, including re-lending instruments in support of policy goals. As part of an effort to shore up the economy, new policy finance tools have been launched to boost technological development, private consumption and international trade, providing low-cost credit to targeted industries and regions.

Asset prices now command greater importance in the PBC's decision matrix, reflecting a recognition of their importance to the economy through wealth effects. The central bank has brought down mortgage rates significantly to stimulate housing demand (in order to stabilize housing prices) and has launched two relending tools to bolster activity in the stock market. The government has also motivated insurance companies and pension funds to increase their equity investments.

Government bond issuance will play a pivotal role in credit expansion and is on course to reach a historic high in 2025. Apart from the positive effects of monetary easing on credit demand, the capital replenishment of China's large state-owned banks — financed by the issuance of special sovereign bonds — will reinforce their lending capacity.

Foreign Trade and Current Account

Export growth will slow over 2025-2026, following strong performance in 2024, owing to the US's tariff and non-tariff measures on China, as well as their second-order impact on US demand and global economy. Direct exports to the US will drop amid still-high US tariffs, but these factors will be offset by growth in shipments to other markets (both as part of transshipment activities for goods ultimately destined for the US, and as Chinese exporters cultivate alternative sources of demand).

Nevertheless, these other markets may also be frustrated by the inflow of cheap Chinese goods and could launch trade defense measures in response. Services exports, especially related to travel, will achieve solid growth as a result of China's visa-free policies for a handful of countries, although these will remain small compared to goods exports.

Import growth will remain subdued over 2025-2029. China's imports of aluminum and iron ore are constrained by a sluggish construction sector. The country's appetite for crude oil is limited by its green transition, which is creating sustained demand for copper instead. Efforts to build strategic reserves of major commodities will not fully offset such slack in demand. Meanwhile, domestic substitution efforts (and, in some cases, trade restrictions imposed by the US and its allies) will reduce China's reliance on imported capital and intermediate goods, including machinery, semiconductors and cars. China will instead import more low-value-added goods, given the decline in related domestic production in areas such as garments. Services imports will grow, with outbound tourist departures expected to gradually revert to pre-pandemic levels by 2026.

Exchange Rates

The EIU expects the renminbi to be largely capable of fending off the pressure from US tariffs. The government will prioritize currency stability over offsetting the impact of trade tariffs and other punitive trade measures via currency depreciation. The Chinese authorities have effectively maintained a peg

between the renminbi and the US dollar over much of the past two years, when sell-off pressures were high, and seem determined to intervene in the foreign-exchange market in the event of significant shocks. The recent weakness in the US dollar, reflecting the shrinking global investor confidence in the currency, has lent further support to the renminbi. The EIU forecasts that the renminbi will trade at an average of RMB7.18:US\$1 in 2025. The Chinese currency will appreciate moderately against its US counterpart in 2026-2029, driven by more aggressive monetary easing in the US and persistent investor doubt over US policy under Mr Trump. The central bank will also ensure the broader stability of the renminbi exchange rate against major non-US currencies.

China will promote the use of the renminbi as a reliable international currency, at a time when confidence in the US dollar is eroded by Mr Trump's policies. The outlook for renminbi internationalization is mixed, however. China could commit a greater amount of overseas aid and loans, along with other forms of support (including in supply chains), in exchange for the expansion of the renminbi's role in global settlement and financing, as well as a reserve currency. Nevertheless, despite the growing percentage of the renminbi in trade settlement, the currency continues to make up less than 2% of global trade invoicing — suggesting limited (albeit growing) uptake among international trade partners. The renminbi will not emerge as a credible substitute to the US dollar in the medium term, reflecting both China's capital account restrictions as well as the enduring safe-haven status of the US dollar (and the attractiveness of long-term US government bond yields). These factors will persist despite moderating investor appetite for the greenback tied to Mr Trump's presidency, which the EIU expects to be most pronounced in the short term.

Key Macroeconomic Indicators

	2025	2026	2027	2028	2029
Real GDP 2010 RMB (billion)	105266	110003	114623	118979	123310
Annual Growth (%)	4.7	4.5	4.2	3.8	3.6
Nominal GDP RMB (billion)	140541	146134	152053	157983	163986
Annual Growth (%)	4.2	4.0	4.1	3.9	3.8
Consumer Price Index 2010 = 100	132.1	132.3	132.9	134.3	135.9
Annual Growth (%)	-0.3	0.1	0.5	1.0	1.2
Real Private Consumption 2010 RMB (billion)	39846	41719	43596	45427	47274
Annual Growth (%)	4.8	4.7	4.5	4.2	4.1
Real Government Consumption 2010 RMB (billion)	16663	17563	18476	19381	20273
Annual Growth (%)	5.2	5.4	5.2	4.9	4.6
Exchange Rate RMB : US\$	7.18	7.01	7.12	7.09	7.09
Annual Growth (%)	-0.2	-2.4	1.6	-0.4	0.0

Source: Economist Intelligence Unit, July 2025

Population by Age Distribution (million)

	2025	2026	2027	2028	2029
0-14	222.69	214.92	206.24	197.14	188.59
Annual Growth (%)	-3.2	-3.5	-4.0	-4.4	-4.3
15-64	973.06	965.75	958.26	950.75	942.49
Annual Growth (%)	-1.2	-0.8	-0.8	-0.8	-0.9
65+	220.34	232.24	245.18	258.25	271.20
Annual Growth (%)	7.8	5.4	5.6	5.3	5.0
Total Population	1416.10	1412.91	1409.68	1406.14	1402.28
Annual Growth (%)	-0.2	-0.2	-0.2	-0.3	-0.3

Source: Economist Intelligence Unit, July 2025

Political Environment

Political Institutions

The People's Republic of China is a single-party state, governed by the Chinese Communist Party (CCP). The CCP wields broad executive, legislative and judicial powers, which are exercised jointly by the party itself, the Central People's Government, and by their respective provincial and local branches.

All executive and legislative appointments are for five-year terms, and officials may serve a maximum of two consecutive terms in office, although term limits for the president were abolished via a revision to the constitution in 2018. The CPP Congress, which meets at least once every five years, elects a Central Committee numbering 204 members. In turn, the Central Committee is responsible for selecting a 25-strong Politburo. Executive power is concentrated in the hands of the Politburo's standing committee (PSC), which controls all senior political and judicial appointments.

While they are effectively chosen by the PSC, executive appointments are approved by the National People's Congress (NPC) – a unicameral legislature whose close to 3,000 delegates are selected variously by provinces, municipalities, autonomous regions, and the armed forces. The NPC is in full session for only two to three weeks a year, and its 200-strong standing committee is responsible for handling day-to-day parliamentary affairs.

The official government comprises the president and a cabinet-like State Council, which is headed by the prime minister, and which includes a 15-member executive committee responsible for overseeing government ministries, commissions, and major state-owned enterprises.

Moves to establish a functioning legal system began in the late 1970s – largely in response to financial reforms and the demands of a growing economy. Civil, criminal, and commercial codes have all been updated regularly since then, while more transparent, adversarial trial procedures have been introduced. The judicial system comprises four tiers, from Basic People's Courts to the Supreme People's Court, which is the highest court in the land. Notwithstanding recent reforms, the judiciary is not independent of the CCP, which approves all court appointments, and which has the power to intervene in the deliberations of any court.

Current Political Environment

Domestic politics will be stable in 2025-2029. The dominance of the president and general secretary of the Chinese Communist Party (CCP), Xi Jinping, has reshaped China's political dynamics. Mr Xi's allies hold major roles in the CCP, armed forces and state bureaucracy. The president's decision-making powers have been cemented through several CCP central commissions under the direct leadership of Mr Xi and his confidants. This has weakened the policy influence of the State Council (cabinet). Other forces within the CCP, including the more liberal China Youth League faction, have been sidelined. Mr Xi may remain in power into the 2030s.

China's economic challenges pose a risk, highlighting a mixed record in tackling the structural issues that are frustrating business and household confidence. The likelihood of social unrest is limited by the CCP's tight control of the military and the public security apparatus, but could nonetheless rise if the government mishandles the economy in a way that leads to a deterioration in living standards

The authorities are responding to risks to social instability by increasing government spending and enhancing policy support in areas such as social security and income inequality.

Policy Trends

The shocks from Mr Trump's second term as US president, alongside persistently soft domestic demand, will cause Chinese policymakers to adopt a more supportive stance towards the economy in 2025-2026. This pivot had already begun to materialize by late 2024, which was ultimately followed by the passage of a stimulus package in March 2025. Officials are preserving the option to roll out additional support packages, which will address three broad aims around reviving domestic demand; reflating goods and asset prices (namely housing and stock market); and alleviating debt burdens facing local governments. Even so, recent signs of resilience in China's economy will keep policymakers cautious about excess stimulus; officials tend to make policy decisions based on current economic conditions, rather than forward-looking assessments around potential shocks. The EIU consequently does not expect more expansionary policies to materialize until the end of the third quarter.

Boosting private consumption now commands a higher priority. China has expanded its consumer goods subsidies, providing more income transfers (including nationwide birth and child-rearing subsidies) and enhancing social security benefits, although the total size of direct financial support is moderate relative to China's GDP.

The government has also offered policy support for consumption loans and spending on services. Non-financial measures have been rolled out as well, including the curtailment of overtime in the workplace (increased leisure time is perceived to incentivize consumer spending) and supply-side efforts to boost demand by providing "high quality" goods and services. That said, promoting investment remains a main policy focus that will receive a higher amount of public financial support through government bonds.

The Chinese leadership is increasingly aware of the role of asset prices in the economy through wealth effects, with the housing and equity markets so far receiving the most attention. The EIU expects government policy to result in a milder slump in residential property prices in 2025 and an eventual stabilization in 2026. However, policymakers will no longer use property as a means to stimulate the economy. New home sales — and, by extension, construction activity — will continue to decline in 2025-2026, influenced by the government's directive to curb new supply. The contribution of the housing sector to the broader economy will consequently shrink further.

Structural economic reform will remain selective and gradual. Among other priorities, the government will: improve the social safety net for disadvantaged groups (such as the older rural population); enhance the economic rights of China's rural population and encourage their urbanization; enhance fiscal sustainability and guide the market as part of its "anti-involution campaign", which has exacerbated existing overcapacity. However, decisive reforms will still be lacking in several important areas. For instance, the EIU does not foresee a significant liberation of policy towards foreign enterprises (which would grant them many more market opportunities). Concerns over social stability and local employment will prevent drastic efforts to consolidate sectors with overcapacity, such as automotives.

Officials are inclined to balance economic growth with national security, which is now subject to a much wider definition than in the 2010s. Apart from the build-up of military capabilities and anti-

espionage efforts, hefty investments will be made in bolstering the country's self-sufficiency and resilience against external shocks in major areas; these include food, energy, medical products, supply chains and technology.

The state's emphasis on developing “new productive forces” – a term coined by Mr Xi – captures the desire for technological breakthroughs and their diffusion in the economy. Relative to other markets, China's innovation model will continue to be defined by the heavy role of the state in guiding the market and providing capital. The emergence of homegrown technological alternatives will help to address major security concerns (for example, by gradually reducing reliance on imported semiconductors), but at the expense of foreign businesses that will see their domestic market share decline relative to homegrown competitors (partly as a result of government directives, via procurement and licensing decisions, as well as explicit and implicit policy guidance to phase out foreign products).

This initiative has become a priority on China's policy agenda amid the rapid deterioration in US-China relations, which the EIU expects will ultimately curtail Chinese access to vital US technologies. In the long run, China also hopes that these investments will translate into value added for the economy and higher productivity, offsetting the drag from a withering property market and shrinkage of the labor force tied to demographic ageing.

International Relations

Relations with the US will remain under heavy strain in 2025-2029, particularly as a result of the tariff actions imposed by the US administration under Mr Trump. The bilateral weighted average tariff rate (WATR) between the US and China briefly soared above 100% in April-May 2025, in effect creating a trade embargo between the two economies. A truce — struck in May, then reaffirmed in June and extended in August — has temporarily reduced tariffs on Chinese firms to around 40%, but this rate remains significantly higher than those faced by other Asian countries.

Further US tariff reductions are possible, but the risk of fresh escalation is also high. The trade détente will also do little to reverse a loss of mutual trust. The US is also increasing its scrutiny over Chinese export transshipments via third countries, such as Mexico and Vietnam, against which it has threatened 40% tariffs. Even so, the EIU expects US authorities to struggle to stop such illicit trade activities (ranging from product rerouting, rebranding, mislabeling and smuggling), given the lack of guidance on what counts as a transshipment; this, in turn, could keep Chinese trade flows relatively resilient over the coming quarters.

Conflict between the US and China will be intense in other areas, including supply chains for critical minerals, cross-border investment and advanced technologies such as AI chips. Even if the two countries reach a temporary agreement over their tariff regime, shipments of fentanyl (a synthetic opioid) and export controls, the EIU would not expect any such accord to prevent an eventual reignition in bilateral tensions. The disagreements between the two sides will prove too wide to be bridged. A deal that is broad in scope, secures significant compromises from either side and can be effectively enforced, will be impossible to strike.

An attempt by China to claim Taiwan militarily remains outside the EIU's forecast in 2025-2029. The likelihood of US intervention and coordinated economic and financial sanctions from the West will act as the main deterrent. Nevertheless, the risk profile for the EIU's forecast period remains elevated, given the strong hostility towards mainland China by the current Taiwanese government, led by Lai

Ching-te. The frequency of Chinese military exercises, as well as economic coercive measures against Taiwan, will intensify over 2025-2029.

Mr Trump's penchant for isolationism will provide an opportunity for China to redefine diplomatic relations with the US's traditional allies. It is already seeking to de-escalate tensions with the EU, where strains have emerged owing to Europe's perceived Chinese support of Russia's invasion of Ukraine, as well as the bloc's tariffs and other punitive trade measures against specific Chinese goods. Limited progress has been made so far; the EU remains concerned about China's market openness and the potential redirection of Chinese exports (which have been shut out of the US) into the European market.

China will see some success in these diplomatic overtures. However, while cross-border travel, cultural exchanges and certain areas of trade are likely to benefit, regional security issues and trade diversion risks will continue to cast a shadow on EU-China relations. For example, despite Chinese efforts to mend ties with the EU, the recent imposition of tit-for-tat bilateral banking sanctions affirms that their diverging positions on the Russia-Ukraine war has derailed progress on improving their relationship. This combination of factors informs the EIU's view that EU-China economic and political tensions will worsen again in the second half of 2025.

China is courting emerging economies to counteract the deterioration in its relations with the West, in order to safeguard its supply chains for energy and other commodities and to generate new commercial opportunities. China maintains close ties with Russia, and has refrained from joining US efforts to reach a peace deal with Ukraine. That said, China-Russia relations are not rock-solid — they are hampered by pressure from the EU, China's reluctance to import more Russian energy and some emerging trade disputes (including obstacles facing Chinese automotive exports to Russia).

China is increasing bilateral trade and investment with many member states of the Association of South-East Asian Nations (ASEAN) and Gulf Co-operation Council (GCC). Those countries will generally take a pragmatic approach to their ties with China, prioritizing economic development. Some countries will levy tariffs on inflows of Chinese goods, however, which will be rerouted to third markets in the face of heightened US tariff barriers. This could be seen by China as them "choosing sides" in the face of US protectionism.

The Belt and Road Initiative, designed to bolster China's economic and diplomatic influence abroad, will remain a major foreign policy focus in 2025-2029. Greater isolationism by the US — made evident in its slashing of foreign aid — creates room for China to expand its influence in regions such as Africa and Latin America. China has limited capacity to fill the gap left by the US, but it could leverage the opportunity to secure more strategic benefits by guiding its aid and investment flows. Meanwhile, China's overseas financial support will continue to be dominated by commercial lending, rather than aid and grants. Driven increasingly by considerations over financial returns, China now focuses on projects that are smaller in scale, capable of delivering higher returns on investment and less prone to geopolitical risks. As its relations with the US deteriorate, China is poised to accelerate the adoption of its currency, the renminbi, in overseas lending and international trade settlements, although progress will be gradual; China's wide current-account surplus and capital-account restrictions remain visible obstacles, even as the currency's adoption in global trade continue to rise.

Source: Economist Intelligence Unit, China five-year forecast report, published August 2025

KEY DEVELOPMENTS AND TRENDS

Published: March 2025

Healthcare Provision

Healthcare System

China has achieved basic medical insurance coverage of around 95%. However, significant disparities remain in the availability, accessibility and quality of healthcare services, and the system is struggling to cope with rising demand. Policymakers are pursuing further reforms, designed to improve the capacity, quality, organization and prioritization of healthcare provision, expand insurance coverage, harmonize access to healthcare services, and encourage more efficient use of available health resources.

With growing demand and a slowing economy, funding pressures on the healthcare system are increasing. While the healthcare system remains strongly oriented to larger hospitals, moving healthcare out of hospitals, especially at the tertiary level, and into primary care facilities, community-level healthcare institutions and other grassroots outlets will remain the overarching objective. Further structural reform will be sought by establishing national and regional medical centers, encouraging care integration, strengthening the health workforce and prioritizing capacity-building at grassroots level.

Key Developments and Trends

Demand for Care Rebounded Strongly in 2023

- Following subdued demand under the impact of the COVID-19 pandemic on patient access to healthcare (see Country Context: Healthcare Provision: COVID-19 Pandemic), demand rebounded strongly in 2023. Both inpatient admissions and outpatient visits to healthcare facilities increased significantly in 2023. Inpatient admissions to all healthcare facilities reached 301.9 million in 2023, 22.2% higher than in 2022. There were 9.5 billion outpatient visits (an average 6.8 per head of population) in 2023, 13.3% more than in the previous year.
- Growth in inpatient admissions was particularly marked (+21.9% in hospitals, +25.6% in grassroots institutions), with patient numbers exceeding previous peaks in 2019, prior to the pandemic. There was also double-digit growth in outpatient consultations at both hospital and grassroots level, with patient numbers some way ahead of previous highs in 2021 and 2019, respectively.
- While the strong growth in 2023 represented a recovery in demand following the end of the zero-COVID policy, healthcare demand will continue rising over the forecast period, with accelerated population ageing also weighing heavily on resources.

Growing Healthcare Funding Pressures

- At the same time, pressures on government healthcare funding are growing, as the pandemic's impact on funding is aggravated by slowing GDP growth. The financial situation of the basic medical insurance (BMI) schemes is deteriorating, hitting city and hospital finances (see Health Insurance).
- In the urban and rural residents' BMI (URRBMI) scheme, the current balance between income and expenditure reached declined to just 1% in 2023 (versus 8% in 2022). This

mostly reflected strong year-on-year growth in expenditure, which was driven partly by pent-up demand from restricted access under pandemic conditions.

Reforms Continue to Target Tiered Healthcare System

- Healthcare delivery remains strongly oriented to larger hospitals, where patients expect to find superior expertise and resources. While outpatient consultations managed by primary care institutions (including community health centers, township health centers and clinics) increased by 15.7% to 4.9 billion in 2023, representing 52% of outpatient consultations, there were still 4.3 billion hospital outpatient consultations (+11.5% year on year). Almost 84% of these occurred in public hospitals, and 62% in tier 3 institutions.
- Moving healthcare out of hospitals, especially at the tertiary level, and into primary care facilities, community-level healthcare institutions and other grassroots outlets, will remain a central plank of health policy. The overarching goal is to establish a tiered healthcare system, with a solid network of family doctors and functioning referral system. More healthcare will be delivered locally where possible, while delivery will be better aligned with the level and severity of patient need, as well as demographics and socio-economic parameters.
- Alongside the establishment of national and regional medical centers, and encouragement for integrated medical consortia and close-knit medical communities, capacity-building in grassroots medical services will be supported by leveraging resources from tier 2 and tier 3 hospitals. This may be through staff transfers, pairing programs, 'medical tours' or digitally aided remote consultations.
- Underpinning these structural shifts will be continuing initiatives to expand workforce capacity, particularly at grassroots level. According to National Health Commission (NHC) data, the total health workforce in China reached 15.2 million in 2023, an increase of 5.7% over 2022. This included 4.8 million licensed and assistant physicians (+7.8%) and 5.6 million registered nurses (+7.9%).
- While improving, the ratio of practicing nurses per 1,000 population (4.0) in China still lags far behind the OECD average of 9.2 (in 2021). In terms of physicians, the NHC data show that there were 3.4 licensed and assistant physician per 1,000 in 2023. This compares to an OECD average for doctor density of 3.7 per 1,000 (in 2021). General practitioners remain relatively thin on the ground, though, with only 4.0 per 10,000 population in 2023.

Health Policy

Health policy has been shaped for the last 15 years by the comprehensive reform program launched by the government in 2009. With basic medical insurance (BMI) coverage now covering most of the population, the focus has shifted to the expansion and harmonization of insurance benefits, coupled with improvements in the quality and efficiency of healthcare provision. This entails hospital management and financing reforms, implementing a tiered diagnosis and treatment system, and measures to limit costs associated with enhanced BMI coverage.

Health policy will continue to pursue a multi-tiered, integrated healthcare system. Short to medium-term objectives include improving health service capabilities and infrastructure, especially in the primary care sector. Priorities also include further reform of public hospitals; expansion of health

insurance coverage; and encouraging innovation in the pharmaceutical sector. Health policy will also seek to improve prevention and treatment of chronic and rare diseases. The clampdown on corrupt practices in the health sector will be sustained.

Key Developments and Trends

Key Health Policy Directions

- Key health policy priorities highlighted by the State Council in June 2024 to deepen the health system reform process in the second half of the year included strengthening health service capabilities and the healthcare infrastructure; further improvement and reform of public hospitals; development of the health insurance system; and deepening reform and innovation in the pharmaceutical sector.
- Further strengthening the primary care infrastructure, and with it a progressive shift towards a tiered diagnosis and treatment system remains a key objective. This involves, for example, improving the primary care infrastructure, as well as enhancing grassroots capabilities by encouraging secondary and tertiary hospitals to organize staff transfers, training and medical tours (see Primary Care).
- Digital resources, such as telemedicine, mobile health, intelligent auxiliary-diagnosis systems and public data assets, will also help to extend high-quality medical services to grassroots facilities. These initiatives go hand-in-hand with efforts to drive care integration and consolidation by, for example, building more national or regional medical centers ('dual centers'), co-ordinating medical and preventive services, and establishing close-knit medical consortia and communities at urban or county level (see Hospital Sector).
- Continuing reforms at public hospitals will include improving accessibility; stepping up clinical research at high-level facilities; reforming and promoting the revision and dynamic adjustment of medical service prices; strictly prohibiting revenue-generating targets linked to medical products or services; and optimizing hospital-based nursing (see Hospital Sector).
- A significant component of health policy in public hospitals is the progressive application of diagnosis-related group (DRG) and 'disease intervention package' (DIP) mechanisms to BMI provider payments (see Hospital Sector). Case-based reimbursement models have already been piloted extensively at city level, in an effort to promote cost efficiency.
- In the private hospital sector, restrictions on foreign investment in Chinese hospitals are being eased through pilot programs allowing wholly foreign-owned hospitals in nine areas of China, including the capital Beijing (see Hospital Sector). The new initiative expands exemptions previously applied to investors from Hong Kong, Macau, and Taiwan, and in the Shanghai Free-Trade Zone.
- In parallel, efforts will continue to expand and align health insurance coverage, both through the BMI program and alternative providers, such as city-level *huiminbao* schemes and commercial insurers (see Health Insurance). Policy priorities for 2024 included raising the per capita subsidy for residents' BMI scheme, standardizing *huiminbao* insurance, and widening the scope for commercial health insurers to reimburse innovative medicines.

Continuing Anti-Corruption Drive

- The anti-corruption-drive in the healthcare and pharmaceutical sectors, launched in July 2023, continues to be a significant influence on interactions between the pharmaceutical industry and its customers. While the clampdown applies throughout the pharmaceutical value chain, from clinical trials to approvals, manufacturing, procurement, formularies, prescribing and promotion, much of the impact has been felt in drug detailing, medical affairs and conference organization (see Pharmaceutical Business Environment: Sales and Promotion).
- Fraudulent medical reimbursement claims have been another focal point. That has involved stepping up inspections of medical institutions and leveraging big data resources. On the latter front, the National Healthcare Security Administration (NHSA) has put pressure on hospitals to ensure that their healthcare information systems are compatible with the NHSA's own, so that the Administration and the city authorities that control healthcare reimbursement can audit hospitals at any time.
- In January 2025, the NHSA said that it had recovered nearly RMB27.5 billion of medical insurance funds in 2024, having worked with public security authorities to investigate 3,018 medical insurance fraud cases, which led to the arrest of 10,741 suspects. In addition, by October 2024, designated medical institutions had returned around RMB3 billion in insurance funds after self-assessing their practices based on an NHSA checklist. These investigations have also led to more conservative hospital prescribing (see Prescribing and Dispensing: Prescribing Influences).

Enhancing Health Status

- The overriding objective of China's healthcare reform program is enhancing the population's health status and achieving targeted improvements in key health indicators. Under the 14th Five-Year Plan for National Health, the life expectancy target is 78.9 years by the end of 2025. According to the latest NHC data, average life expectancy at birth in China was 78.6 years in 2023, up from 77.9 years 2022. The infant mortality rate fell to 4.5 per 1,000 and maternal mortality to 15.1 per 100,000 in 2023, an all-time best in China that matched the levels of advanced middle- to high-income countries worldwide, the NHC said.
- More effective screening, diagnosis and treatment of chronic NCDs will be an important tool in managing population aging. In 2023, over 4.0 million high-risk groups were screened, diagnosed and treated for gastric, esophageal, colorectal, breast, lung and other key cancers, while over 1.5 million people were screened for cardiovascular and cerebrovascular diseases. The premature death rate for cardiovascular or respiratory diseases, cancer and diabetes in people aged 30 to 70 years with chronic conditions fell from 18.5% in 2015 to 15% in 2023, according to the NHC.
- At the same time, infectious diseases are on the rise. In 2023, the NHC reported 2.8 million cases of Class A and B infectious diseases and 27,000 associated deaths (excluding COVID infections), up by 14.9% and 23.1%, respectively, over 2022. The bulk of these infections were viral hepatitis, tuberculosis, syphilis, gonorrhea and brucellosis. While officials report that hepatitis infections, for example, have declined steadily due to better monitoring and intervention, 75 million people are still living with chronic hepatitis B.

Action on Rare Diseases

- Rare diseases, which affect an estimated 20 million people in China, are another health-policy focus, including the establishment of a dedicated medical service network, comprising 419 institutions nationwide by mid-2024, geared to improvement in and co-operation on the diagnosis and treatment of rare diseases. The average time to definitive diagnosis for rare diseases has fallen from four years to less than four weeks, according to press reports.
- A February 2024 report by the Illness Challenge Foundation and Frost & Sullivan noted that, as of February 2023, 199 drugs for 87 rare diseases had been launched worldwide and only 103 for 47 rare diseases in China. By the end of the year, 165 types of rare disease drugs addressing 92 different conditions were available in China. The government has also established special trial zones to facilitate imports of orphan drugs not yet approved or launched locally.
- Despite priority reviews for critically needed orphan drugs, and the growing inclusion of orphan products on the National Reimbursement Drug List (NRDL), though, treatment access still lags behind global levels, and the high cost of orphan drugs remains problematic. Although 15 more rare disease drugs were incorporated into the updated NRDL in early 2024, China still lacked a dedicated system to absorb the costs of rare-disease treatments.

Pharmaceutical Policy

- The State Council's key policy directions for health system reform in 2024 included promoting co-ordination between the Essential Drug List (EDL) and the NRDL; timely adjustment and optimization of the EDL; expanding centralized procurement of multisource medicines (see Drug Procurement); and stepping up generic quality and consistency evaluations (GQCE) (see Pharmaceutical Business Environment: Generics and Biosimilars).
- The State Council also strongly emphasized drug innovation. Among its priorities for 2024 were intensifying the comprehensive clinical evaluation of innovative medicines and encouraging faster, albeit rational, uptake of new drugs by the healthcare system. This would involve further reforms of the drug approval system, as well as prioritizing rare disease therapies, drugs urgently needed in clinical practice, and products for epidemic prevention and control.
- Further reforms to the regulatory system will be targeted, in line with the State Council's January 2025 *Opinions on Comprehensively Deepening the Reform of Regulation of Drugs and Medical Devices to Promote the High-Quality Development of the Pharmaceutical Industry*, which call for further enhancements to the regulatory system (Regulatory Environment: Drug Administration Law).

Healthcare Financing and Expenditure

National expenditure on health increased at double-digit rates from the launch of the government's healthcare reform program in 2009 to 2021. This drove a progressive increase in the proportion of GDP invested in healthcare, from less than 5% to more than 7%, and a shift in the structure of national health expenditure, with patient exposure to healthcare costs progressively reduced.

Growth in national healthcare expenditure slowed to 6.2% in 2023, as costs associated with COVID-19 containment measures reduced and government spending on healthcare slowed. Social and personal health expenditure grew under the impact of delayed demand for treatment and a recovery in screening activity in the wake of the pandemic, however. In the medium to long term, demographic and epidemiological trends will put growing pressure on healthcare finances. In parallel, slowing economic growth will constrain both public health budgets and revenues in the BMI system.

Key Developments and Trends

Growth in National Healthcare Spend Slows as COVID-19 Eases

- National healthcare expenditure continued to rise in 2023, by 6.2% to RMB9.1 trillion (or RMB6,425 per capita), albeit growing at a markedly slower rate year-on-year than in 2022 (+11.0%), when costs were inflated significantly by COVID-19 containment measures. Despite the slowdown in national healthcare expenditure growth for 2023, its share of GDP rose to 7.2%, up from 7.1% in the previous year, according to preliminary data published by the NHC in mid-2024.
- Whereas government expenditure was a strong driver (+16.3%) of overall spending growth in 2022, in 2023 it rose by only 0.4% to RMB2.4 trillion, making up 26.7% of the total, compared with 28.2% in 2022. Social and personal healthcare expenditure grew by 8.7% and 7.9% (+9.7% and +8.2% in 2022), respectively, accounting for 46.0% and 27.3% of overall spending. Under its *Healthy China 2030* initiative, the government is targeting a reduction in the share of healthcare costs covered by out-of-pocket payment, to 25% by 2030.
- The data for 2023 suggest that, with the decline in costs associated with COVID-19 containment measures, such as the costs of vaccinations and testing, direct government allocations for healthcare have returned to a more subdued pattern of growth. At the same time, the relatively strong continued rise in social and personal health expenditure highlights the financial pressures on BMI schemes and patients, as delayed demand for treatment and a recovery in screening activity pushed up costs in the wake of the pandemic.

Healthcare Spending Outlook

- While the impact of the COVID pandemic on demand is waning, in the medium to long term demand will be driven by demographic and epidemiological trends as older segments of the population, typically with more expensive care needs, expand rapidly, and as the incidence of NCDs continues to rise. In parallel, slowing economic growth will constrain both public health budgets and revenues in the BMI system, which is already under considerable strain, especially the URRBMI (see Health Insurance).
- The growing financial pressures on the BMI system suggest that healthcare cost containment under the BMI schemes is likely to be stepped up. NHC data show that efforts to date have begun to show some effect on hospital inpatient costs (where case-based reimbursement is increasingly applied). In 2023, the average hospitalization cost per visit was RMB10,316, down 5.2% year-on-year at comparable prices, while the average daily hospitalization cost was down by 1.4% (to RMB1172). By contrast, the average outpatient cost per visit rose by 5.3% at comparable prices, to RMB362.

- The strain on BMI finances challenges the government's ability to expand reimbursement, coverage, including reimbursement of costly innovative drugs through the NRDL, even with deep price cuts. The government will continue to provide incentives for private sector healthcare investment and encourage multi-tier health insurance incorporating supplementary and commercial schemes (see Health Insurance).

Pharmaceutical Expenditure

- Annual updates of the NRDL have progressively improved reimbursement of innovative medicines through BMI schemes. At the same time, VBP of multisource drugs at both national and provincial level has helped to maintain tight control over pharmaceutical expenditure and its contribution to overall treatment costs. The broader implementation of case-based provider payment systems is expected to encourage a more cost-conscious approach to prescribing (see Prescribing and Dispensing: Prescribing Influences).
- NHC data show that pharmaceuticals accounted for 22.9% of average hospital inpatient treatment costs in 2023, down from 24.3% in 2022, and for 29.0% of average inpatient costs in primary care facilities (down from 29.4%). Medicines generate a significantly higher proportion of outpatient treatment costs, although their share in 2023 was slightly reduced in both the hospital and primary care settings. In hospitals, they accounted for 36.9% of total outpatient spending, versus 38.0% in 2022. In primary care facilities, medicines represented 74.4% of outpatient treatment costs, compared with 75.0% in 2022.
- Although the broader application of VBP to multisource purchases under BMI, and regular updates of the NRDL, will continue to reduce patient exposure to drug costs in the coming years, co-payments for more expensive NRDL-listed innovative medicines will remain challenging for many patients. For drugs not reimbursed under the BMI schemes, subsidized access may be available through *huiminbao* schemes or commercial health insurance for patients able to afford to top up their BMI coverage.

Health Insurance

More than 95% of the population is enrolled in BMI schemes rolled out as part of the government's healthcare reform program. Policymakers have now turned their attention to the expansion and harmonization of BMI benefit schedules, and to improving affiliate access to those benefits. Acknowledging the need for private payers to play a larger role in the sector, the government is promoting supplementary health insurance schemes and measures to encourage uptake of private health coverage.

Financial pressure on the basic medical insurance (BMI) schemes is building, aggravated by the slowing economy and the ageing population. The government will continue to promote an integrated multi-level health insurance system, including *huiminbao* city-level supplementary insurance schemes and commercial health insurance. While many *huiminbao* schemes have struggled financially, in the medium to longer term, improving regulation should help to stabilize the sector.

Key Developments and Trends

Total BMI Affiliate Numbers Decline, While Pressure on Finances Is Growing

- Total BMI affiliate numbers continued to fall in 2023, after declining for the first time in 2022. According to the NHSA, 1.33 billion people were covered by BMI at the end of 2023, down by 0.9% from 2022, when the year-on-year decline was 1.2%.
- Membership of the urban employee scheme (UEBMI) was up by 2.4%, after a 2.3% increase in 2022, reaching 371 million affiliates. In the much larger URRBMI scheme, though, the number of affiliates declined by 2.0% to 963 million (from 984 million in 2022), after declining by 2.5% year-on-year in 2022.
- The NHSA dismissed claims of a wave of URRBMI drop-outs in some rural areas of China, prompted by an increase in the annual per capita URRBMI premium from RMB350 in 2022 to RMB380 in 2023. Officials said the decline was due to the unification of the BMI information system, leading to the removal of affiliates with duplicate enrollment and the transfer of affiliates from the URRBMI to the UEBMI upon graduation or employment. Demographic trends will impact enrolment negatively in the medium term, however, as China's population begins to decline from 2026.
- The financial situation of the BMI system is deteriorating. Total BMI revenues for 2023 were RMB3.4 trillion, 10.7% higher than in 2022. But BMI expenditure outstripped revenue growth, rising by 14.7% in 2023 to RMB2.8 trillion, considerably more than the 2.3% spending increase in 2022. That left the BMI with a cumulative balance of RMB3.4 trillion at year-end, down 20.3% from RMB4.3 trillion at the close of 2022. While both the UEBMI and the URRBMI saw expenditure growth outstrip revenue growth in 2023, the URRBMI's financial situation is particularly precarious.
- The growing challenges to healthcare financing through the BMI, which are compounded by the slowing economy, were reflected in reports of reimbursement funding shortages during the second half of 2024, with an impact on city and hospital finances as well as hospital drug prescribing. The deteriorating financial situation will increase pressure for cost-containment.

"Because the economy is slowing down, I think companies cannot help to pay for the insurance for their employees, or the level of insurance is lowered. [...] Several key provinces are facing a huge problem, and hospitals cannot get funding from the government. [...] Starting from the second half of this year [2024] a lot of hospitals actually stopped prescribing or they basically asked patients to pay out of pocket. There's no funding." (Local Company Executive)
- A July 2024 State Council notice on improving the long-term mechanism for BMI participation suggested further relaxing the household registration restrictions on participating in the BMI system at the place of permanent residence and employment, including for migrants; linking residents' medical insurance premiums to per capita disposable income; raising maximum reimbursement limits for critical illness insurance where affiliates maintain coverage over several years and/or did not make reimbursement claims; and improving conditions for real-time medical insurance settlement by eligible village clinics.
- The government is also looking at provincial pooling of the BMI funds with a view to improving risk pooling and reducing regional disparities. The government's work report presented at the 14th National People's Congress session in March 2024 proposed promoting

provincial central pooling, while some provinces have already adopted the approach. The UEBMI has also been pursuing fund pooling at provincial level, integrating funding pools managed by different cities to smooth out regional disparities in fund balances and strengthen resilience to risk.

UEBMI Under Growing Pressure Despite Boost to Pooled Funds

- Despite its growing membership and a nominally healthy balance of funds, the UEBMI is seeing financial pressure grow as expenditure rises more quickly than revenues. The scheme brought in revenues of RMB2.3 trillion in 2023, 10.3% more than in 2022, while expenditure rose by 16.4% year-on-year, to RMB1.8 trillion.
- The cumulative surplus for mutual (pooled) funds was RMB2.6 trillion in 2023, as mutual fund revenues increased by 26.0% over 2022, against expenditure growth of 21.9%. This partly reflected UEBMI funding reforms, whereby employer payments were being transferred from personal medical accounts to mutual pools. Separately, affiliates' personal accounts held a cumulative surplus of almost RMB1.4 trillion in 2023.
- The funding reforms were intended to boost the pooled insurance funds to support spending on inpatient treatment, and to help to fund improvements in the reimbursement of outpatient treatment where elderly and chronically ill affiliates are exposed to high out-of-pocket payments. At the same time, UEBMI affiliates were given more leeway to pay for treatment of family members out of personal accounts.
- Despite the funding transfers, the UEBMI scheme will come under growing pressure in the medium to long term as more employees retire, the population ages and demand for treatment for age-related conditions escalates. NHA data show that almost 100 million (27%) of the UEBMI's 371 million affiliates in 2023 were retirees, and their number is growing faster (+3.7%) than the number of active employees (+1.9%). Highlighting the growing pressure on the scheme, the hospitalization rate for UEBMI affiliates overall was 21.9%, but the rate for retirees was 49.0%, whereas for active employees it was 11.3%.

URRBMI Financial Pressures More Acute

- Financial pressures on the URRBMI scheme are much more acute. NHA data show that between 2018 and 2023, expenditure in the URRBMI increased at a compound annual growth rate (CAGR) of 10.7%, reaching RMB1,046 billion in 2023. Meanwhile, URRBMI income since 2018 has grown at a compound 8.7% annually to RMB1,057 billion, leaving the fund with a cumulative balance of RMB766 billion – and a current balance of just 1% (RMB11.2 billion), its lowest ever, in 2023, down from 10% in 2018.
- In 2023, URRBMI accounted for around 72% of total BMI affiliate numbers, yet with access to only 39% of BMI funds. The cumulative URRBMI funding balance of RMB766 billion for 2023 was only 23% of the overall BMI surplus. Moreover, the URRBMI provides cover for two-thirds of BMI affiliates aged 60 years and over.

Huiminbao City-Level Supplementary Health Insurance

- As pressure builds on BMI finances, policymakers continue to advocate for city-level supplementary health insurance schemes (*huiminbao*). These schemes charge very low premiums in return for coverage including pre-existing conditions and access to benefits not

available under BMI, including certain innovative new medicines not listed for reimbursement on the NDRL.

- Supplementary health insurance provided by *huiminbao* schemes has grown rapidly since 2020. Some 300 million people were reported to have been enrolled in *huiminbao* schemes in 2023, according to data cited by the Asia Insurance Review. As of February 2024, 21 provinces/regions and 142 cities across China were offering *huiminbao* products that year, according to KPMG China.
- At the same time, there are indications that the initial rapid growth of the schemes has stalled, with many schemes struggling financially. Progress is being hampered by adverse selection, low participation rates, inadequate publicity, and low *huiminbao* reimbursement rates for innovative medicines, which may require substantial patient co-payments.

“The participation rate is lower. Either the pricing is not attractive, or the drug list is not fully updated. Also, I think the public have concerns about whether they can really be reimbursed for a lot of expensive treatment at only RMB120 or 150 per year.” (Pharmaceutical Industry Consultant)

- In line with most commercial health insurance schemes, *huiminbao* schemes are usually short-term (usually one-year) policies. Critics argue that, with schemes largely focused on minimizing claims, there is little incentive to embrace long-term health management, engage in risk classification, or improve the risk profile of affiliates with pre-existing conditions.

“Most of the city-level insurers don’t run profitably. The majority are losing money. They try to include really high-priced products, like CAR-T therapies, to make their program more attractive to subscribers. But the subscription fee is very low, RMB100 or 150 for the whole year, it’s crazy.” (Local Company Executive)

- Nonetheless, *huiminbao* schemes have emerged as an option for innovative drugs, whether as an alternative to NDRL listing or to gather real-world experience that can be used to facilitate NDRL uptake. According to IQVIA China research, the cumulative number of drugs covered by commercial health insurance schemes rose from 84 in 2020 to 550 in 2023 (see Pricing and Reimbursement: Reimbursement).

NFRA Seeks to Improve *Huiminbao* Regulation and Reimbursement

- Seeking to improve regulation of *huiminbao* schemes and encourage enrolment, the National Financial Regulatory Administration (NFRA) in January 2024 issued a draft *Notice on Promoting the Smooth and Orderly Development of Urban Customized Commercial Medical Insurance*. If adopted, the NFRA’s proposals could stabilize *huiminbao* provision with more flexible, risk-sensitive plans and a more holistic health management approach.
- The draft NFRA notice recommended designing *huiminbao* as long-term plans with adjustable premiums that could be modified on a risk-classification basis. The notice suggested that while participation should remain voluntary, employers should utilize preferential tax policies and purchase group *huiminbao* insurance.
- The draft NFRA notice also emphasized that each city or region should offer no more than one *huiminbao* scheme. There are already some province-level schemes, based on the premise that a larger patient pool should enable insurers to balance better premium income, claims expenditure and the risk profile of affiliates.

- Seeking to diversify payment mechanism for innovative drugs beyond the NRDL, the NHSA in December 2024 called for the development of a national-level list of innovative drugs to be reimbursed under *huiminbao* schemes, which would later be extended to other commercial health insurance schemes (see Pricing and Reimbursement: Reimbursement).
- The new list - dubbed 'List C' – would include innovative drugs reimbursed under *huiminbao* schemes, to complement List A and List B of the NRDL which provide the framework for reimbursement under the BMI schemes. The new list would regulate and standardize drug reimbursement under *huiminbao* schemes and has been welcomed by the industry.

Commercial Insurers Show Signs of Recovery and Are Being Encouraged

- In recent years, commercial health insurers in China have faced not only tighter regulatory scrutiny, but also growing competition from low-premium *huiminbao* schemes, slowing growth in the sector. After a 15.8% downturn in 2023, the market showed signs of recovery in 2024. Income from original health insurance premiums rose by 6.1% over 2023, to reach RMB773.1 billion, according to NFRA data. However, the Healthy China 2030 roadmap's target of RMB2 trillion in commercial health insurance premium income by 2025 will not be met.

"I don't think they [commercial health insurance] are growing as originally expected. One reason is that these kinds of insurance policies are not actually very profitable compared with other forms of life insurance. Secondly, they lack the data and the expertise to design some of these unique disease specific insurance policies." (Pharmaceutical Industry Consultant)

- The establishment of an integrated multi-level health insurance system that can ease pressures on BMI finances and expand coverage beyond basic medical needs, remains a key government objective. In a September 2024 *Opinions* document on promotion of a high-quality insurance industry, the State Council called for further diversification of commercial health insurance products, offering a broader scope of protection covering new drugs and technologies.
- The NHSA in September 2024 called for a '1+3+N' system, comprising a comprehensive health data platform (the '1') offering a unified personal health record for anyone covered by health insurance, a coded drug tracking system, as well as centralized claims, invoices and settlements; the core pillars of the system (BMI, medical assistance and critical illness insurance - the '3'); and supplementary providers or 'N' (*huiminbao*, other commercial health insurers, charities, mutual aid through trade unions, other funds).
- The commercial insurance sector currently has limited access to data. The health data platform should help commercial insurers access public hospital data, aiding actuarial calculations, pricing strategies, and commercial insurance policy design. The platform, along with the introduction of case-based reimbursement and the reform of medical service pricing, should allow commercial insurers to design more informed and comprehensive policies that complement BMI provision.
- In May 2024, the NFRA issued *Guiding Opinions on Promoting the High-Quality Development of All-Inclusive Commercial Insurance*, with an implementation target of the next five years. The *Opinions* require commercial insurers to include the development of all-inclusive insurance in their performance evaluations. All-inclusive commercial health insurance would be given support by the government, including access to government data,

while premium rates for these policies should be lower than those for comparable insurance products under similar conditions.

- The *Opinions* specifically call for development of health insurance products for rural residents and improvement of protection for county residents. The NFRA also stressed the need for commercial health insurance products geared to specific patient groups, such as the elderly, people with pre-existing conditions and chronic diseases, as well as products providing improved protection for cervical and breast cancer, congenital conditions or rare diseases.

Hospital Sector

Policymakers recognized that the public hospital sector would need to be overhauled substantially when they launched the healthcare reform program in 2009. A raft of measures has been introduced, in an effort to discourage profit-driven overtreatment, improve the quality and efficiency of hospital care, and shift a growing proportion of demand away from major public hospitals into lower-tier facilities. Care quality and efficiency in the sector remain variable, however, while public hospital financing issues have not been fully resolved. Although the private hospital sector has continued to expand, public facilities still bear the brunt of demand for secondary and tertiary care.

The government continues to pursue reforms in the hospital sector geared to decentralization, cost efficiency and accessibility. Measures include the establishment of national and regional medical centers. Reforms of medical service pricing and hospital payment via case-based payment models will be implemented more widely. The easing of restrictions on foreign investment should help drive improvements in the private hospital sector.

Key Developments and Trends

Hospitals Remain Under Pressure as Demand Bounces Back Strongly

- Following the end of China's zero-COVID restrictions in December 2022, inpatient admissions surged by 21.9% in 2023, to 245 million. The increase was stronger in the public sector, with admissions up 22.7%, while private sector admissions rose by 18.4%. Outpatient consultations also bounced back strongly, rising by 11.5% to reach 4.3 billion in 2023. Outpatient visits in public hospitals grew by 11.6%, while private sector visits grew by 11.1%.
- These figures evidence the release of pent-up demand following COVID-related lockdowns, as well as a surge in hospitalizations caused by non-COVID respiratory disease outbreaks in the final quarter of 2023. The national bed occupancy rate showed a similar pattern: after a decline to 71.0% in 2022, in 2023, it was back up at 79.4%, and even at 86% in public hospitals (75.6% in 2022). The bed occupancy rate in tier 3 facilities was highest at 91.1% (79.8% in 2022), compared with 74.3% and 54.1% for tier 2 and 1 facilities, respectively.
- This suggests there is still a long way to go before the goal of a tiered healthcare system can be achieved, and patient pressure on top-level public hospitals can be relieved. While the impact of COVID-19 pandemic has largely been absorbed in terms of delayed diagnoses and treatments, population aging and the rise chronic diseases will put further strain on public hospitals over the medium to long term.

Reforms Continue to Pursue Infrastructure Upgrades and Decentralization

- The total number of hospitals across China grew by 3.7% to 38,355 in 2023, adding 1379 facilities. Growth was driven by the private sector (+5.4%), although the decade-long decline in the number of public hospitals stabilized, with their number growing marginally (+0.2%) to 11,772, to account for 31% of hospitals, according to NHC data. Despite its smaller share, the public hospital sector accounted for 82% of inpatient admissions, and 84% of outpatient consultations.
- The number of tertiary care hospitals continued to grow strongly, increasing by 334 (+9.4%), versus growth of 7.2% and 3.4%, respectively, for tier 2 and 1 facilities. Overall bed capacity rose by 4.5%, to 8.0 million, with the public sector adding over 172,000 beds (+3.2%) and the private sector over 169,000 (+7.4%). Bed capacity continues to be dominated by the public sector, which accounted for 69% of beds in 2023.
- In line with government objectives under the 14th Five-Year Plan to expand and decentralize high-quality medical resources, 13 national medical centers had been established across China by mid-2024, providing first-class diagnosis and treatment in a special medical field. In addition, 125 national-level regional medical centers, and 114 provincial-level regional medical centers have been set up, to provide high-quality medical services in different regions. According to the NHC, 92% of county-level hospitals have reached the service capacity and standards of tier 2 hospitals or above.

Medical Service Pricing and Hospital Payment and Reforms Are Gaining Pace

- Reforms to hospital payment systems, including better calibrated, more standardized and more dynamic medical service pricing and a shift to flat-sum or case-based reimbursement, are gaining pace. The reforms aim to replace a system in which medical service prices were based largely on hospital expenditure, with drugs and medical consumables accounting for the majority of costs, while labor costs were substantially underestimated.
- Standardized medical service pricing at national or provincial level aims to minimize regional variations, more accurately reflect actual clinical input (particularly for more complex medical procedures), stabilize staff income and discourage reliance on mark-ups for medicines, tests or other medical consumables/services. By late 2024, the NHSA had issued guidelines on standardized pricing for eight specific types of medical service, including organ transplantation, dental implants and obstetrics.
- Payment models based on Diagnosis-Related Groups (DRG) and Diagnosis Intervention Packet (DIPs) have been piloted in public hospitals across China since 2019 and 2021, respectively. Under the DRG/DIP reforms, reimbursement is shifted from a fee-for-service model to disease-based groupings. Patients are classified by clinical characteristics including diagnosis, co-morbidities and complications, alongside treatment. This should encourage more cost-efficient use of resources by tying reimbursement more closely to clinical guidelines.
- The NHSA said in 2021 it wants the two models to be rolled out across the whole of the public hospital sector by the end of 2025. By the end of 2023, the NHSA noted, over 90% of BMI-linked institutions had adopted the reforms, while the proportion of hospital payments based solely on treatment processes had declined by one quarter. Medical institutions were

managing costs more effectively and the co-payment burden on patients had reduced. By encouraging greater attention to cost-effective treatment, DRG/DIP payment models will affect hospital prescribing practice, leading to a more cost-effective approach to prescribing (see Prescribing and Dispensing: Prescribing Influences).

- In July 2024, the NHSA introduced an updated version of the DRG/DIP framework, with more scientifically driven disease groupings. Fewer diseases were listed in the Version 2.0 framework, but the number of DRG groups increased from 376 to 634, bringing them more into line with clinical standards and advances in technology. The aim was for regions implementing DRG/DIP reforms during 2024 to use the updated version. Regions that had already introduced DRG/DIP payments would switch to Version 2.0 from 2025.

Recovery in the Private Sector and Encouragement of Foreign Investment

- The number of private hospitals has continued to grow - by 5.4% year-on-year in 2023, reaching 26,583 facilities and accounting for 69% of all hospitals in China. Bed capacity in private hospitals was up by 7.4% to 2.5 million, while inpatient admissions were 18.4% higher at 44.9 million. Following a slowdown in demand for private hospital treatment during the pandemic, this indicates a post-COVID recovery in demand among better-off patients and/or those with private health insurance.
- Seeking to encourage further growth in the sector, in September 2024, the Ministry of Commerce, the NHC and the National Medical Products Administration (NMPA) announced that a pilot program would permit wholly foreign-owned hospitals in eight relatively wealthy Chinese cities, including Beijing and Shanghai, as well as Hainan province.
- The NHC, the Ministry of Commerce and other departments released a *Pilot Work Plan for Expanding Openings in the Wholly Foreign-Owned Hospital Sector* in November 2024, clarifying the specific conditions. The program excludes hospitals focused on traditional Chinese medicine (TCM), while foreign investors may not acquire public hospitals in the denoted regions.
- The pilot program replaces the requirement for foreign investors in the private hospital sector to find Chinese partners, with the latter holding at least 30% of the equity. Exceptions were made, for investors from Hong Kong, Macau, and Taiwan, and in the Shanghai Free-Trade Zone. Moreover, foreign investors could fully acquire health clinics. China had over 60 joint-venture hospitals in 2024, according to press reports.
- The pilot program was welcomed by foreign healthcare providers, including Raffles Medical Group and IHH Healthcare, and the first investments under the program were announced as early as December 2024. Singapore-based Luye Medical Group announced that it will establish an international medical headquarters and a wholly-owned specialty tertiary hospital in Shenzhen.
- Fellow Singaporean Perennial Holdings said that it had been given approval to open its first wholly-owned general hospital in 2025, with an investment of RMB1 billion. The 500-bed multi-disciplinary tertiary hospital will be part of an integrated medical hub developed by Perennial, which also includes other joint-venture medical and eldercare facilities. The group has announced plans for similar facilities in Kunming, Xi'an and Guangzhou.

- The new wholly-owned foreign hospitals are expected to provide more diversified, specialized and patient-centric services than the public sector, with associated opportunities for advanced technologies, including medical equipment and innovative medicines. By introducing premium medical services, the hospitals are expected to serve as a model for local providers, while also promoting the development of commercial health insurance and driving demand for policies tailored to cover premium services.

Primary Care

The 'grassroots' healthcare system comprises an extensive network of urban community health centers and rural township health centers, village clinics and other clinics/infirmaries, although county-level hospitals are also defined as primary care facilities. The sector has traditionally been underused, mainly because it has been poorly resourced, limiting the range and quality of care available. Strengthening primary care provision is a core health policy goal, though, and will be a prerequisite for achieving targets outlined in the *Healthy China 2030* policy initiative. Key goals include establishing family doctor networks and a referral system, encouraging a tiered approach to diagnosis and treatment.

Efforts to shift more demand into primary care facilities are making gradual progress. Further improvements are being targeted in the range and quality of services available in grassroots facilities, and a push is being made to establish closely integrated medical and health consortia in all counties, with the objective of establishing a tiered diagnosis and treatment system. Progress varies significantly between provinces and individual cities, and the government's decentralization program will have to overcome negative perceptions of expertise in primary care.

Key Developments and Trends

Continued Increase in Grassroots Healthcare Capacity and Utilization

- Following a slowdown in growth in 2022, growth in the number of grassroots healthcare facilities in China picked up again. There were 1.0 million such facilities in 2023, 3.7% more than in 2022. Growth was driven by an increase in community health centers (CHCs, +2.0% year-on-year) and clinics/infirmaries (+12.9%), while the number of rural township health centers and village clinics continued to decline, reflecting the adjustment of administrative divisions and moves to consolidate primary care into larger, better-resourced facilities.
- In the aftermath of the pandemic, outpatient consultations at grassroots facilities increased by 15.7% to 4.9 billion in 2023, outpacing the 11.5% growth in hospital outpatient consultations over the same period. Inpatient admissions at grassroots facilities rose even more sharply, by 25.6% to 45.5 million (while hospital inpatient admissions were up by 21.9%). This shows that efforts to move more outpatient consultations and inpatient admissions to local level are gaining traction, even if the strain on tier 3 hospitals especially remains considerable. Nearly 52% of all outpatient consultations were at grassroot level in 2023.

Primary Care Is Being Strengthened, Including in Rural Areas

- The 14th Five-Year Plan targets improvements in the range and quality of services available in grassroots facilities, with the objective of establishing a tiered diagnosis and treatment

system. This includes the establishment of a network of family doctors, as gatekeepers to tiered healthcare. The NHC has set a target to have 75% of the population covered by family doctor contracts by 2035.

- A NHC policy notice published in January 2024 called for the establishment of 500 community hospitals in 2024 and 2025, so that by 2025 30% of community health centers would be supported by hospitals. The notice also called for further improvements in service capacity standards of community and township health centers, as well as village clinics, and further strengthening staffing and training of grassroots health personnel.
- As part of the drive to improve primary care services in rural areas and facilitate two-way referrals, the government aims to establish closely integrated medical and health consortia in more than 90% of China's counties by the end of 2025, and in all county-level regions by 2027. Some 91% of counties had already sent staff from county hospitals to township health centers and village clinics for extended stays as of September 2024, according to the NHC. NHC statistics also showed that 40% to date of newly built medical facilities offering key clinical specialties were located outside provincial capital cities.
- Physician resources continue to be concentrated largely in urban areas. To alleviate imbalance, the State Council's health reform priorities for 2024 called for an additional 8,000 rural undergraduate medical students to be recruited for township health centers in central and western China, under programs that provide free medical education to rural students who commit to working locally, as well as subsidies for undergraduates in village clinics.
- While efforts to shift demand into primary care facilities are making gradual progress, and the NHC reported a drop in patients travelling across regions when seeking care, progress varies significantly between provinces and individual cities, and patient preference for seeking care at higher tier facilities remains engrained, especially in urban areas.

"I see these very fancy community healthcare centers, but they're empty. [...] So, there's an intention to do this, but it's not strictly enforced. [...] Also, Chinese patients in general have very limited trust in doctors, and when you get to these primary care places, even lower trust. Moreover, the competency level in many of them is extremely low. You don't want to go there unless it's very simple." (Multinational Company Executive)

Drug Procurement

Pharmaceuticals used in public hospitals were traditionally purchased largely through competitive tenders at provincial level, city level, or for groups of hospitals at local level. Following a pilot in December 2018, national tenders have been issued for a growing number of multisource drugs, with significant reductions in the price of affected products. This volume-based procurement (VBP) model is also being used more widely at provincial level, and by multi-provincial alliances.

Further expansion of VBP will be pursued, with 600 active ingredients targeted at national and provincial level by 2025. The results of the 10th round of national VBP were announced in January 2025, covering 62 active ingredients, and reported to have resulted in an average price reduction of over 70%, while a second round of insulin VBP was concluded in May 2024. Most VBP contracts will be taken up by local companies. Increased multi-province VBP will mean more complexity and uncertainty for bidders. National VBP for multisource biologics may be introduced in 2025 or 2026.

Key Developments and Trends

10th National VBP Round Adding Another 62 Active Ingredients

- With the conclusion of the ninth round in September 2023, products incorporating 374 active ingredients in total had been tendered through VBP since 2018, with average price reductions of over 50%. The government's target is for over 600 active ingredients to be covered by VBP, at both national and provincial level, by 2025, with contracting rounds occurring around twice a year.
- A 10th round of centralized VBP was announced in September 2024, including 62 active ingredients in 263 presentations, a significant increase on the 41 active ingredients in 195 presentations tendered in the ninth round, according to local media reports. More than 56% of the active ingredients on the new list were injectables. Among specific active ingredients were the ovarian cancer drug doxorubicin hydrochloride, the spasmolytic phloroglucinol, and the diabetes treatment sitagliptin, all of which are widely used in China.
- The results were announced in January 2025, with a total of 439 companies having submitted bids for 778 products, and 234 companies awarded procurement contracts for 385 products. Competition in the 10th round was reported to have been particularly intense, with 31 companies qualified to compete for sitagliptin, and 30 companies for phloroglucinol injections. Price reductions were significant, with the average reduction reported to have been over 70%, and with 134 products reported to have seen price reductions of over 90%.
- The 10th round of centralized VBP brings the total number of multisource products covered under national contracts to 436.

Concerns About the Evolution of VBP

- With multiple rounds of national VBP now completed, some stakeholders say that the rules and conditions for national tendering are now more transparent and predictable, even if VBP implementation still needs to improve.
- At the same time, there are concerns about the increasing fragmentation of VBP, as provinces take on growing responsibilities, including in renewal and renegotiation of VBP contracts, and as a growing number of winners share VBP contracts.

"It is unpredictable, you cannot predict how much you can get. The essence of VBP is that you know the price you put in and you know the volume you can get back – but if the rule allows multiple players to win the bid, you still have to deal with the competition in the hospital and convince the doctor to prescribe yours instead of others." (Local Company Executive)

"The VBP renegotiation, extension, implementation – it's truly chaotic between the different therapy areas, different provinces, alliances!" (Multinational Company Executive)

- There also concerns that the NHA will step up regulation of price variation. Companies cite a new policy direction in late 2024, whereby additional price ceilings and multiples might apply to selected products that have already been through VBP and have agreed pricing trade-offs for guaranteed contract volumes. For example, where there were multiple winning bids for a particular active ingredient, the ceiling price could not exceed three times the lowest-priced winner, narrowing the spread of prices among bid winners.

“So what happened is that two weeks ago, the China government issued an updated policy for VBP products, related to the update on VBP policy, in relation to the list of products that was attached to that update. [...] And basically what it said is that in each of the provinces where the product is a multi-win, the ceiling for the winning price should be no more than three times the lowest price.” (Multinational Company Executive)

- As of late 2024, it remained unclear how this new approach might be applied at provincial level, particularly with the threat of winners that flout the three-times price ceiling being excluded from volume shares and treated as reserve options. How this might affect their pricing in the 20% or so of the hospital market not covered by VBP also remains to be seen.
- Another worrying aspect of the new policy direction is that, for selected active ingredients, products that failed to win bids would have to maintain prices at up to one-and-a-half times the lowest winning price in the tender. There are fears that these price ceilings might eventually serve as a basis for aligning retail drug prices with the outcomes of VBP rounds.

Multinationals Continue to Shun Small Molecule VBP Tenders

- Where multinational companies have participated in early rounds of VBP, they have mostly lost out to local suppliers and low-cost generics, and they have increasingly lost their appetite for submitting mature off-patent products for national or provincial procurement, as prices are driven down to unsustainable levels. Instead, multinationals have increasingly focused on their innovative portfolios, while outsourcing mature products to local companies or contract sales organizations (see Pharmaceutical Industry Environment: Sales and Promotion).
- They are also exploring opportunities in the 20% or so of the public hospital market not covered by VBP or, where possible, putting more resources behind affected brands in retail outlets or online channels.

“I think it’s clear you get outliers who find [VBP] beneficial from a P&L perspective. But the bulk of [multinationals] choose to lose, and that’s the right thing to do from a business perspective. We have an innovative portfolio coming in the next three to five years. It doesn’t make sense still to invest in a VBP portfolio, but rather to outsource it.” (Multinational Company Executive)

- This may not be the right strategy for every mature product. Hospital sales of non-VBP brands are complicated by increasingly strict enforcement of agreed contract volumes for procured products, as well as closer scrutiny of procurement practices under the government’s anti-corruption drive (see Health Policy).

New Round of Insulin VBP Completed

- Insulin products have been subject to national procurement since May 2022. These contracts have been more flexible, with broader purchasing options for hospitals and more leeway to include originator products, along with a price threshold that enables all companies offering cuts of at least 40% to win a share of supply contracts.
- The first insulin VBP contracts were for a two-year term. Follow-up procurement was launched in the provinces in April 2024, with the winning bidders awarded three-year contracts that took effect from May 2024. More than 35,000 medical institutions took part in the new round, with demand for over 240 million units of second- and third-generation insulins. According to the NHSA, all bidding companies had all or some of their products selected.

- The bidding involved 53 products from 13 companies in total, with 49 products selected for procurement. Where prices of selected products had fallen by an average of 48% in the initial insulin round, prices in the second round reduced by a further 3.8%, according to the NHTA. Price differentials for third-generation premixed insulins, which were the widest in the May 2022 insulin VBP, narrowed from 2.3 times to 1.6 times the lowest price accepted, the NHTA noted.
- Moreover, the proportion of Class A winners (those awarded the largest share of supply contracts) with lower price quotes increased from 40% in the previous round to 71% in the follow-up VBP. These included products from multinationals and newly listed companies.
- In the latest round, some attention was paid to how much contract volumes awarded in the May 2022 round had been taken up by hospitals over the previous two years, with the realized volumes used as a basis for contract renewal. That not only led to further shifts in contract volumes but also underlined the importance of hospitals managing contract volumes effectively.
- This has been challenging in some cases, especially for Class A and B winners (which are more likely to be local generics). There may, for example, be sensitivities around switching patients established on one form of insulin to another form, in an effort to use up allocated contract volumes.
- This means hospitals have to rely on new insulin patients to shift the necessary contract volumes. If they cannot fulfil the quotas, these hospitals are likely to impose stringent prescribing restrictions, essentially locking out Class C winners and non-VBP products, even if there may be viable medical reasons for giving the patient a different type of insulin.
- This is one reason for companies with potential Class C winners (usually the original brand) to start building brand loyalty in hospitals very early after launch, and certainly well in advance of VBP negotiations. Another reason is to have a better chance of accessing the remaining part of the public hospital market not covered by VBP, as well as other 'free market' options (e.g. private hospitals, retail channels, online sales), when a product is excluded from VBP, or a company decides to withdraw it strategically from bidding as the competitive stakes are too high.
- At the same time, the dynamics for injectable insulin products and other biologics spilling over from VBP into the retail market may be different from those in small-molecule VBP. In particular the insulin tenders are structured for A, B and C winners, whereas the small-molecule VBPs operate essentially on an all-or-nothing basis. In the latter scenario, brand loyalty is more likely to transfer to pharmacy level, especially as small-molecule products are easier to manage, with a better risk profile and typically lower prices than biologics.

"For insulin, it is different, because we can still get free market volume as C winners. The patients can still be served in the hospitals, but not all of them. There has been some outflow to the pharmacies, but also some challenges in terms of whether patients could actually get reimbursement for these products when they bought them in the pharmacy." (Multinational Company Executive)

National Biologics VBP Expected in 2025 or 2026

- The insulins tender was conceived as a pilot program that would set a precedent for national VBP of other multisource biologics, under similar bidding conditions (including less acute

price erosion than for small molecule drugs), to enable supply of both originator biologics and biosimilars. By late 2024, though, VBP had not expanded to biologics generally.

- Stakeholders expect that VBP for multisource biologics, as well as for related products such as human growth hormone, could be introduced in 2025 or 2026, and could mean steeper price cuts than for multisource insulin, with the government reportedly considering price cuts averaging around 48-50% achieved under the insulin tenders not tough enough to achieve the necessary cost savings.

Provinces to Take the Lead in Continuing VBP

- National VBP will continue to focus on multisource medicines that have obtained generic quality consistency evaluation (GQCE) certification, but the government is also encouraging selected provinces to take the lead in continuing procurement of drugs previously subject to national VBP contracts, as well as organizing tenders for products not previously included in national VBP rounds.
- In a May 2024 notice, the NHSA called on provinces to take the lead in national procurement of small molecule drugs without GQCE, TCMs and commonly used medicines purchased in significant quantities for broad patient populations.
- For example, Hubei and Shandong provinces will be responsible for procuring new batches of TCMs, and Sanming city for centralized procurement of respiratory and anticancer drugs. The NHSA wanted each of these provinces to have completed at least one round of centralized procurement by the end of 2024. This would draw more than 500 drugs into the VBP process, including those contracted through multi-province alliances.
- Provincial involvement in VBP, typically in the form of multi-province regional and cross-regional alliances, will expand as more national procurement contracts expire. This has even extended to VBP of biologics, which is not yet subject to any official national VBP program. For example, the Guangdong multi-province alliance included rituximab among the drugs it solicited VBP bids for in 2022.
- Further decentralization of national VBP to the provinces will present challenges to industry. Some companies complain of significant existing disparities between provinces, hospitals and therapeutic categories in the way VBP is applied, extended and renegotiated. This dilutes the primary attraction of VBP: large, guaranteed supply volumes in return for substantial price reduction. In a highly fragmented system, volume gains are more modest.
"I know there is an intention to synchronize all of this and make the system more efficient, more predictable. For the time being, it's not happening. [...] You don't have guaranteed volume. I asked in one of the provinces, and they have 15 winners in the winning category!" (Multinational Company Executive)
- There is also likely to be continuing price erosion, as provinces gain more leverage in centralized procurement and exercise buying power through large alliances. The Guangdong procurement alliance, for example started with around eight provinces and is expected to expand to 12 or 13 in 2025.
- Despite the growing consolidation of provincial VBP, companies are also worried about increasing complexity and unpredictability, as more and more centralized procurement is handled at provincial level. This is likely to prompt more rigorous assessments internally, to

determine whether products are important enough strategically to justify the risk of uncertain price-volume relationships.

- Companies also report that there have been indications that provinces, or provincial alliances, may be prepared to use VBP prices established by other provincial alliances as reference prices, including for products beyond the scope of their own VBP – and without even guaranteeing supply volumes in return.

“Beijing, Tianjin and Hebei are very close to each other. [...] if product A is already in the Guangdong VBP alliance, then the Beijing alliance says, “We are also going to do VBP. But we are just going to take whatever the VBP price is in the other alliance and use the same price for our alliance”. And there’s no volume allocation at all. There are no winners. And now Fujian is following suit.” (Local Company Executive)

E-Health

The government is keen to encourage e-health initiatives and has recognized healthcare big data as a strategic resource. Rapid developments in the field have been spearheaded by private players, which have developed online platforms supporting appointment bookings, remote consultations, digital prescriptions and e-pharmacy. Other stakeholders, including payers, distributors, online pharmacies and pharmaceutical manufacturers, have also begun harnessing digital technologies to refine their operations.

Further expansion and improvement of e-health services will be pursued with a view to relieving pressure on the healthcare infrastructure. The government has called for a consolidation in nationwide information sharing for medical and health institutions; further development and use of public data resources in healthcare; encouragement of mobile and online interaction with medical services and integration of medical and pharmaceutical data. Commercial players will continue to expand and broaden capacity and capability of services.

Key Developments and Trends

E-Health Services Continue to Grow

- The onset of COVID-19 boosted uptake of e-health services significantly in China, and while investment in China’s digital healthcare industry slowed in 2022, partly reflecting a downturn in the Chinese economy, there has been continued growth in the e-health services market, driven by key players such as Alibaba, WeDoctor, JD.com and Ping An Health.
- By December 2024 the user base for internet medical services in China had reached close to 420 million people, up from 414 million at the end of 2023, and accounting for 38% of all internet users nationwide, according to data from the government-affiliated China Internet Network Information Center (CNNIC). Some 70% of primary care health centers had established telemedicine cooperation with hospitals. The number of internet hospitals had grown to 3,340 in 2024, delivering over 100 million online diagnoses and treatments annually, according to the NHC.
- Further expansion and improvement of e-health services will be pursued with a view to relieving pressure on the healthcare infrastructure. The State Council’s healthcare reform

priorities for 2024 included consolidating nationwide information sharing for medical and health institutions; further promoting the development and use of public data resources in healthcare; encouraging mobile and online interaction with medical services; and integrating medical and pharmaceutical data.

- In October 2024, the Chinese Medical Association, in collaboration with leading medical institutions and the National Foundation for Australia-China Relations, announced a pilot program to develop, publish and implement new standards for internet-based family doctor services in China. While no details were announced, the pilot program will be based on Ping An's Family Doctor service, according to media reports.

Commercial Players Remain Strong Drivers

- Local players such as Ping An Health, JD Health, Alibaba Health, Miao Health, Ding Xiang Yuan and WeDoctor (Tencent Holdings) took an early lead in the commercial market for digital healthcare. Their online platforms have expanded to cover a range of services and products, from basic telemedicine to hospital appointments, online consultations, disease management, health checks, electronic medical records, internet hospitals and online pharmacies.
- Ping An Healthcare and Technology Co Ltd (Ping An Health), for example, reported its first full-year profit in 2024, driven by an expansion of its health and senior care services. Ping An Health's medical network included approximately 50,000 doctors across 29 specialties, partnerships with nearly 4,000 hospitals, and collaborations with more than 105,000 health service providers by the end of 2024. The company leverages artificial intelligence to enhance its healthcare services.
- The company introduced an upgraded version of its family doctor membership program in 2024, offering one-stop, proactive health management services. The company claims around 14 million users had access to its 24-hour, AI-assisted Family Doctor service in 2024. Ping An Health's family doctor team has received certifications from institutions such as Peking University International Hospital and the World Organization of Family Doctors.
- Commercial players will continue to expand and broaden capacity and capability of services over the forecast period, including 24-hour access to online consultations and the delivery of prescribed drugs, but also corporate employee health management platforms, for example.

Prescribing and Dispensing

Prescribing Influences

Hospital financing mechanisms and patient expectations traditionally encouraged widespread overtreatment and irrational prescribing, inflating healthcare costs. Since the launch of the 2009 reform program, government policies have sought to address the issue through hospital financing reforms and controls on drug budgets, as well as measures favoring rational prescribing. The elimination of hospital drug mark-ups, the introduction of volume-based procurement (VBP) contracts for a growing number of multisource drugs, and efforts to curb corrupt practices have also had a growing impact on prescribing behavior.

The government's anti-corruption campaign has led to more conservative prescribing behavior, while hospital audits and claims controls have been stepped up. At the same time, VBP is exerting a growing impact on prescribing choice for hospital doctors, with opportunity for non-winners limited. The rollout of flat-sum provider payment models will also contribute to more cost-sensitive prescribing, while growing pressures on BMI finances can affect prescribing, especially towards the end of the year.

Key Developments and Trends

Anti-corruption Measures Drive More Conservative Prescribing

- The government's anti-corruption drive, launched in July 2023 (see also Pharmaceutical Business Environment: Sales and Promotion), has led to more conservative prescribing behavior, as well as more widespread adherence to hospital prescribing guidelines and protocols. In parallel, the NHSA has been stepping up hospital audits and claims controls, utilizing hospital information systems to identify and eliminate fraudulent claims.

"There are a lot of investigations, I think which trigger that the prescription volume will not grow as aggressively and prescriptions become very, very compliant. The NHSA, the asked the hospital to build an HIS [hospital information system], a payment system, and the city, the NHSA, they say they can audit the hospital at any time. [...] With the healthcare information system, investigations and audits can be very regular." (Pharmaceutical Industry Consultant)

- Controls of prescribing for outpatients are also being stepped up. Paper prescriptions dispensed to outpatients through retail pharmacies under the BMI system were to be more closely monitored from 1 January 2025. This applies especially to chronic-disease therapies and drugs dispensed under the dual-channel system (see Dispensing). The move followed claims that four pharmacies in Harbin were processing thousands of fraudulent handwritten prescriptions to file false insurance claims. This has made doctors in hospitals even more cautious about behavior that might put them under suspicion.

VBP Is Exerting a Growing Impact on Prescribing Choice

- As VBP contracts cover an increasing number of molecules, VBP is exerting a growing impact on prescribing choice for hospital doctors. With the conclusion of the 10th round of national VBP in January 2025, products containing 436 active ingredients in total have been tendered through national VBP, and additional molecules are covered by provincial VBP. The

government is aiming at 600 active ingredients covered by VBP, at both national and provincial level, by the end of 2025.

- With large shares of the hospital market delivered to winning bidders, opportunities for prescribing of non-winners are limited, especially as hospitals sometimes struggle to meet contract quotas, and as pressure on hospitals to fulfil contract quotas for drugs tendered through VBP contracts is being stepped up.
- For drugs sidelined in the VBP process, generating any usage in hospitals can be particularly challenging, with dual-channel or retail sales the only really viable options (see Healthcare Provision: Drug Procurement). There may be leeway for hospital use of drugs on a named-patient or temporary procurement basis, according to need, but these strategies do not guarantee stable supply and depend on budget availability.

Hospital Payment Reforms Expected to Encourage Cost-conscious Prescribing

- More than 90% of BMI-linked institutions had adopted the DRG/DIP reforms by the close of 2023 (see Healthcare Provision: Hospital Sector). Defined budgets under case-based payment models are expected to drive stricter adherence to prescribing guidelines and protocols in the hospital setting. The outcome will be more cost-sensitive prescribing of at least established medicines, and quite possibly of more innovative therapies too, if they cannot find a place within DRG/DIP frameworks.

“There is some impact: maybe not so significant compared with the NRDL or VBP. For example, if an oncology drug can only be prescribed and injected within the hospital, the cost will be managed by the DRG. If the key part of the cost relates to one drug, then the doctors prefer to choose a cheaper one.”
(Multinational Company Executive)

- The exception to these constraints might be if breakthrough therapies with demonstrable evidence of clinical improvement and pharmacoeconomic value may be prioritized under the DRG/DIP system, especially if less cost-effective alternatives have been overprescribed. Conversely, in a high budget-impact category where a number of comparable treatment options are available, DGPs/DIPs are unlikely to favor prescribing of more expensive originator medicines.
- Some cities and provinces, such as Beijing and Zhejiang, are exploring exemptions or add-on payments for costly innovative technologies, including medicines, that would otherwise be caught by the DRG/DIP system. Eligibility is determined by criteria such as level of innovation and clinical improvement, efficacy, safety, cost, budget impact, and annual case load.
- In the meantime, some specialist hospitals have already found ways to work around the DRG/DIP system and maintain prescribing of more expensive therapies usually confined to inpatients. For example, specialist hospitals (e.g. for oncology) may establish day wards where patients are technically outpatients and can access more expensive drugs, such as high-cost injectables, under the same reimbursement criteria as an inpatient, without falling within the scope of a DRG/DIP.

Growing Pressures on BMI Finances Affect Hospital Prescribing

- Growing pressures on BMI finances (see Health Provision: Healthcare Financing and Expenditure) are also trickling down to hospitals, with restraints on prescribing typically felt as hospitals run short of funds towards the end of the year.

“At the end of the year, a lot of hospitals have run out of funds, and everyone’s insurance card has run out of funds. So, instead of giving three months’ prescription, they give you a month’s prescription and hope you come back early next year.” (Multinational Company Executive)

Essential Drug List

Establishing an essential drug system was one goal of the 2009 healthcare reforms, and the first national Essential Drug List (EDL) was published during that year. It has been updated twice since then. The latest version, unveiled by the NHC in October 2018, contains 685 drugs (417 western medicines and 268 TCMs). The list was initially intended to guide drug usage in CHCs, which had to restrict prescribing exclusively to molecules on the EDL. This discouraged patients from using primary care facilities, however, and restrictions on prescribing outside the list have been relaxed over time.

The NHC issued draft measures for the administration of the EDL in November 2021, while a new approach to EDL management was outlined by the National Medical Products Administration (NMPA) in May 2022, as part of a draft regulation designed to implement changes to the Drug Administration Law (DAL). The proposals envisaged efficacy and pharmacoeconomic considerations being taken into account during the selection of candidates for EDL inclusion. They also sought a more proactive approach to management of the list, involving comprehensive updates every three years, and ad hoc inclusion of newly approved drugs, as long as these were available at “appropriate” prices. As of the end of 2024, however, there had been no announcements about the next update of the EDL.

National Reimbursement Drug List

The NRDL, which provides a framework for reimbursement of drugs prescribed to BMI beneficiaries, comprises two sub-lists. Products in List A, which includes drugs deemed clinically essential and available at low prices, must be reimbursed fully by all provinces. Products in List B, which contains more expensive drugs, are partially reimbursed, with patients making percentage-based contributions to their cost. Regulators have pursued a more proactive approach to updates of the NRDL since 2017, leading to a progressive increase in the number of drugs eligible for reimbursement by BMI schemes.

The latest annual update of the NRDL, announced in November 2024, added 91 new drugs to the list – fewer than the 126 drugs added in 2023. A record number of drugs were innovative drugs, with 90 drugs launched in the previous five years, and 38 new innovative drugs globally. The presence of domestically developed drugs is growing, with 65 newly added drugs from domestic companies, accounting for more than 70%. Newly added drugs include 26 oncology drugs and 13 rare disease therapies.

Key Developments and Trends

NRDL Listing Slows with 2024 Update, Prioritizes Innovation

- The latest and seventh annual update of the NRDL, announced by the NHSA in November 2024 and implemented on 1 January 2025, added 91 new drugs to the list. This was considerably fewer than the 126 drugs added to the NRDL in 2023. Of the drugs added in 2024, 89 were included following a negotiation/bidding process, while two products were incorporated directly into the NRDL after being selected for national VBP.
- At the same time, 43 drugs, which the NHSA said had either been superseded in clinical practice or had not been manufactured and supplied for a long time, were taken off the list. In the 2023 NRDL update, only one drug was removed from reimbursement.
- The 91 drugs added to the list in November 2024 included 90 innovative drugs launched in the last five years, and 38 new innovative drugs globally - a record high both in terms of percentage and absolute number. They included 26 anticancers; 13 rare disease therapies; 15 chronic disease therapies, such as anti-diabetics; seven anti-infectives; four psychiatric drugs; and 11 TCMs, the NHSA said. Some of these drugs have multiple indications.
- With 43 drugs removed from the NRDL, the total number of drugs on the list increased from 3,088 in the 2023 update to 3,159 as of December 2024, including 1,765 western medicines and 1,394 TCMs. According to the NHSA, options for reimbursed medicines were significantly expanded in the fields of cancer, chronic diseases, rare diseases and pediatric medicines.
- The presence of domestically developed medicines on the NRDL has also grown, the NHSA pointed out. Of the 91 newly added drugs, 65 were from domestic companies, accounting for more than 70% of all listings. These included the first dual-target immunotherapy for advanced cervical cancer and the first targeted drug for advanced exon 20-mutated lung cancer with exon 20 insertion.

“One thing we feel is that the government really encourages local Chinese innovative products. You can actually get a real premium on your price if your product is locally developed. If you already have multinational brands in the list, so theoretically your product is a latecomer, your price should be lower than [those of] existing imported brands. But, to some extent, you can still get a premium over them.” (Local Company Executive)

- Nonetheless, there were some notable additions from multinational players to this year’s NRDL. Among these were Novo Nordisk’s GLP-1 agonist Ozempic (semaglutide) for type 2 diabetes; Eli Lilly’s Verzenio (abemaciclib) for breast cancer; Janssen’s Darzalex (daratumumab) for multiple myeloma; Bristol-Myers Squibb’s Zeposia (ozanimod) for multiple sclerosis; Everest Medicines’ Tarpeyo for IgA nephropathy; GSK’s Benlysta (belimumab) for systemic lupus erythematosus; and Alexion Pharmaceuticals’ Soliris (eculizumab) for paroxysmal nocturnal hemoglobinuria.

Hospital Formularies

Gaining access to public hospital formularies is essential for prescription generation, even when drugs are included on the NRDL. Formulary listing can be challenging and lacks transparency. Recognizing

that patients will not reap fully the benefits of NRDL listing unless hospitals are more proactive about uptake, the government now mandates the inclusion of newly listed products in hospital formularies. Previous caps on the hospital drug-to-revenue ratio have been relaxed in a bid to encourage listing of more costly medicines.

While there have been selective improvements in hospital formulary listings following NRDL updates, barriers persist. These include delayed or even non-existent formulary-committee meetings. More established NRDL-listed drugs face further difficulties accessing formularies if they lose out on VBP contracts. With the number of drugs on the NRDL increasing, and with more favorable listing criteria for breakthrough therapies, competition for hospital formulary listing will remain fierce.

Key Developments and Trends

Some Improvement for Hospital Formulary Listing

- Access to hospital formularies for NRDL-listed medicines, and the associated timing and frequency of formulary committee meetings, are still variable and often unpredictable. IQVIA analyses have shown that it takes two to three years on average for drugs added to the NRDL to achieve listings on 80% of targeted hospital formularies.
- Barriers to hospital formulary updates include hard caps on permissible listings, medical insurance coverage and controls, the limited nature of DRG/DIP standards, controls under the BMI system, the anti-corruption campaign, as well as concerns about the contribution of drugs to hospital income or their impact on overall treatment costs.

“On average, we probably take one to two years to complete more than 60% of the market we need to cover.” (Multinational Company Executive)

- The NHSA’s November 2024 *Notice on Issuing the National Basic Medical Insurance, Work-related Injury Insurance and Maternity Insurance Drug List* seeks to speed up formulary listing of drugs newly included in the NRDL.
- The notice stipulates that designated medical institutions should, “in principle”, hold pharmacy committee meetings before the end of February 2025, and ensure timely adjustments to their formularies. Reportedly, Guangxi now requires all regional hospitals to organize formulary committee meetings within one month of NRDL listing. In Fujian, Jilin, Inner Mongolia and other areas, there is a two-month window, while Guangdong and Hainan require meetings being held within three months of the NRDL update.
- Anecdotal evidence suggests that some hospitals, such as those in Shanghai, have significantly improved the timeliness of formulary listing procedures over the last couple of years, with almost 100% of NRDL-listed innovation added to formularies after one month. Some local authorities have also allowed for newly NRDL-listed drugs to be excluded from global hospital budgets or audits to encourage uptake.

“In Shanghai, there’s significant improvement in the past two years. [...] They ask the hospital and the doctors that they should not set any barrier for prescription use of these innovations.” (Pharmaceutical Industry Consultant)

- Strategies for accommodating NRDL-listed medicines left off, or waiting for access to, hospital formularies, also include the dual-channel mechanism, allowing reimbursement of these products in designated retail pharmacies (see Dispensing). This initiative has been

relatively slow to take off, though, and is now the focus of an anti-corruption clampdown (see Dispensing).

Challenges Remain, Amplified by VBP and Growing Competition

- For all of these efforts, however, formulary listing remains a challenge. There are reports of some hospitals never holding formulary committee meetings, of substantial listing delays, or of uncertainties about when committee meetings are due, and who will be chairing them. Moreover, there may be limited slots on hospital formularies where a number of comparable drugs have been NRDL-listed for the same condition. In some cases, hospitals request donations or other support in exchange for listing.

“More and more we face issues with the formulary committees. When are they meeting, Who is chairing? Is there someone to chair that committee? That’s probably one of the main questions.” (Multinational Company Executive)

“Some of Zhejiang’s hospitals have a lot of limitations. They also rely on some engagement with pharmaceutical companies before product listing. We are a company with a high compliance standard, we are not allowed to exchange anything with the hospital. [...] For example, they might ask you to donate something for their events, and then probably you can enter a priority list.” (Multinational Company Executive)

- Centralized VBP of multisource medicines used in hospitals adds another layer of complexity and difficulty for more established NRDL-listed drugs vying for formulary placement, particularly as hospitals will prioritize VBP-selected drugs with contractual obligations to meet volume quotas.
- With progressive increases in the number of drugs added to the NRDL, and a growing emphasis on listing of more costly breakthrough innovations (albeit with prices moderated by listing concessions), competition for formulary placement will only intensify in the coming years. The onus will be on companies either to present compelling cost-effectiveness data on new medicines to formulary committees/key opinion leaders, or to seek alternative outlets in the dual-channel/retail or private-sector markets.

Dispensing

Hospitals dominate dispensing activity, although government policy favors the progressive separation of prescribing and dispensing. This was encouraged by the elimination of hospital drug mark-ups and the imposition (since relaxed) of hospital drug-to-revenue caps. The COVID-19 pandemic shifted some dispensing activity, as patients avoided visiting healthcare facilities, purchasing prescription drugs instead in retail pharmacies or via online pharmacy platforms. Nevertheless, IQVIA data show that hospitals generated over 80% of drug sales in 2024.

Dispensing of outpatient prescriptions outside hospitals will increase, supported by the dual-channel policy, which enables reimbursement of selected NRDL drugs in designated retail pharmacies. Progress with the policy has been slower than expected, however, and controls on dual-channel dispensing will be tightened. Many companies have increased investment in the retail channel in recent years, especially for products that lost out on VBP contracts. Shifts in dispensing activity will be gradual, however.

Key Developments and Trends

Gradual Shift to Retail Outpatient Drug Dispensing

- Policy initiatives to move more outpatient drug dispensing out of hospitals, which received an unexpected boost from the COVID-19 pandemic, have encouraged manufacturers to pay more attention to retail and online pharmacy channels. The trend gained momentum as multinational companies in particular struggle to secure public hospital listings for drugs that miss out in VBP bidding and focus more attention on the retail pharmacy sector (see Healthcare Provision: Drug Procurement).
- Many companies have increased investment in the retail channel in recent years, especially for products that lost out on VBP contracts. At the same time, companies report that close engagement with the retail trade may be necessary to avoid generic substitution, as patients attempting to refill prescriptions for branded drugs in retail pharmacies may be steered towards cheaper generics.

"We found that in some areas of that market, whenever the patient tried to refill a prescription, the pharmacist would say, "Come on, this product is too expensive. You need to pay 30 to 50% of the drug cost." They can choose cheaper generics with 100% self-pay and their cost is pretty low, lower than our out-of-pocket part. So, if we can establish some engagement with retail pharmacists, of course, we can avoid that kind of situation." (Multinational Company Executive)

Dual Channel Is Growing, Albeit Slowly

- Uptake of the 'dual-channel' policy, which enables the reimbursement of selected NRDL drugs in designated retail pharmacies, and is intended to divert the flow of dispensing from hospitals to retail pharmacies, has been slower than expected to date.
- Uptake has been hampered by a number of factors, including significant disparities between provinces in the number of designated dual-channel retail outlets; variations in the contents of dual-channel drug lists (DCDLs), based on assessments of local clinical need; funding restrictions on dual-channel uptake at city level; formulary listing policies in hospitals, some of which prefer not to list drugs obtainable in retail pharmacies under dual-channel conditions; and the substantial investments in technology and personnel needed for pharmacies to supply dual-channel drugs, for which they cannot charge mark-ups.
- To address some of the barriers to prescription outflow from hospitals to retail pharmacies under the dual-channel scheme, provinces were asked to establish online platforms by the end of 2023 that would link BMI-designated hospitals with participating pharmacies. Hospital prescriptions could then be transmitted automatically to designated dual-channel pharmacies, which could dispense the prescriptions and claim BMI reimbursement simultaneously. Most provinces had launched these platforms by the end of 2023.
- At the same time, provinces issued directives on DCDLs, to standardize hospitals' dual-channel policies - for example, by integrating dual-channel prescribing into the activities of the hospital's pharmacy and therapeutics committee.
- Manufacturers are exploring dual-channel opportunities, particularly for NRDL-listed multisource originators shut out from VBP contracts, or even as a prelude to hospital-formulary listing. However, more clarity is needed around the criteria for inclusion on dual-

channel drug lists. These are managed at city level. However, while DCDLs are based on NRDL listing, they can vary markedly according to local factors, such as disease profiles or budgetary capacity.

“Dual-channel reimbursement is based at city level. In China we have 339 prefecture cities, in which they have different policy capacities. The central government decides which products can enter the NRDL. However, the policy landing is totally different in each city or each province, purely based on their attention to some diseases or their budget capacity.” (Multinational Company Executive)

- For branded companies, leveraging the dual-channel policy also means dual-channel marketing, to reinforce linkage between hospital prescribing and retail-pharmacy dispensing, even if the designated pharmacy is hospital-affiliated. Hospital doctors may have an originator drug listed by generic name in their prescribing software, meaning that the doctor needs to be persuaded to write the brand name on the prescription, while the patient must pick up the originator in the designated retail pharmacy, and with a higher co-payment than for the same drug as a hospital inpatient.

“That’s why we still need warm bodies on the ground to promote the drug to the physicians, to say that, “There is a generic version in the hospital, but if you want the originator, it is available in the dual-channel pharmacy outside of the hospital.” Our headquarters sometimes find it difficult to understand that. If you have it in a retail pharmacy, why do you still need to have someone to promote it in the hospital?” (Multinational Company Executive)

Controls on Dual-Channel Dispensing to Be Stepped Up

- There are indications that controls on BMI spending for outpatient dispensing of listed medicines under the dual-channel policy will be stepped up. The NHA’s November 2024 notice accompanying the publication of the new NRDL highlights the need for local BMI departments to step up supervision of dual-channel activity to combat fraud, while ensuring that BMI affiliates have ease of access to designated medicines.
- As of 1 January 2025, designated retail pharmacies will have to process dual-channel prescriptions through a new centralized online system. This will tie outpatient expenditure more closely to reimbursement restrictions imposed on dual-channel medicines in hospitals, avoiding too much budgetary overspill at provincial or city level (see Prescribing Influences).

“For some of the products, they are trying to start connecting the hospital reimbursement with the dual-channel pharmacy, to make sure that, if the hospital reimbursement is limited, they will not overspend in the dual-channel pharmacy at the provincial or the city level.” (Multinational Company Executive)

- The government will nevertheless continue to support the strategy as a means of easing budgetary pressures on hospitals and broadening treatment choices for patients. Originator companies in particular will also step up their dual-channel efforts, as they look for new sales outlets that can accommodate both a narrower focus on more costly innovative medicines, and pragmatic adjustments to a VBP environment in which originator products are increasingly marginalized, unless they are prepared to accept potentially unsustainable prices.

Pricing and Reimbursement

Pricing System

The removal of explicit drug price controls in 2015 marked a shift towards a largely market-based pricing system. While companies must still file pricing information with regulators at launch, the actual prices achieved by drugs qualifying for BMI reimbursement are now determined almost exclusively through either negotiation (for patented and other single-source products seeking access to the NRDL), national VBP tenders (for multisource drugs available in bioequivalent-certified generic form), or tender-based procurement at provincial, city or hospital level.

Pricing pressure on both new and more established medicines will be maintained, as the government looks for new ways to balance encouragement for pharmaceutical innovation with relieving pressure on BMI budgets. A more interventionist approach to launch prices issued by the NHSA in early 2024, caused some concern in industry, but appeared to have been withdrawn as of early 2025.

Key Developments and Trends

NHSA Proposal for New Launch Pricing Mechanism

- In February 2024, the NHSA issued a draft *Notice on Establishing an Initial Pricing Mechanism for Newly Listed Chemical Drugs to Encourage High-quality Innovation*, for comment from industry associations. On paper, the NHSA's draft *Notice* was intended to allow innovative companies more flexibility in setting procurement prices, while also facilitating the procurement process for innovative drugs.
- The draft proposal called for a value-based self-evaluation process for chemical drugs to help determine initial pricing. Self-evaluation would be guided by a comprehensive form, with different point scores assigned to different aspects, while also considering whether a drug is a breakthrough treatment, is indicated for rare diseases, or addresses supply shortages. A quantified self-evaluation table would categorize products into 'high', 'middle', and 'low', based on scores for pharmacology, clinical value, and clinical evidence of safety and efficacy.
- For drugs with high self-evaluation scores, the draft proposal would have allowed an expedited application process to provincial procurement platforms; exemption from providing detailed pricing component materials; early marketing price application; and a price stability period of one to five years, during which their price would remain unaffected by government interventions, including VBP.
- The draft proposal is understood to have been designed to link with the new value rating system for innovative medicines used in national price negotiations for NRDL listing (see Health Technology Assessment). The expectation was that drugs with self-evaluation scores of 'middle' or 'low' would likewise be priced lower in NRDL negotiations. At the same time, a truly innovative medicine would still have to negotiate a reimbursement price (inevitably much lower than the self-determined launch price) if it wanted to be available through the NRDL.

Proposal Withdrawn for The Time Being Following Industry Concerns

- The draft proposal was seen by companies as trying to restrict the current free market-oriented launch pricing, available for newly launched medicines without the burden of self-

evaluation. While price controls are currently enacted through mechanisms such as national price negotiations for NRDL listing and VBP, these relate not to launch prices, but to negotiating acceptable prices for hospital procurement and reimbursement through the BMI system.

- Another concern was the proposed benchmarking of new products that offer only moderate or low levels of innovation to the prices of equivalent drugs already on the market, without any indication of how that equivalence would be defined; doubts about the rigor and precision of the mooted self-evaluation scales; and the limited value of a potential one- to-five-year price freeze for newly launched innovative drugs.
- Furthermore, there were indications – worrying for industry – that the NHTA’s conception of drug innovation might differ significantly from the criteria applied by companies themselves or by the NMPA. For example, the NHTA would place more emphasis on different mechanisms of action and chemical structures, as well as novel benefits for patients, generally imposing a higher barrier to entry.
- The draft proposal elicited strong objections from the industry, especially from local companies. As of December 2024, there were reports that the draft proposal had been withdrawn, at least for the time being, and was not expected to be implemented in the short term.

“The whole industry was against it. [...] I think the draft policy was withdrawn, so they didn’t implement this. But this actually placed a signal that the government not only want to guide the NRDL price, they actually also want to guide the new launch price, which is in the private market. So the whole industry actually stopped it. But it is not a good sign, I would say.” (Local Company Executive)

- While the NHTA’s draft notice appears to have been withdrawn for the time being, it gives some flavor of the government’s current thinking on drug pricing. The proposal implies a degree more control over free pricing of medicines at launch, whether or not that involves a self-assessment process and/or is geared to encouraging and supporting genuine drug innovation.

National Price Negotiations

Since 2016, national price negotiations have been used to lever substantial price cuts from innovative new drugs in return for inclusion on the NRDL. While the number of innovative medicines on the list has grown significantly, manufacturers have made substantial concessions to secure reimbursement, as regulators have applied increasingly sophisticated negotiating tools to NRDL updates.

Manufacturers have also been forced to cut prices further during the renegotiation of initial two-year listings for innovative products.

Of the 91 drugs included in the NRDL in November 2024, 89 were listed following a national price negotiation or bidding process, with an average price reduction of 63% - the sharpest average cuts since annual updates of the NRDL started in 2017. Since 2023, the introduction of product-specific value ratings, and incremental cost-effectiveness ratio (ICER) thresholds for innovative new drugs has resulted in more generous envelope prices for innovative products, although only a relatively small percentage of drugs are likely to be classified as truly breakthrough therapies. New rules on the

renewal of existing NRDL listings, introduced in 2022 and 2023, have reduced price erosion on listed products.

Key Developments and Trends

The 2024 Round of National Price Negotiations

- A total of 574 drugs entered the formal review process for NRDL listing in June 2024, and 445 were long-listed in August. Of the long-listed drugs, 249 were seeking initial NRDL listing and 196 were up for renewal or re-negotiation. Following the expert review process, only 47% of the drugs seeking initial listing were shortlisted for negotiations, compared with 64% in 2023 and 67% in 2022.
- A total of 117 drugs not previously listed on the NRDL entered national price negotiations in November 2024, of which 89 were listed following a national price negotiation or bidding process (a success rate of 76%), with an average price reduction of 63%, the sharpest average cuts since annual updates of the NRDL started in 2017, although comparable to the 62% seen in the 2023 update.

Innovation Prioritized with Value Rating System and ICERs

- The NHSA pointed out that more than 90% of innovative drugs successfully completed negotiations in 2024, with 90 of the 91 newly added drugs launched in the last five years, and 38 of them new innovative drugs worldwide. This highlights the more innovation-friendly climate for pharmaceuticals in China generally, and more specifically in pricing and reimbursement.

“We also have the news from the government that, for this [2024] round, they will try their best to include innovative drugs [on the NRDL] and will further reinforce the VBP part to achieve budget control and make the room for innovative drugs.” (Pharmaceutical Industry Consultant)

- The relatively high profile of innovative drugs in the 2024 NRDL update (and in 2023, when their success rate at the negotiation stage was 92%) also testifies to the NHSA's adoption in 2023 of a pilot 'value rating' system and the application of ICER (incremental cost-effectiveness ratio) thresholds to breakthrough products offering significant improvements over current therapies (see Health Technology Assessment).
- With the introduction of product-specific value ratings, and ICER thresholds for innovative new drugs, more generous envelope prices are now available to these products (see Health Technology Assessment). This has led to more successful price negotiations for innovative new drugs, such as Ozempic (semaglutide), Darzalex (daratumumab), Enhertu (trastuzumab deruxtecan) or Zeposia (ozanimod) in 2024. While over 90% of innovative medicines entering negotiations were added to the NRDL in 2024, the success rate for all other drugs was 76%.
- The NHSA insists it has been allocating more 'real money' to support innovative-drug development. In its December 2024 press conference, the NHSA said that 835 drugs, including 149 innovative medicines, had been added to the list over seven rounds of negotiations. As of October 2024, the BMI had spent more than RMB350 billion on these products, with the outlay increasing each year. In the first 10 months of 2024, for example,

the BMI paid around RMB92 billion for negotiated drugs within the latest agreement period, 21 times more than in the same period of 2019.

- The flipside to more innovation-friendly conditions is lower tolerance for ‘me-too’ drugs deemed to offer no meaningful improvement over currently available therapies, unless these products have potential to broaden clinical choices within a therapeutic category. IQVIA China analysis estimates that around 80% of new drugs submitted for inclusion on the NRDL may be classed as merely equivalent to existing therapies.
- Companies report that, overall, the process has become clearer, even though price pressures remain considerable. Companies are also wary about NRDL-agreed reimbursement prices being disclosed and used for price referencing, although the NHSA did introduce a price confidentiality mechanism for selected drugs in the 2023 NRDL update.

“The rules are much clearer now than before, and it gets clearer as the government has multiple rounds, they learn from the previous times, and companies like us plan much better how we want to get into NRDL. The question is not about the clarity of the rules, but more about whether the rules are something multinationals are willing to partake in. For example, giving lowest prices in the world, multinationals are always careful about whether the prices are going to be referenced by the other parts of the world.”
(Multinational Company Executive)

More Modest Price Cuts Under Amended Renewal Conditions

- New rules on the renewal of existing NRDL listings, introduced in 2022 and 2023, have reduced price erosion on listed products. Under the 2022 rules, downward price adjustments are calculated in accordance with an explicit algorithm, which has halved price cuts for drugs on the NRDL for more than four years. In an additional change, introduced in 2023, drugs listed via price negotiation that have been on the NRDL for more than eight years are incorporated automatically into List B of the NRDL, without any need for further price revisions.
- The new rules have already led to more modest price adjustments for listed drugs, while reducing the number of products that need to renegotiate prices. In the 2023 NRDL update, some 70% of 100 drugs were renewed at their original prices, while 31 products had their prices reduced by an average of 6.7%, after exceeding anticipated sales volumes, according to the NHSA. A total of 17 drugs with new indications were renewed at their original prices.
- In the 2024 NRDL round 44 drugs came up for renewal, with the resulting price cuts averaging 11.6%. Despite the new rules, renewal conditions were competitive, and several drugs failed to secure listings for new indications.

Reimbursement

A National Reimbursement Drug List (NRDL) was first established in 2000. Historically, updates to the list were sporadic, while the scope of provincial reimbursement lists varied widely. Since 2017, though, the NRDL has been updated at least once a year, and in 2020, the NHSA confirmed that the list would be revised annually going forward. With provincial reimbursement lists now required to mirror the national list, the NRDL is the sole determinant of BMI funding for prescription drugs.

In recognition of the limits to reimbursement of costly innovative drugs under the BMI schemes, the NHSA in December 2024 called for a diversification of payment mechanisms beyond the NRDL, proposing the development of a new reimbursement list – dubbed ‘List C’ – which would include innovative drugs to be reimbursed under *huiminbao* schemes initially, and which would later be extended to other commercial health insurance schemes.

Key Developments and Trends

Delays in Securing Reimbursement Have Significantly Improved

- Since the NHSA started updating the NRDL annually, delays in securing reimbursement for drugs approved by the NMPA have reduced significantly, following the extension of potential listing to any new drug approved during the first half of the calendar year in which annual NRDL revisions occur.
- The scope of the NRDL update for 2024 included medicines approved by the NMPA up to 30 June 2024. Among the 91 drugs added to the NRDL, 90 had been launched in China during the previous five years. In the 2023 NRDL update, 108 of the 224 drugs included on the long list for negotiations had been approved in China during the previous 12 months, and 73 since the beginning of 2023.

Limits to NRDL Reimbursement

- For all the benefits of the new value rating system, affordability is still front of mind for regulators where the BMI system is concerned. An (undeclared) annual price cap of around RMB500,000 prior to national price negotiations, and RMB300,000 thereafter, applies to breakthrough therapies considered for inclusion on the NRDL. This has repeatedly excluded a number of CAR-T products from reimbursement and will make listing difficult for ultra-expensive cell and gene therapies.
- Originator companies have called for adjustments of the price negotiation rules in a bid to accommodate more expensive breakthrough medicines, such as price-volume agreements. Nonetheless, these and other more advanced value-based reimbursement pricing models (e.g. payment by results drawing on real-world data) are unlikely to find a footing in China for some time, particularly with growing pressures on BMI funds (see Healthcare Provision: Health Insurance).
- Perceived ‘lifestyle’ drugs, such as of GLP-1 receptor agonists for obesity, are expected to have to generate convincing evidence of more far-reaching health benefits (e.g. cardioprotection) before they can qualify for BMI reimbursement. The new anti-obesity treatments, such as Novo Nordisk’s Wegovy (semaglutide, approved in June 2024 and marketed as NovoCare in China) and Eli Lilly’s Mounjaro (tirzepatide, approved for the treatment of obesity in July 2024), will initially have to pursue opportunities in the out-of-pocket market or through group-insurance platforms. With some 9% of adults classified as obese in China, according to the World Obesity Federation, there could be a sizeable population, at least in wealthier urban areas, ready to pay for anti-obesity treatments out of pocket.

“For treating or preventing indications like cardiovascular disease or kidney failure, I think, [weight-loss therapies] can definitely be reimbursed in hospitals. We’ll see how soon and how effectively these companies develop their clinical data and submit new drug applications, also involving the national price negotiations.” (Pharmaceutical Industry Consultant)

Reimbursement Opportunities Outside the NRDL Remain Limited

- Commercial health insurance (CHI) schemes, and *huiminbao* schemes in particular, have begun to offer some potential for drug reimbursement outside the NRDL, although growth has slowed in recent years (see Healthcare Provision: Health Insurance). According to IQVIA China analysis, the absolute number of drugs covered by CHI schemes increased by 87%, from 84 in 2020 to 550 in 2023, while the average number of drugs covered by CHI schemes rose by 35%, from 18 in 2020 to 44 in 2023. CHI coverage has to date been limited mostly to cancer, rare disease and other specialty medicines.
- Competition for listing by *huiminbao* schemes can be fierce, and the sheer number of schemes complicates efforts. The listing process may involve dealing with third-party administrators such as Meixin, Yuanxin Technology and Chenxi, which tend to be affiliated with major distributors. Moreover, the listing criteria may lack clarity, while some companies complain of preferential treatment given to local players based in a particular city or province.

“Several of our brands joined some of the city-level insurance [schemes], but we don’t know if it’s going to be sustainable. And the second thing is, the competition to get listed in these city-level insurance [schemes] is furious. You have to work with third-party TPAs to get your product in, and the criteria actually is not clear at all.” (Local Company Executive)

- Despite the difficulties, companies continue to pursue inclusion of new drugs on *huiminbao* formularies prior to NRDL inclusion, with many seeking to pave the way for faster take-up of drugs that are later included on the NRDL. Company strategies are portfolio-specific, however, and depend on patient volumes, with NRDL listing usually the preferred option where large patient volumes compensate for negotiated price reductions.

“If I were in rare diseases, I would consider commercial health insurance, because the population is really small and it doesn’t make sense cutting the prices to the world’s lowest prices to get into the NRDL, and then struggling to find those patients. But with the portfolio that we have, there’s a sizable amount of population. It’s a very competitive market that if I don’t get into the NRDL, I risk not being competitive.” (Multinational Company Executive)

- In a recent example, in October 2024, a *huiminbao* scheme in Zibo, Shandong province, started covering Eisai/Biogen’s Alzheimer’s treatment Leqembi (lecanemab) through the scheme’s high-value drug list for outpatients. The annual deductible was RMB20,000, while cumulative maximum compensation was set at RMB1.5 million. Eisai said it is testing the waters with Leqembi in the Chinese private market before deciding whether to seek reimbursement through the NRDL.

NHSA Call for Diversification of Payment Mechanisms for Innovative Drugs: List C

- In recognition of the limits to reimbursement of costly innovative drugs under the BMI schemes, the NHSA in December 2024 called for a diversification of payment mechanisms for innovative drugs beyond the NRDL. It proposed the development of a new national-level reimbursement list - dubbed ‘List C’ – which would include innovative drugs to be reimbursed

under *huiminbao* schemes initially, and which would later be extended to other commercial health insurance schemes.

- List C would supplement Lists A and B of the NRDL governing reimbursement under the BMI schemes, to expand coverage to high-value, innovative drugs offering significant clinical benefits, but which do not qualify for reimbursement under the BMI schemes for cost reasons. While selection criteria remained unclear as of early 2025, priority may be given to drugs which passed formal expert review for national price negotiations, but which did not reach the negotiation stage, such as for example CAR-T therapies, with treatment costs much higher than the benchmark for national negotiations.
- The publication of a draft NHA proposal outlining details for comment by the industry was expected as of early 2025. It was understood that the application, evaluation and publication procedures for List C would take place in parallel to procedures for the 2025 update of the NRDL, with manufacturers given the choice of applying for NRDL or List C listing. The list was expected to include around 10-20 products initially, and while listing procedures were expected to involve price negotiations, these were anticipated to be less drastic than those for NRDL List B, according to press reports.
- List C, which would represent the first national catalog of drugs reimbursed by commercial health insurance, would regulate and standardize drug reimbursement under *huiminbao* schemes (and eventually other commercial health insurance schemes) and has been welcomed by the industry. By aligning reimbursement under BMI and commercial health insurance schemes more closely, the publication of List C will represent a step towards the government's objective to establish a comprehensive, multi-level health insurance scheme.

Payment Standards

Introducing the 2015 pricing reforms, policymakers signaled that their longer-term plans included establishing a reference pricing mechanism for drugs reimbursed by BMI schemes. This would involve determining reimbursement 'payment standards' applicable to all versions of multisource drugs included on the NRDL. Patients would have to pay the difference between the payment standard and the price of any more expensive versions dispensed to them.

In its 14th Five-Year Plan, the State Council underlined its commitment to establishing reimbursement payment standards for a growing number of multisource drugs on the NRDL. While prices secured through national price negotiations and VBE tenders already function as de facto payment standards, a competitive bidding procedure for multisource drugs seeking access to the list, introduced with the 2022 NRDL update, has begun to create payment standards for additional molecules. While progress has been gradual, payment standards for multisource products favor drugs priced at or below the standard.

Key Developments and Trends

Gradual Progress with Payment Standards for Multisource Products

- In its 14th Five-Year Plan, the State Council underlined its commitment to establishing reimbursement payment standards for a growing number of multisource drugs on the NRDL, including chronic-disease treatments prescribed on an outpatient basis.
- A new competitive bidding procedure was introduced for the first time with the 2022 NRDL update, opening up listing to non-reimbursed multisource products. Under the procedure, the NHSA calculates an envelope price, with multiple manufacturers to submit price offers. Where one or more offers are equivalent to or below the envelope price, the drug is added to List B of the NRDL, with the lowest offered price the BMI reimbursement payment standard for the molecule, subject to a floor price of 70% of the envelope price.
- The first 17 products were admitted to reimbursement on the basis of successful bidding procedures following the 2022 NRDL update, while a further five drugs were admitted to the NRDL following competitive bidding in the 2023 update. As of early 2025, the NHSA had not indicated how many products were admitted under competitive bidding procedures in the 2024 update.
- Payment standards for multisource drugs are expected to influence prescribing for affected molecules, to the benefit of versions that do not require patient co-payments above the agreed standard.

Health Technology Assessment

The NHSA has overseen moves towards the formal integration of health technology assessment (HTA) into drug pricing and reimbursement procedures. The submission of pharmacoeconomic data and budget-impact analyses became mandatory in 2018 for sponsors of high-cost medicines invited to enter into national price negotiations. The National Center for Evaluation of Medicines and Health Technologies was established in October of that year. Updated guidance on the use of pharmacoeconomic analysis in pricing and reimbursement assessments was published in August 2020.

The NHSA's new value-rating system for NRDL candidates will continue to be refined over the prognosis period. Only a relatively small percentage of drugs are likely to be categorized as 'breakthrough', which requires for China to the global first-launch market, while the vast majority will be judged as 'equivalent'. Innovative medicines will still be subject to overall cost ceilings, while reliance on low-priced comparator drugs will be a particular source of frustration to companies.

Key Developments and Trends

Value-based Classification at the Expert Review Stage

- A new 'value rating' system, piloted during the 2023 NRDL update, introduced the possibility of premium pricing for innovative new drugs recognized as addressing unmet medical need or offering significant improvements over existing therapies.

- The value rating system comprises four categories based on a product's perceived clinical value:
 - **Breakthrough:** a first-in-class drug with no obvious price or other comparator, launched first in China.
 - **Improvement:** a drug showing a high level of innovation and a significant improvement compared to existing therapies, with benefits not previously seen in the class.
 - **Equivalence:** a drug showing no evidence of additional benefit but potential to broaden clinical choices in the therapy class.
 - **Inferiority:** a drug's clinical value (efficacy, safety or patient compliance) is inferior to that of existing products on the NRDL, and where there is a high level of competition in the therapy area. These products would be low priority for NRDL listing.

- The assigned value-based classification determines an innovative drug's price in relation to reference products. IQVIA China estimates that up to 5% of new drugs seeking NRDL listing will be categorized as breakthrough therapies, with the proviso that China needs to be their first launch market. Between 5-10% of drugs, including the majority of MNC originator drugs, could fall into the 'improvement' category, while as many as 80% are likely to be judged as equivalent.

"An MNC may have a really good drug with an innovative mode of action. But if you want to get into the 'breakthrough' category, we don't think it depends on how excellent the materials [are], or what value story you prepare. You must put China as a first-round global launch." (Pharmaceutical Industry Consultant)

- The basic aim of the value rating system is to enhance equity and efficiency in NRDL decision-making, in terms of determining envelope prices. Innovation is prioritized by awarding qualifying drugs price premiums versus comparator products. While drugs in the 'breakthrough' category will be allowed a high price premium, those in the 'inferiority' category will be expected to provide a discount on already-listed drugs.
- Categorization is decided during the NRDL selection process, as part of the integrated review by the two expert groups, which score products against a range of criteria and which also evaluate health economics and budget impact and decide whether a product will be eligible for national price negotiation.
- Originator companies may disagree with the NHSA the classification of innovation, as well as with the NHSA's selection of comparator drugs for NRDL pricing negotiations. For example, a first-generation drug may be selected as a comparator for a second-generation therapy, or the comparator may be similar to the applicant drug but with different indications.

"[Comparator selection] is very critical during the negotiation, and every manufacturer, they want to strive for non-comparator, which means that they can have their own pricing benchmark. But in most of the cases, the government just lists a very cheap comparator and they need to have the health economic analysis based on some comparator with a very low price. This is a very critical issue." (Pharmaceutical Industry Consultant)

Discounts and Margins

Consumer drug prices in China have historically been inflated by tariffs, value-added tax, complex distribution channels and, until 2017, the imposition of hospital mark-ups. The gap between ex-manufacturer and consumer prices narrowed following the elimination of hospital mark-ups, however. The reduction of tariffs on some drugs and the rollout of the 'two-invoice' system across the distribution chain from 2017 have also reduced overall mark-ups.

Hospitals often still attempt to lever discounts in return for formulary listing, especially for products excluded from VBP contracts. Demand for discounts on non-VBP brands and products not included on the NRDL is also rising in the retail sector, as well as in crowded segments of the market, reflecting the rapid expansion of pharmacy chains that wield considerable purchasing power. Retail margins and discounts are under growing scrutiny, however, and many provinces and cities have imposed maximum pharmacy mark-ups on drugs subject to VBP tender contracts.

Regulatory Environment

Regulatory Agency

As part of a broader institutional restructuring in 2018, the China Food and Drug Administration (CFDA) was replaced by the National Medicinal Products Administration (NMPA). While the CFDA's remit included food products, the NMPA's focus is narrower. Reporting directly to the State Administration for Market Regulation (SAMR), the agency is responsible for oversight of the drug, medical device and cosmetics sectors, from development through to regulatory approval, production, distribution and consumption.

The NMPA will continue to build on the reform process that has markedly improved regulatory efficiency in recent years, driven by a growing emphasis on innovation and by harmonization with international regulatory standards through collaborative channels such as the International Conference on Harmonization (ICH). The NMPA has completed all 68 current ICH guidelines. The agency is also seeking to strengthen bilateral cooperation with selected countries.

Key Developments and Trends

Growing NMPA Involvement in Global Regulatory Initiatives

- Progressive reform of the NMPA will be driven partly by the agency's involvement in global regulatory initiatives, such as the ICH. ICH membership will drive alignment of the NMPA's regulatory standards with international best practices, facilitating global collaboration and improving market access for medicines. The China Drug Evaluation Report 2023, issued by the NMPA's Center for Drug Evaluation (CDE), noted that the NMPA had completed the implementation of all 68 current ICH guidelines.
- The NMPA is also seeking to strengthen bilateral cooperation with a growing number of regulators. The NMPA has been reaching out to regulators in Vietnam and Thailand, and in November 2024, the agency signed memoranda of understanding with regulators from both countries, covering areas such as information exchange, inspections and quality control.

Drug Registration

The regulatory climate for pharmaceuticals has improved markedly since 2015. The pace of change, driven by an overhaul both of drug regulations and the agencies responsible for administering them, has exceeded expectations. This has had a major impact on new drug approval timelines, and on the number of applications filed by both multinationals and local innovators. The entry into force of a revised Drug Registration Regulation (DRR) in 2020 has paved the way for further improvements in regulatory efficiency. Guidance has also been published on conditional approval and breakthrough products.

Drug approvals will continue to pick up pace over the prognosis period, with regulators making further efforts to streamline and facilitate registration of innovative medicines especially, and of foreign drugs addressing urgent unmet needs. Oncology products and biologics will be particular beneficiaries of these trends.

Key Developments and Trends

Drug Approvals Back on Growth Track

- The NMPA saw a significant increase in approval applications during 2023, according to the CDE's 2023 China Drug Evaluation Report, published in February 2024. The CDE accepted 16,898 applications in all (including clinical trial and generic applications), an increase of 36.6% over 2022.
- Drug approvals in general, and approvals of innovative vaccines and medicines in particular, continue to pick up pace. The CDE approved 1,448 applications for chemical medicines in 2023, 12.6% more than in 2022. Oncology remained by far the leading category for chemical drug approvals, at 521, even though this represented an 8% decline over 2022.
- The CDE approved 891 applications for therapeutic biologics, including 695 innovative products, an increase of 26% over 2022. Again, the vast majority (478) of approvals was of oncology drugs. The CDE also cleared 59 approval applications for preventive biologics during 2023, 31 of which were new innovative products. The approval rate marked a year-on-year increase of 72%, following a 25% decline in 2022. In addition, the CDE approved 45 drugs for rare diseases, 92 pediatric drugs, and 3 CAR-T cell therapy drugs in 2023.
- The growth trend continued into 2024, with the NMPA reporting that 31 innovative new drugs were approved between January and August 2024, a gain of nearly 20% on the same period in 2023. As of June 2024, a total of 82 innovative new drugs had been approved since 2022.
- The NMPA's efforts to facilitate approvals of innovative medicines have also been good for the local industry: from the launch of the 14th Five-Year Plan in 2021, the agency had approved 113 innovative drugs developed in China, according to press reports in November 2024, nearly three times the number approved during the 13th Five-Year Plan.

Faster Approvals of Innovative and Urgently Needed Drugs

- The NMPA continues to push for more streamlined approval pathways, both for innovative drugs in general and for medicines produced outside China, where these are the subject of urgent clinical needs on the domestic market. Companies continue to view the NMPA positively.

"In short,[our experience has been] positive, very positive. [...] I can give a very successful example, this year one of our products, a biologic, obtained the first approval in the world in China, even ahead of the US FDA. [...] I am of the opinion that the NMPA has done a fantastic job in fast tracking the approval where your compound really brings benefit." (Multinational Company Executive)
- The CDE approved 40 Class 1 innovative drugs in 2023, up from 21 in 2022. These included 34 drugs approved under special pathways:
 - Priority review: 9
 - Conditional approval: 13
 - Breakthrough therapy designation: 8
 - Special approval: 4 (COVID-19 drugs)
- In October 2024, the CDE announced a new policy whereby innovative drug applications and submissions already granted priority review or conditional approval status would, from

November onwards, be moved to the front of the approvals queue for immediate processing. This 'front-end acceptance' service does not, however, include technical review issues.

- Potentially more consequential are the NMPA's moves to accelerate approvals of foreign drugs urgently needed in China. A draft *Notice on Further Optimizing the Review and Approval of Clinically Urgent Overseas Drugs*, issued in June 2024, proposes that urgently needed foreign medicines, including originator compounds, generics and biosimilars, that meet the agency's specifications should qualify for priority review procedures in China. For example:
 - Clinical trials could be waived, with the agreement of the CDE, following submission of comprehensive trial documentation. This would include including literature reviews and data from international multi-center trials encompassing Chinese participants and showing no differences in effect between Chinese and other patients.
 - Pre-registration testing would be available for urgently needed rare disease therapies, with the standard 90-day inspection period reduced to 70 days, and further reduced from 60 to 40 days if the necessary review involved only product samples.
 - Streamlined overseas registration verification, aligned and integrated with post-marketing overseas inspections according to risk assessment, would be applied to urgently needed drugs in the priority review category.

Clinical Trials

Historically, the requirement to conduct local clinical trials as part of the approval process was a significant contributing factor to delayed market access for most new drugs in China. This was aggravated by lengthy clinical trial application (CTA) approval times and the lack of a prioritization system. In line with the government's desire to improve and accelerate the drug approval process, steps have been taken since 2015 to simplify and clarify clinical trial regulations, and to improve the quality of trials undertaken in the country.

With clinical trials recovering strongly after a COVID-associated lull, a stream of new guidelines and measures will enhance the attractiveness of China as a base for clinical research, in particular by shortening approval times for trials of innovative medicines. While the regulatory framework for large-scale data transfers out of China is now more accommodating, companies may still struggle with grey areas around sensitive or important data, and with persistent restrictions on data exports involving human genetic material.

Key Developments and Trends

Clinical Trials Gain Traction

- China is an increasingly favored destination for clinical trials. Study registration in the domestic market bounced back in 2023, after momentum slowed in 2022 due to the COVID-19 pandemic and a sluggish economy. The number of registered clinical trials rose by 26.1% year on year to 4,300 in 2023, lifting the five-year growth trend to an average 16% per annum.

- According to CDE data, small-molecule drugs still accounted for the bulk (76.0%) of newly registered trials, although registration of studies involving biologics was up by 22.2% over 2022. Among trials for Class 1 innovative medicines, small molecules made up 53.7% of new registrations and biologics 44.8%.
- Domestic sponsors dominated the trial landscape, with 91.7% of new trial registrations, while the number of overseas sponsors fell by 3.2% in 2023. Oncology was a key focus of interest, generating more than 70% of registered trials, 44.8% of which were for biologics.

Further Measures to Accelerate Clinical Trial Approvals

- A number of new guidelines and other provisions governing the approval and conduct of clinical trials in China took effect during 2024. These included:
 - Technical guidelines for clinical trials of gene therapies addressing rare diseases (January 2024)
 - Measures for the supervision and inspection of clinical trial institutions (effective March 2024)
 - Technical guidelines for decentralized clinical trials of rare disease drugs (May 2024)
 - Technical guidelines evaluating the correlation between adverse events and study drugs in clinical trials (June 2024)
 - Technical guidelines covering changes in biological products during clinical trials (June 2024)
- Prominent among these new measures was a pilot program, implemented in July 2024, to speed up and optimize the review and approval of innovative medicines. The pilot, which applies both to domestic and multinational trial sponsors, is restricted to trials of Class 1 innovative products, excluding certain categories, such as cell and gene therapies as well as vaccines.
- The main objective is to reduce to 30 working days, from 60 working days currently, the time taken to review and approve applications for clinical trials with innovative new drugs. The pilot program will last for a year. Beijing and Shanghai, which have a strong industry presence and research capabilities, were selected as the first regions to take part in the program.

Regulatory Challenges on Data Transfer Eased

- Restrictions on personal data exports from China, imposed by the 2023 Standard Contract for Cross-Border Transfer of Personal Information, required regulatory approval where personal health data transfers involved 10,000 or more individuals. That threshold potentially applied to a number of international players conducting multiple clinical trials in China.
- The restrictions were generally eased by the Cyberspace Administration of China's (CAC) final *Regulations on Promoting and Regulating Cross-border Data Flows*, published in March 2024. These raised from 10,000 to 100,000 individuals the threshold for outbound transfers of non-sensitive personal information requiring a standard contract/personal information protection certification from the CAC. Regulatory clearance could be mandated, though, if patient data from clinical trials were regarded as "important" data for privacy and security purposes, or as sensitive personal data, requiring a standard contract or certifications for data on fewer than 10,000 individuals.

- Conditions for defining “important” data (whose transfer out of China requires a CAC security assessment) are now more relaxed, with the new regulations enabling data handlers to treat their data as non-important unless expressly informed otherwise. Whether clinical trial data fall into any of these more restrictive categories is likely to depend on the nature and source of the personal data, as well as the degree of data anonymization.
- For the time being, it remains unclear what exactly is exempt from CAC approvals. For example, under the CAC’s final rules, free trade zones (FTZs) can draw up negative lists for their own jurisdiction determining which types of cross-border data transfer are subject to CAC security assessments, standard contracts or certification.

Human Genetic Material Data Are More Complicated

- Further complications arise when clinical trial data involve human genetic material (HGM). Despite the more permissive environment introduced by the CAC’s final *Regulations on Promoting and Regulating Cross-border Data Flows*, China’s data protection laws usually classify human genetic and genomic information as sensitive personal data. Any cross-border transfers of these data are subject to CAC clearance.
- In September 2024, a government circular was issued on pilot programs easing restrictions on foreign investment in the healthcare sector (see Healthcare Provision: Hospital Sector). The circular included provisions for foreign-invested enterprises (FIEs) to develop gene and human stem-cell technologies in the FTZs of Beijing, Shanghai, Guangdong, as well as the Hainan Free Trade Port.
- These activities would include the registration, marketing and production of cell and gene therapies (CGTs). Previously, foreign investment in CGT development within China was restricted, or expressly prohibited, for reasons including biosecurity and ethical concerns, particularly in relation to the management of genetic and genomic data control.
- Although the pilot programs open up possibilities for multinational company involvement in local CGT businesses on a limited basis, the circular also made clear that FIEs would have to comply with regulations on HGM management. Those included, for example, pre-approval from the Ministry of Science and Technology (MST) for HGM collection, use or export.

Drug Administration Law

The Drug Administration Law (DAL) is the principal piece of legislation governing the pharmaceutical industry in China. Amendments to the DAL since 2015 have provided a framework for subsequent reform of rules governing all stages in the development and commercialization of medicines. Further changes were outlined in a series of DAL amendments ratified in December 2019. Among other things, these marked a shift away from manufacturer-based authorization to a marketing authorization holder (MAH) system.

The General Office of the State Council in January 2025 called for further improvements to the legal and regulatory frameworks governing drugs and medical devices, aiming to enhance the regulatory system, with more detailed measures on data exclusivity, market exclusivity, expedited review pathways, and enhanced compliance, expected to be covered in revisions to the implementing regulations of the Drug Administration Law.

Key Developments and Trends

General Office of the State Council Call for Further Reforms

- Further reforms to the regulatory system will be sought over the forecast period. In January 2025, the General Office of the State Council issued *Opinions on Comprehensively Deepening the Reform of Regulation of Drugs and Medical Devices to Promote the High-Quality Development of the Pharmaceutical Industry*. The Opinions aim to enhance the regulatory system to foster innovation and to transform China from a major pharmaceutical manufacturer into a pharmaceutical powerhouse.
- The Opinions call for further improvements of the legal and regulatory frameworks governing drugs and medical devices, in particular where the quality and efficiency of review and approval processes for innovative drugs and devices are concerned.
- The document outlines 24 reform measures across five key areas:
 - Increasing support for innovation in R&D.
 - Improving review and approval efficiency.
 - Enhancing the compliance level of the pharmaceutical industry via efficient and strict supervision.
 - Expanding opening up and cooperation.
 - Building a regulatory system that meets the needs of industrial development and safety.
- The 2025 Opinions set out a framework, with more detailed measures on data exclusivity, market exclusivity, expedited review pathways and enhanced compliance expected to be covered in revisions of the implementing regulations of the DAL.

Quality Standards and Drug Safety

The imposition and enforcement of tighter quality standards governing drug development, manufacture and distribution is a key health policy goal. Improvements in the quality of locally manufactured drugs are driving down the prices of patent-expired molecules, delivering significant savings to patients and payers. They are also seen as a prerequisite for the emergence of China as a more important exporter of finished pharmaceuticals.

The NMPA will maintain its drive to tighten up standards for drug quality and manufacturing, particularly where production is outsourced. It will also seek a level playing field for domestic manufacturers and multinationals by relaxing requirements for end-to-end production of biologics. The NMPA will encourage multinationals to manufacture previously imported drugs within China by piloting expedited review processes for associated approval applications.

Key Developments and Trends

Oversight of Quality and Safety in the Pharmaceutical Market

- The NMPA has stepped up its oversight of quality and safety in the pharmaceutical market, while dealing pragmatically with growing strains on its inspection resources by introducing a risk-based approach to on-site inspections from 2022. While problems with manufacturing

standards do arise from time to time, the agency reported that during 2024, drug safety in China was on a stable footing, with a pass rate of 99.3% from some 21,000 spot checks on drug manufacturers.

- The NMPA has generally tightened quality and equivalence standards for generic drugs. This has included guidance imposing tighter deadlines for compliance with generic quality consistency evaluation (GQCE) certification. A growing number of new generics are bioequivalent, while established products increasingly achieve GQCE certification.
- With the success of VBP heavily reliant on competition from low-priced generics, these quality assurance programs are essential in mitigating any suggestions that low-cost generics fall short on equivalence or safety.

Relaxation of Position on End-to-End Production of Biologics

- The rapid growth of contract manufacturing by marketing authorization holders (MAHs) in China over recent years has raised some concerns, particularly around the contract manufacture of biologics. The NMPA has addressed this through measures such as making MAHs more explicitly responsible for the safety and efficacy of manufactured products, along with mandatory liability insurance to cover the risk of contract manufacturing.
- At the same time, the NMPA is taking a more relaxed position on end-to-end production of biologics. In October 2024, the agency launched a pilot program allowing biologics companies to manufacture their drug substances and products at separate locations. The NMPA had previously insisted on end-to-end production of biologics at a single manufacturing site, which was seen as disadvantageous to domestic producers subjected to stricter compliance than their multinational counterparts.
- The pilot program, which runs until 31 December 2026, applies to provinces selected by the NMPA or those with experience of supervising biologics manufacturers. Eligible products include innovative and urgently needed biologics, monoclonal antibodies, insulin products, antibody-drug conjugates, GLP-1 agonists and multivalent vaccines.

Further Encouragement of Local Manufacturing of Imported Drugs

- There has also been further encouragement for local manufacturing of drugs usually supplied from outside China. The NMPA set new rules for the local production of imported drugs in its April 2024 *Notice on Optimizing Marketing Authorization Applications for Transferring Manufacturing Sites of Approved Overseas Drugs to China*.
- This clarifies that, if a drug has approval in China but is currently manufactured overseas, and the MAH wants to transfer production to China, it must follow the requirements and procedures for submitting marketing authorization applications in China. However, the applicant may now submit the original marketing authorization dossier from overseas, where applicable, along with studies on the transfer of the manufacturing site, as supporting documentation for the marketing authorization application in China.
- The new policy is expected to speed up the review process for drugs approved in China but manufactured elsewhere. Moreover, approval applications for originator drugs and biological in these circumstances will be eligible for priority NMPA review.

- All the same, challenges remain to the expansion of local manufacturing for previously imported drugs, such as economic feasibility, intellectual property protection during technology transfers, quality assurance, and variable local policies for public procurement of originator and generic products.

Intellectual Property

In line with commitments made under the 2020 US-China Economic and Trade Agreement, policymakers have taken steps to strengthen intellectual property protection for pharmaceuticals. Amendments to the Patent Law, ratified later in 2020, paved the way for establishing an explicit patent linkage mechanism and introducing patent-term restoration and adjustment systems.

With final rules on patent term extensions (PTEs) and patent term adjustments implemented in January 2024, the intellectual property landscape for originator companies is clearer. However, originator companies' access to PTE provisions will be severely restricted unless they choose to conduct early clinical trials in China and file for approval in China before anywhere else. Even where PTEs are granted, they will have a relatively limited scope, such as applying only to a single patent and for one indication.

Key Developments and Trends

IP Situation is Still Challenging Despite Patent Linkage

- Originator companies have met with mixed success in recent years when bringing patent infringement cases against generic challengers. Despite the introduction of a patent linkage mechanism from June 2021, originators have generally fared even less well in cases fought on that basis.
- In a case in 2024, China's Supreme People's Court upheld decisions by the China National Intellectual Property Administration (CNIPA), to invalidate, on grounds of obviousness, formulation patents held by an originator company. Before they were challenged, the patents were set to expire in September 2030.
- The Supreme People's Court judgments are expected to result in an influx of generic competitors into the Chinese market and significant price erosion on the brand, which previously dominated the market. Marketing authorizations had reportedly already been granted to at least two generic contenders by late 2024.

Final Rules for Patent Term Extensions Implemented

- While patent term extensions (PTEs), compensating for unreasonable delays in granting drug patents or marketing approvals, have been available in China since an amended Patent Law took effect in June 2021, eligibility for these provisions remained uncertain until final rules were published on PTEs and patent adjustments. The situation is somewhat clearer since the CNIPA issued the final version of its revised *Implementing Rules for the Patent Law and the Guidelines for Patent Examination*, effective from 20 January 2024.
- The final rules and guidelines provided some welcome predictability for patent applicants in China. But they also confirmed the fears of originator companies that the CNIPA would take a

narrow view of “new”, “innovative” and “improved” drugs eligible for extended patent terms, ruling out most multinational brands currently on the market.

- Specifically, PTEs are available for compound, formulation, crystal, method-of-manufacture and medical-use patents covering “new drugs”, i.e., innovative drugs and certain improved new drugs. However, these categories are interpreted in accordance with existing drug regulations and NMPA provisions. In that sense, both innovative and improved new drugs are understood as “new to the world” not just new to China. In other words, an approval application must have been filed for the drug in China before it is cleared for marketing in any other country – although “new drugs” are not defined specifically in either the patent law or the implementing regulations, leaving open the possibility of future legal challenges.
- Originator companies wishing to take advantage of PTE will have to think about conducting clinical trials in China from an early stage, to support a worldwide first approval filing in that market. Another limitation is that PTE is available for only one patent (if a drug is covered by multiple patents) and for only one drug (if the relevant patent covers multiple drugs).
- Moreover, and unlike patent term restoration provisions in the US or the EU, PTE in China is indication-specific. This could give generic or biosimilar challengers the opportunity to market competitor products with narrow labeling (i.e. avoiding the indication granted PTE) and then to promote off-label use covering the patent-extended indication, even if this might be technically illegal and a barrier to reimbursement.
- Eligible companies must file for a PTE within three months of marketing approval in China. The extension term is determined by subtracting five years from the period between patent filing and marketing approval in China. The maximum PTE term is five years after regular patent expiry, and the resulting overall patent term should not be more than 14 years from the drug’s marketing approval date.

Conditions for Patent Term Adjustments Clarified

- The final *Implementing Rules for the Patent Law and the Guidelines for Patent Examination* have also further clarified conditions for patent term adjustments (PTAs). Moreover, they specify the procedure for submitting PTA requests (within three months of the patent grant being published), together with the associated fees.
- For example, the number of days qualifying as unreasonable delays would be the number passed since four years from an invention patent application date, and since three years from a request for substantive examination, up to the date when the patents were granted. Subtracted from this figure would be the number of days of reasonable delays (e.g. arising from re-examination procedures) and of unreasonable delays caused by the applicant (e.g. failure to respond on time to CNIPA notifications).

Pharmaceutical Business Environment

Industry Structure

China's pharmaceutical industry is highly fragmented, with over 4,000 local companies operating in a market that has also been the subject of significant foreign interest. Consolidation of the local industry has been actively encouraged by policymakers, who are keen to improve the quality of locally manufactured generics, promote the emergence of a more substantive local research base and, in the longer term, position China as a major supplier of finished drugs as well as APIs. Medium-term ambitions for the local industry were outlined in a five-year plan published in 2021.

The government will continue to encourage drug innovation, tempering liberalization with price erosion and strong reliance on locally manufactured generics. With limited funding sources locally, domestic innovators will rely largely on out-licensing of early-stage assets to generate funding for sustainable R&D, while exploring alternative financing mechanisms, such as spin-off 'NewCos'. Some MNCS may rein in their Chinese ambitions, while companies will increasingly hand over mature brands to local partners. Many leading multinationals, though, remain committed to China as an innovation source for the long term.

Key Developments and Trends

Policy Seeks to Balance Innovation Catalysts with Budget Management

- The Chinese government continues to deliver industrial policies and regulations favoring inward investment by pharmaceutical multinationals, coupled with an emphasis on clearing space for innovative therapies both in approval procedures and reimbursement strategies. The government has also nurtured a more innovative domestic industry, alongside pragmatic encouragement for low-cost and better-quality generics.
- Recent innovation-friendly policies include various investment and financing incentives; the use of value ratings in drug assessments for NRDL listing; leeway for multinationals to pursue cell and gene therapy activities in selected free trade zones; and a more lenient stance on cross-border transfers of clinical trial data.

"There's a tsunami of policies stimulating R&D innovation and the shift, especially in Chinese biopharma, from being a me-too type of generic player, leveraging the manufacturing capability delivered over the last decade, to the more innovative side of biopharma, which is now catching up - not only by licensing things, but also by creating in China." (Multinational Company Executive)

- At the same time, the government seeks to address the growing strain on BMI funds, particularly related to the aging population and the increase of chronic diseases. Consequently, pricing pressure has increased, with substantial price reductions a condition of access to the NRDL and to volume-based procurement, to a point where many multinationals will no longer participate in bidding for centralized VBP in the public hospital sector.
- There are further challenges to market access in China, such as the anti-corruption drive, launched in mid-2023, which extends all the way from import strategies to relationships with

doctors and retail dispensing. Although the peak impact appears to have passed, companies have still had to rethink the way they interact with their customers and other stakeholders.

- Another source of disruption is geopolitical tensions, and notably the Biosecure Act, which puts US pharmaceutical multinationals at risk if they maintain relationships with certain named Chinese partners. These tensions have played out differently across the multinational presence in China. Some companies have reduced their local exposure, while others have doubled down on investments in China and re-affirmed their commitment to leveraging the country's innovative capabilities while addressing high levels of unmet patient need.

Local Manufacturers Extend Reach Through Out-licensing, NewCos

- Domestic manufacturers dominate the generics market in China, and locally produced generics secure the vast majority of VBP contracts, albeit at sometimes barely sustainable prices. Over the past decade, a growing number of Chinese companies have, with government backing, sought alternative opportunities in the development and commercialization of more innovative medicines.
- Progress in this respect has been hampered by a shortage of venture capital and related funding for local industry in recent years, particularly in the face of a slowing economy. Returns on local company investment in the development and commercialization of novel drugs have not always met expectations, reflecting the impact of tough negotiations on the price of products seeking access to the NRDL, while the government has imposed stricter conditions for initial public offerings (IPOs), in an effort to recharge the stock market.
- With limited sources of finance on home ground, Chinese companies have responded with a surge in out-licensing deals to generate funds for further innovative drug development, often releasing assets when they are only at the preclinical or early clinical stage (see Research and Development). This tends to dilute product valuations, due to uncertainty over prospects in more advanced clinical trials and eventual commercial potential.
- One way for these companies to extract more value from their R&D assets, tapping into broader sources of funding to ready product candidates for eventual sale or out-licensing globally, is the fast-emerging 'NewCo' model. This involves spinning off selected pipeline assets into a newly established independent entity, typically incorporated overseas and supported by a mixture of equity interests. The founder company either holds a controlling/minority stake in the NewCo or engages with it through a parallel shareholding structure.
- The NewCo model, which is attracting strong government support (at least at the local level), is more than a source of out-licensing revenues. As a spin-off, the NewCo has the flexibility to secure further input from private equity, partnerships, public offerings, complete or reverse mergers, and other financing strategies. Moreover, NewCos incorporated overseas can take advantage of cross-border investment opportunities on less restrictive financial and regulatory terms than apply to direct foreign investment in a Chinese company.

“It’s more like a PE or VC company. They try to put together different assets from Chinese companies and to form a completely new company with some industry veterans, or with very rich experience globally. We see this as a new trend when considering how to go global. You don’t have just one option, like product-by-product out-licensing. You can also think about forming a completely new company. And maybe, several years later, this company can be independently IPO-ed, or even acquired by a multinational company.”
(Local Company Executive)

- One example is the Kailera Therapeutics NewCo, originally established in the state of Delaware, USA, as Hercules CM in May 2024. The initial investment came from Bain Capital Life Sciences, Atlas Venture, RTW Investments and Lyra Capital. Kailera has licensed a portfolio of GLP-1/GIP agonists, with potential in obesity management and other metabolic conditions, from China’s Jiangsu Hengrui Pharmaceuticals Co., which also acquired a near-20% equity stake in Hercules.
- Leading Chinese companies will also pursue the establishment and expansion of overseas businesses. BeiGene, for example, is already generating substantial product revenues overseas and establishing a physical presence in key markets (BeiGene is building a US clinical research and manufacturing center in Hopewell, New Jersey).
- In parallel, the next few years could see a continued stream of out-licensing deals, plus growing reliance on alternative financing mechanisms like NewCos, as domestic players specialized in antibody-drug conjugates (ADCs), anti-PD1 antibodies, GLP-1 agonists or other areas of global interest look to attract fresh funding streams and to bolster their international ambitions.

Continuing but More Nuanced Investment from Multinationals

- Multinational companies, on the other hand, have been reconsidering their options in China, confronted with persistent downward pressure on prices, and market-access restraints such as fierce competition in VBP bidding, variable and/or static hospital formularies or limited uptake of dual-channel medicines. There are also indications that even non-reimbursed drugs in retail channels, and even online channels, are subject to increasing price pressures (see Healthcare Provision: Drug Procurement).
- To these challenges can be added geopolitical tensions, typified by the punitive measures proposed for US companies in relationships with Chinese biotechs (e.g. WuXi AppTec) whose involvement in US federal government procurement is targeted by the US Biosecure Act. Nor have multinationals escaped the clutches of the government’s renewed anti-corruption drive, which has chilled traditional sales and marketing strategies such as medical meetings and detailing to hospitals (see Sales and Promotion).
- The impact of these complications on MNC strategies is by no means consistent, though. It inevitably depends on broader global strategies, historical commitment to China, and where individual drug portfolios fit in the country’s hierarchy of needs and its balancing act between embracing innovation and controlling costs.
- In response multinationals increasingly out-license more mature original brands that struggle to compete in VBP, and/or to channel non-VBP products through retail outlets, including online pharmacies.

- This enables MNCs to focus more on innovation, whether self-generated or accessed through partnerships with forward-looking domestic players. Once again, though, the choice of strategy is partly a function of portfolio management, such as how concentrated mature products are in a particular therapeutic category, and whether there are competitive risks in selective outsourcing.
- More generally, the demanding operating environment for multinationals in China has not resulted in any large-scale withdrawal from the market, even if some companies have downsized their China businesses or passed on previously core assets. Among these are Pfizer's pneumococcal vaccine Prevnar, which has been distributed and promoted exclusively in China by Keyuan Pharma since November 2023. Around a month earlier, GSK engaged Zhifei Biological Products to distribute its shingles vaccine, Shingrix, in China.
- In fact, some companies, such as Eli Lilly, Pfizer, GlaxoSmithKline, Boehringer Ingelheim and Sanofi, have emphasized their continuing commitment to China amid rising trade and security tensions with other major markets. These companies are investing for the long term in manufacturing, R&D, locally sourced innovation and partnerships throughout the pharmaceutical value chain. Pfizer, for example, aims to bring 60 innovative medicines and new indications to the Chinese market by 2030.
- These trends suggest that multinational companies will continue to pursue opportunities in China, albeit more selectively and with more emphasis on breakthrough therapies in high-need categories, whether from their own laboratories or those of an increasingly innovation-oriented domestic industry.

Research and Development

The government envisages China emerging as a hub for international pharmaceutical product development. Its 'Made in China 2025' strategy, unveiled in 2015, set a series of five- and ten-year targets for the local industry in the fields of drug innovation, import substitution and export development. Its 14th Five-Year Plan for the Development of the Pharmaceutical Industry calls for annual increases of 10% in local industry R&D spending through to December 2025.

Innovation pioneers in the domestic industry will continue to rely quite heavily on out-licensing of early-stage assets to generate funding for sustainable R&D in a challenging climate for financing, aggravated by a slowing economy and pricing pressures in the domestic market. Emerging investment platforms such as spin-off 'NewCos' will broaden the options for local innovators. Government policies will continue to encourage R&D investment in breakthrough therapies from local companies and multinationals alike.

Key Developments and Trends

Local Industry May Struggle to Maintain Innovative R&D without Out-licensing

- In view of the recent downturn in venture capital funding for the biopharmaceutical sector, doubts remain about local industry's ability to maintain investment in innovative R&D without selling off potentially valuable assets at an early stage to MNCs with deeper pockets.

According to Caixin Global, Chinese companies concluded more than 80 out-licensing agreements with foreign partners in 2023, to a record value of US\$45 billion. In January 2024 alone, nearly 20 such transactions were announced, worth over US\$10 billion. Of the 80 out-licensing deals in 2023, almost 20 were valued at more than US\$1 billion each.

“The capital market is still very closed for fundraising. Most of the biotech companies cannot be IPO-ed in the list A market. If they decide to go for Hong Kong listing, their market cap is likely to depreciate. So, a lot of companies are now going the out-licensing or BD [business development] way. And the funding they raise through BD programs is higher than the IPO fund-raising, compared with several years ago.” (Local Company Executive)

- That year also witnessed the first ever full acquisition of a Chinese biotechnology company by a pharmaceutical multinational, with AstraZeneca paying US\$1.2 billion in December 2023 for Gracell Biotechnologies, a specialist in cell therapies for oncology and autoimmune diseases. Shortly afterwards, in January 2024, Novartis took over China’s SanReno Therapeutics, a biotech developing innovative therapies for kidney diseases, for an undisclosed sum.
- In terms of out-licensing, there is a particular emphasis on locally developed ADCs. For example, one of the largest such deals in 2023 was when Bristol Myers Squibb paid US\$800 million upfront, and up to US\$8.4 billion overall, to Biokin Pharmaceutical Co. Ltd. for rights outside China to an early-stage ADC with potential in a range of solid tumors.
- During the year, more than 10 Chinese companies, including Biokin, Hansoh Pharmaceutical Group and CSPC Pharmaceutical Group, signed 15 out-licensing agreements for ADC projects, with multinational partners such as Pfizer, Bristol Myers Squibb, AstraZeneca and Merck. Most of these deals were for products not yet in Phase II clinical trials, suggesting that Chinese asset-holders are settling for limited licensing fees in their need to generate cash.
- There is some degree of additional risk, not to mention potential market saturation, in multinationals pursuing in-demand drug formulations or mechanisms such as ADCs, anti-PD1/PDL1 antibodies or GLP1 agonists. Policymakers in China do not want to see widespread duplicative investment in narrow fields of research.

Local R&D Capacity Ramps Up, Along with Multinational Investment

- In the meantime, locally sourced pharmaceutical R&D is going from strength to strength. According to according to Bi Jingquan, executive vice-chairman of the China Center for International Economic Exchanges (and former director of the CFDA), 4,774 or 35% of the 13,537 drugs in clinical research worldwide as of March 2024 were developed by, or included participation from, Chinese companies, placing China second only to the US as an R&D engine.
- An August 2024 report from the US Information Technology and Innovation Foundation (ITIF), *How Innovative is China in Biotechnology*, suggested that, in biotech specifically, there were several indicators that China was closing the innovation gap with global leaders in western markets. This was helped by government subsidies, financial incentives, new science parks, start-up incubators, talent-recruitment schemes, public-private partnerships, and regulatory reforms to expedite reviews of drug approval applications and strengthen IP protection.

- The US Food and Drug Administration approved three new Chinese drugs during 2023, according to ITIF. These were Loqtorzi (toripalimab), the first FDA-approved therapy for nasopharyngeal cancer; Fruzaqla (fruquintinib) for metastatic colorectal cancer; and Ryzneuta (efbemalenograstim) for chemotherapy-associated neutropenia.
- Moreover, the NMPA cleared five first-in-class therapies from domestic companies for marketing in 2023: glumetinib from Haihe Biopharma, for advanced NSCLC with MET alterations; Raynovent's leritrelvir, for treating mild to moderate COVID-19; Xuanzhu Biopharmaceutical's proton pump inhibitor, anaprazole tablets; Jiangsu Hansoh Pharmaceuticals' pegol-sihematide for anemia in chronic kidney disease; and BioRay Biopharmaceutical's Anruixi (zuberitamab injection) for diffuse large B-cell lymphoma.
- Nonetheless, China still lags behind the US in its capacity to transform science into products, the report added. If it really wants to become a 'global biotech superpower', then China needs to develop a comprehensive biotech ecosystem comparable to that of the US, with its mix of national funding sources, VC start-up funding, pharmaceutical R&D investment, robust IP protection and powerful commercialization capabilities.
- As China progressively evolves and refines its R&D capacity, multinational companies continue to tap into these new sources of innovation, whether through in-licensing of early-stage Chinese assets or direct investment in local R&D resources. Companies such as Pfizer, AstraZeneca, Sanofi and Roche have been particularly active in this respect, including integration of Chinese trials into their global clinical-development programs.

Generics and Biosimilars

At ex-manufacturer prices, generics and copy products account for around 80% of sales by value in the market audited by IQVIA. Overcrowding and unfavorable pricing policies limit the appeal of the market for most of the world's leading international generics houses, and the sector is dominated by domestic companies marketing branded generics. GQCE-certified generics have begun to wrest market shares from a growing number of patent-expired brands since the introduction of the national VBP initiative – although at prices that are squeezing local manufacturer margins. China's first 'true' biosimilar was granted marketing authorization as recently as February 2019, but the sector has gathered momentum rapidly since then.

Policymakers will promote further development of the domestic generics industry, through measures such as 'encouraged generics' lists and large volume uptake of local generics in VBP. Pricing pressures and more rigorous quality standards could marginalize some smaller players. Biosimilars will gain further traction as a growing number of innovative biologics face patent expiry.

Key Developments and Trends

Further Encouragement for Generic Competition

- A government work plan, issued in December 2018, for policies to stimulate the production of locally sourced generics required the NHC and the NMPA to draw up a list of 'encouraged generics' eligible for priority review and fast-track approval. The first such list was published

in June 2019, followed by a second list in April 2021. The NHC published the third edition of its List of Encouraged Generic Drugs, with updated information on the supply and demand of featured products, in December 2023.

- The third list includes 39 active ingredients spanning 12 categories, such as anti-cancers, anti-infectives and cardiovascular drugs. This is both a larger number of molecules and a broader spread of therapeutic areas than in the previous lists, with all three lists incorporating 50 molecules in total.
- VBP has strongly favored local generics manufacturers, but at prices and in quantities that may be beyond the capacity of smaller local producers. With VBP continuing to expand in frequency and scope as a core strategy for hospital drug supply (see Healthcare Provision: Drug Procurement), successive procurement rounds will drive consolidation in the domestic generics sector.
- GQCE compliance deadlines and closer NMPA scrutiny of quality and safety in the pharmaceutical market will also contribute to consolidation. (see Regulatory Environment: Quality Standards and Drug Safety).

Biosimilars Will Gain Further Traction

- China is regarded as fertile territory for biosimilar development and commercialization, given the growing number of innovative biologics on the market and the patent cliff looming for many well-established products. Early players in the biosimilar market have generated substantial returns and multiple domestically developed biosimilar versions compete for market share with originators
- More recent biosimilar approvals in China included, in July 2024, Simcere Zaiming's Enlितuo, the first biosimilar version of cetuximab (Erbix; Eli Lilly) in China, for RAS/BRAF wild-type metastatic colorectal cancer; and, in April 2024, Jiangsu T-mab Biopharma Co.'s (Mabwell) Maiweijian, the first biosimilar of denosumab (Prolia/Xgeva; Amgen) made available in China with the same bone-disease indications as Xgeva in the USA.
- GLP-1 agonists for weight management could be a substantial opportunity for biosimilar manufacturers in China, following the successful launches of originator brands such as Novo Nordisk's NovoCare (the brand name for Wegovy, semaglutide, in China) in November 2024, and Eli Lilly's Mounjaro (Zepbound, tirzepatide) in January 2025.
- Two locally produced GLP-1 agonists, Benemae's beinaglutide (Feisumi) and Hangzhou Zhongmei Huadong's Liluping (liraglutide), were approved in China for weight loss during 2023. Other domestically developed GLP-1 agonists, such as Innovent Biologics' mazdutide, developed under a licensing agreement with Eli Lilly, are in late development.
- Novel weight-loss therapies such as NovoCare are expected to have to generate evidence of cardioprotective and other broader health benefits before they are accepted for NRDL listing. In the meantime, they are exploring opportunities in the private-pay market, including private hospitals, public hospitals with specialist weight-management centers, and retail pharmacies.

“These products are very popular. Everyone who believes they are overweight will try to reach out for these innovations in the retail channel, and will also pay for themselves. The market is significant. Novo Nordisk and Eli Lilly both have abundant pipelines for treating obesity. And also, the local competitors are following the research. They will be launching similar me-too or me-better versions in the next five years.” (Local Company Consultant)

- If biosimilar contenders are priced at enough of a discount to NovoCare and other originator brands, this could broaden significantly the uptake of GLP-1 agonists for weight management and even position the biosimilars for NRDL listing and/or VBP bidding.

Distribution

The distribution sector remains highly fragmented, with almost 14,800 businesses still in operation, according to government figures. Leading players have expanded rapidly through mergers and acquisitions (M&A) in recent years, however. Consolidation was accelerated by the rollout of the ‘two-invoice’ system from 2017. Together, the country’s 100 biggest distributors now handle over 70% of sales by value, while the combined share of the three leading players in the sector is more than 30%. Further consolidation is targeted under the 14th Five-Year Plan. It envisages the top-100 distributors together accounting for 98% of the pharmaceutical wholesale market by the end of 2025.

Consolidation in the distribution industry continues, while the pace of M&A among pharmaceutical distributors has quickened. Late payments by medical institutions to pharmaceutical wholesalers remain a serious issue in China. Revised regulations governing drug distribution and storage took effect in January 2024. Existing distributors have five years to upgrade to the new standards and renew their licenses. Closer scrutiny of the pharmaceutical distribution chain will put further pressure on operating margins in the sector.

Key Developments and Trends

Distributors Face Further Pressure on Sales and Margins

- The larger pharmaceutical distributors continued to grow their businesses in 2023, albeit with some indication of slowing dynamics. Core sales of the top 100 wholesalers rose by 7.6% year-on-year, nearly one percentage point more than in 2021-2022, but just 0.1 percentage points above the industry average for 2022-2023. Profit margins from pharmaceutical distribution were generally tighter in 2023, according to the Ministry of Commerce.
- This more subdued trend continued into 2024. Among the leading players in the sector, Sinopharm Group’s revenues from pharmaceutical distribution rose by just 0.47% year on year in the first half of 2024. The operating margin from these businesses was 2.75%, 0.21 percentage points lower than in the first half of 2023. China Resources Pharmaceutical reported revenues 4.9% higher for its distribution business in H1 2024, but its operating margin narrowed from 4.0% to 3.7%.
- At the same time, industry consolidation has continued, hastened by policies such as VBP, while the pace of M&A among pharmaceutical distributors has quickened. In 2023, for example, the main business income of the top 10 pharmaceutical wholesalers accounted for almost 60% of the national pharmaceutical market, up by 1.3 percentage points year-on-year.

- The Ministry of Commerce also noted that late payments by medical institutions to pharmaceutical wholesalers remained a serious issue in China. The average time to collect receivables from institutions was 152 days in 2023, two days more than in 2022. In many instances, the report added, direct settlement between medical insurance funds and pharmaceutical companies was now being piloted for national or local national/local centralized drug procurement.

Closer Attention to Quality Management and Compliance

- In an effort to upgrade quality standards, the State Administration of Market Regulation (SAMR) published revised regulations governing drug distribution and storage in late 2023. These took effect in January 2024. The main focus is on administering drug-supply licenses, the respective obligations of drug suppliers and medical institutions, and regulators' supervision of stakeholders in drug supply and usage.
- For example, wholesalers must set up and operate their own warehouses, businesses premises, facilities and equipment, rather than working with contractors. Warehouses must be equipped with modern logistics to manage activities such as warehousing, transfer, sorting, shelving and delivery of medicines.
- Distributors also need to have a quality management system in place, and an information management-system compliant with Good Supply Practice for drugs, and covering the whole process of drug management, quality control and traceability. Existing distributors have five years to upgrade to the new standards and renew their licenses. Closer scrutiny of the pharmaceutical distribution chain will put further pressure on operating margins in the sector, as successive rounds of VBP drive drug prices in the hospital-supply sector even lower.

Retail Pharmacy

Most outpatient prescription drugs are dispensed in the hospital sector, reflecting the traditionally hospital-led nature of the healthcare system, the limited availability of prescription drug reimbursement from BMI funds in retail pharmacies and – until the abolition of hospital mark-ups in 2017 – the ability of public hospitals to profit from the prescription and sale of drugs dispensed on their premises. Low standards and the limited capacity of regulators to monitor retail sector dispensing have posed additional barriers to more widespread separation of prescribing and dispensing functions.

Retail pharmacy numbers has grown strongly in recent years, and there are indications that the sector is reaching saturation point. Chains account for 58% of outlets, and the government is keen to see further consolidation. Larger chains with diversified product ranges will be better placed to ride out challenges, including slower-than-expected progress with dual-channel dispensing and growing pricing pressures. Retail pharmacies are expected to look at offering more specialized, value-added professional services in the medium-longer term.

Key Developments and Trends

Continued Growth in the Number of Retail Outlets in 2023

- The total number of retail pharmacy outlets in China rose by nearly 44,000 (+7.0%) to 666,960 in 2023, according to NMPA data, with growth in the number of independent outlets (+6.9%) only slightly behind that of pharmacy chains (+7.1%). At 58%, the share of chains in the overall retail pharmacy number was broadly unchanged from 2022.
- There were 6,725 chains in China in 2023. While most of the leading chains also operate franchising businesses, the number of stores managed directly by the country's five biggest chains continued to grow rapidly on the back of in-house investments and acquisitions.
- The combined store count managed directly by the five biggest chains rose by nearly 17% in 2023-2024, adding over 6,900 stores, to reach over 48,000. Yifeng overtook Yixintang as the country's leading network, counting 10,264 directly managed outlets, but Yixintang's network also topped the 10,000 mark, with 10,255 outlets. Store counts are significantly higher when including franchised outlets.
- The government is keen to see further consolidation, with the Ministry of Commerce's envisaging five to ten specialized and diversified pharmaceutical retail chains operating across China by the end of the 14th Five-Year Plan in 2025, each with turnover of more than RMB50 billion, and with chains accounting for close to 70% of all retail pharmacies.

Factors Expected to Slow Retail Sector Expansion in the Short Term

- While the number of retail pharmacies in China continues to grow, there are indications that the density of pharmacies is nearing saturation point, however, with around one outlet per 2,000 population. Beyond market saturation and intensified competition, a number of factors have begun to affect sales growth and profit margins for retail pharmacies, which are slowing expansion, including:
 - Structural shifts in BMI funding, notably the transfer of employer payments from personal medical accounts to mutual pools in the UEBMI urban employee scheme (see Country Context: Healthcare Provision: Health Insurance). This has led to a shift of some patients back to hospitals or grassroots medical institutions, affecting retail pharmacies dependent on outpatient reimbursement through personal medical accounts.
 - Slower-than-expected progress with dual-channel dispensing of selected NRDL drugs in designated retail pharmacies (see Prescribing and Dispensing: Dispensing).
 - Increasing controls on spending BMI spending for outpatient dispensing of reimbursed medicines under the dual-channel policy. Since 1 January 2025, designated retail pharmacies have to process dual-channel prescriptions through a new centralized online system (see Prescribing and Dispensing: Dispensing).
 - Growing pricing pressures on prescription medicines, under the influence of multi-channel price-comparison systems promoted by the NHSA and implemented by local BMI branches. BMI branches in Shenzhen, Guangdong, Jiangsu, Hubei and other provinces were reported to have launched price comparison systems during 2024, linking BMI-designated retail pharmacies and pegging prices more closely to those charged in online outlets.
 - Indications of growing pricing pressures even on non-reimbursed drugs in retail channels, and even online channels.

- Closer scrutiny of retail pharmacy practices as a result of the anti-corruption campaign. At least one leading pharmacy chain has been implicated in investigations that found evidence of collusion, excessive prescribing, falsified records or medical insurance claims and other misdemeanors, according to press reports.

Growing Consolidation and Transformation Expected

- While these factors may slow further growth in retail pharmacy numbers, they could work in favor of larger chain pharmacies with stronger compliance procedures and more diversified product ranges or service capabilities, raising quality and driving further consolidation in the sector.
- In a competitive environment, retail pharmacies are expected to look at offering more specialized, value-added professional services, such as examination and screening services, as well as chronic disease management. Retail pharmacies are also expected to expand sales of non-pharmaceutical health products. While they will seek to grow sales via their online platforms, they will also explore strategies for integrating online drug sales with dispensing and patient interaction in the physical retail pharmacy setting.
- Pharmaceutical companies have begun to pay more attention to retail channels in recent years, not only for dual-channel medicines, but particularly as an alternative route for medicines awaiting or rejected for NRDL listing or shut out from VBP contracts, and the sector will remain an important alternative sales channel over the forecast period

“Patients now have much more awareness that, if the originator drugs are not available in the hospitals and they are not willing to use a generic version, there is an option to purchase it in a retail pharmacy, 20, 50, 100 meters from the hospital. We as an organization have that as part of our strategy to extend the lifecycle of our older brands.” (Multinational Company Executive)

Online Drug Sales Slowed in 2023, While Outpacing Retail Pharmacy Sales

- Online drug sales – including business-to-business (B2B), business-to-consumer (B2C) and third-party service platform sales – reached RMB248.9 billion in 2023, up 5.5% over RMB235.8 billion in 2022, according to (incomplete) statistics cited in the Ministry of Commerce’s latest *Statistical Report on the Drug Circulation Industry*, published in August 2024. This represents a further slowdown in growth, from 9.1% in 2022 and 21.6% in 2021, but continued to outpace retail pharmacy sales.
- B2B sales accounted for just over 50% of online drug sales, while third-party service platform sales made up 44%, with B2C accounting for the remainder. While hospitals remain important for generating prescriptions, chronic disease treatments in particular are increasingly purchased through online channels, as well as retail pharmacies.

“People’s behavior changed a lot in the past few years. They can very conveniently get a product from online drugstores. It’s mainly for cardiovascular, diabetes, chronic disease products. Then we move efforts from hospital to retail. The prescriptions are still coming from the hospitals. Then people will move online, move to drugstores, to purchase the drugs.” (Multinational Company Executive)

OTC Market

The OTC market is highly fragmented, comprising sales through hospitals and CHCs as well as drugstores, pharmacies and – for low-risk ('green label') products – supermarkets and other retail outlets. While OTC products account for a substantial proportion of pharmacy and drugstore revenues, around 40% of non-prescription bound sales are generated in public health facilities, reflecting the inclusion of some OTC medicines on the EDL and the NRDL. This means patients can obtain them free of charge when they are dispensed in healthcare facilities.

The OTC market stabilized following the exceptional boost to cough, cold and respiratory products after the end of the zero-COVID policy in December 2022. Growth in the OTC market will be driven by rising incomes, population ageing and the embrace of proactive wellness strategies, especially among the younger population, aided by digital innovation. Clearer direction on Rx-to-OTC switches may also stimulate market growth.

Key Developments and Trends

OTC Market Trends

- With inflation rates remaining low, OTC (non-prescription-bound) sales were essentially flat at ex-manufacturer prices in 2023. They rose by just 0.3% in local currency dollar terms (constant exchange rates) to reach US\$14.24 billion, according to IQVIA's OTC Review 2024.
- That followed 7% year-on-year growth in 2023, when OTC sales were boosted by demand for cough, cold and respiratory (CCR) products in particular, after the government relaxed its zero-COVID policy in December 2022. Volume sales of OTC products declined by 3% in 2023 to 5.98 billion units, reversing the COVID-19 effect in 2022, when volume growth was 9%.
- CCR products remained the largest OTC category in China, accounting for 28.5% of the total non Rx-bound market with sales of US\$4.06 billion. That marked a 6% decline, though, against 2022, which had seen an exceptional sales increase of 27% for the category, due to heightened COVID-related demand for respiratory medicines. In volume terms, CCR sales fell by 10% in 2023.
- Value sales of other top-five OTC categories in 2023 ranged from fairly strong to negative. Digestive and other intestinal products bounced back from a flat performance in 2022 to show year-on-year growth of 8.9%, while pain-relief products were up by 6.2% compared with roughly 2% growth in 2022. Both of these categories registered 5% volume gains in 2023, indicating a real increase in demand. Value sales of vitamin and mineral supplements, tonics and other stimulants fell by 2.5% against 2022, while volume contracted by 2%, with demand stabilizing following a substantial boost to the category during the pandemic.
- Haleon, the demerged consumer healthcare business of GSK, held onto its lead in the OTC market with a 7.5% increase in value sales for 2023. Haleon took only a 3.4% value share of the market, though, while the top ten OTC companies accounted for just 16.3%, underlining the fragmentation of the Chinese market, and reflecting the general dominance TCMs.

- Future growth in China's OTC market may come from rising income levels (albeit selectively), enhanced health awareness, population aging and digital advances, that address the consumer health demands of the younger population with more personalized strategies.
- This shift from a disease- to a health-centered approach, particularly in the face of dwindling BMI finances, was exemplified by Kenvue (formerly Johnson & Johnson Consumer Health) stepping up its strategic partnership with Chinese drugstore chain Shenzhen Nepstar Medical Co. in September 2024, to focus on OTC 'big health'. The concept aims to integrate medical intervention with wellness, disease prevention, healthy living and lifestyle management.

Draft Guidelines on Switches Outline Key Inclusion and Exclusion Criteria

- Chinese regulators have historically taken a conservative view of switches. Flagging a more relaxed attitude, new regulations for Rx-to-OTC switches took effect in November 2022, simplifying the required documentation for switches. In May 2024, the NPMA's Center for Drug Re-evaluation released draft guidelines on the *Scope of Applications for Switching Prescription Drugs to Over-the-Counter (OTC) Drugs* for public consultation.
- The draft outlines the key inclusion and exclusion criteria for Rx-to-OTC switches. The former include broad usage in China and a favorable safety profile, along with suitability for consumer self-awareness, self-judgment, self-medication, and self-monitoring. The guidelines exclude a number of drug categories from switching, such as treatments for serious diseases, systemic antimicrobials and hormones (except contraceptives), or drugs requiring medical supervision and guidance during use, such as anticoagulants or vaccines.

Sales and Promotion

Pharmaceutical sales and marketing practices have evolved significantly over the past decade, in the face of a government crackdown on corruption and, more recently, the emergence of VBP for supply of the public hospital market. Coupled with the introduction of new regulations governing activity in the sector, these developments have prompted manufacturers to re-assess established promotional models, shifting towards more holistic, multi-channel approaches. That trend accelerated dramatically following the onset of the COVID-19 pandemic.

Overall interactive contacts with healthcare professionals remained significantly below their 2019 (pre-COVID) levels in 2024. While the impact of the latest anti-corruption campaign is receding, and draft compliance guidelines issued in October 2024 gave companies some leeway to sponsor meetings, many doctors remain nervous about closer scrutiny of interactions with industry, necessitating a shift away from face-to-face detailing or direct funding of meetings towards informal/virtual communications and third-party sponsorship.

Key Developments and Trends

Overall Interactions with HCPs Continue to Decline in 2024

- Overall interactive contacts with healthcare professionals remained significantly below their 2019 levels in 2024, with combined face-to-face and remote interactions 26% lower than prior to the COVID-19 pandemic. Face-to-face contacts, which accounted for 91% of interactions

in 2019, were responsible for 60% of the total in 2024, with remote interactions making up 40%. A switch back towards more widespread in-person detailing had been widely anticipated following the removal of COVID restrictions in December 2022, but this only happened to a minor extent.

- While the overall volume of interactions had begun to increase slightly in 2022, the anti-corruption campaign launched in 2023, which includes the threat of harsh financial penalties or even imprisonment for illegal activities, drove a renewed decline of -4% in 2023, which intensified in 2024, when overall interactions declined by a further -12% over 2023.

Anti-Corruption Drive Still Shaping Promotional Interactions

- The anti-corruption campaign continued to impact promotional activities through 2024. Many companies have reined in hospital visits, as physicians were often nervous about seeing sales or medical representatives. While they were more amenable to virtual interactions or meeting outside hospital grounds, companies are wary of investing further in hospital sales forces, other than for new launches.

"We didn't reduce the field-force size. But we basically froze everything. We aren't recruiting any new reps. But for new business, we definitely have to."
(Local Company Executive)

- Conference engagements supported by individual companies have also reduced markedly, although physicians were more comfortable with meetings funded by third parties, such as industry or medical associations. There were indications of industry and its stakeholders finding ways to work within the new constraints, of meetings and other affected practices cautiously reviving.
- While doctors in China remain a key channel for education and awareness-raising on innovative therapies in particular, their confidence was shaken by having to return money from speaker fees or related activities. This has discouraged many from asking companies to support academic research or more routine practice needs. The reduced interaction has been particularly challenging for new product launches, compared with established medicines, where physicians have the relative security of reimbursement listings and treatment guidelines.

New SAMR Compliance Guidelines and a Cautious Improvement

- In draft *Compliance Guidelines for Healthcare Companies to Prevent Commercial Bribery Risks*, issued in October 2024, the SAMR emphasized that 'academic visits' to healthcare professionals (HCPs) entailed a high risk of bribery. Sales activities or collecting/requesting prescription information were prohibited in this context.
- At the same time, there are indications that the government recognized that the clampdown's impact on legitimate commercial or educational activities might have been too severe, however. The SAMR's draft guidelines acknowledged that academic conferences sponsored or organized by industry had declined significantly since the second half of 2023. It was acceptable, therefore, for companies to pay HCPs for lecturing or research services, based on 'genuine, reasonable, and lawful business needs'. The draft guidelines also set out specific compliance requirements for engaging HCPs as consultants.

- By late 2024, some activities, such as meetings sponsorship, were showing signs of recovery as industry and its stakeholders adjust to the new scrutiny. As anxiety about the anti-corruption drive gradually began to wane, there has been a cautious return to face-to-face detailing while doctors were said to be more open.

“MNCs do more collaboration with medical associations or foundations to organize meetings by leveraging their platforms. So the doctors receive the invitation not from pharma companies but from the association, and they are more than happy to join.” (Multinational Company Executive)

“Physicians are more open compared to the last 12, 18, 24 months. Physicians are more open to accepting our scientific engagements, our field visits. This has seen significant improvements in the past few months.” (Multinational Company Executive)

Increasing Reliance on Digital Tools

- In a bid to maximize limited opportunities for interaction with HCPs, companies are leveraging AI and other digital tools to train sales reps in advance of visits, and generally enhance their skillsets, so that the right message can be delivered to available hospital doctors in less time.

“We are also starting to explore AI solutions, for example, to enhance the quality of the single field-force visit to a doctor. Previously, maybe the sales reps in the hospital would meet roughly eight or even more than 10 HCPs in a day. But now it's challenging. So, if the sales reps have the chance to talk to a doctor, the challenge for us is how to enhance the quality, the message delivery to the doctors.” (Multinational Company Executive)

- Omnichannel promotional approaches incorporating diversified digital engagement (and at lower cost) are likely to become the standard for industry-HCP interactions as the dust clears from the anti-corruption campaign. The key will be finding the optimal balance between digital strategies and traditional modes of engagement for the Chinese market.

Broader Promotional Trends

- Companies are also looking more closely at the retail sector as a promotional target, in an effort to leverage both dual-channel sales -- albeit with parallel promotions to public-hospital doctors, to secure initial prescribing and brand awareness -- and create new opportunities for established medicines left out of VBP contracts. The strategy may also suit innovative therapies excluded from/awaiting NRDL listing, where there may be some self-pay potential as a potential precursor to eventual listing (e.g. weight management therapies).
- More MNCs are also seeking out commercial partnerships with local companies for mature portfolios, although there may be competitive reasons not to outsource mature brands. If most of a mature portfolio is in one therapeutic category, for example, outsourcing may be more difficult to manage. Outsourcing mature products in the same category selectively risks creating a competitor out of the commercial partner, especially if a company's own sales force may already be achieving synergies with a limited portfolio.
- Companies will also be more selective in their promotional coverage, particularly as BMI funds come under growing strain, targeting those provinces or regions where the financial situation is more stable, or where local policies are more hospitable to innovation.

“In 2025 we will quickly adjust our strategy. We are trying to focus more on our strategic key products, and with fewer resources. We only choose where we can win. In the past, or let's say before 2024, we focused on most of the provinces in China. Usually, we needed to take care of 30 provinces. But, from 2025, I will only deploy my sales team in 17 to 20 provinces overall.”
(Multinational Company Executive)

COUNTRY CONTEXT

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Healthcare Provision

Healthcare System

More than 95% of the population is covered by one of two basic medical insurance (BMI) schemes rolled out as part of the major healthcare reform program launched in 2009 (see Healthcare Reform). The schemes cater, respectively, for urban employees, and for other urban residents and rural dwellers. Benefits are limited, however, and vary not only between the schemes, but also between provinces, and between urban and rural areas. In a bid to address these issues, the National Healthcare Security Administration (NHSA) is pursuing the expansion and harmonization of BMI benefit schedules (see Health Insurance).

Access to Healthcare Services

In urban areas, healthcare is provided through hospitals, community health centers (CHCs) and health stations, while in rural areas the health infrastructure comprises a network of county hospitals, township health centers and village clinics. Public hospitals are the main providers of healthcare, though the private hospital sector is expanding rapidly.

Major public hospitals suffer from serious overcrowding. Patients tend to seek consultations at leading hospitals, even for minor health problems, and those from rural areas are willing to travel to city-based hospitals for treatment. Primary care is expanding but remains underdeveloped, complicating efforts to establish a working referral system. Initiatives to boost the capacity of the healthcare system are struggling to keep pace with demand, which has been driven up significantly by more widespread access to subsidized provision. Overcrowding remains a particular problem in top-tier public hospitals (see Hospital Sector).

Physician and nurse numbers are also rising, but the distribution of qualified healthcare professionals varies widely, with low salaries and unfavorable working conditions hampering recruitment efforts in rural areas. The doctor/patient ratio exceeds the OECD average in urban areas but is significantly lower in rural areas. Nurses are in much shorter supply, with shortages affecting both urban and rural health facilities, and the national nurse/patient ratio standing at less than half of the OECD average.

Traditional Chinese medicine (TCM) is an integral part of China's healthcare system. The health infrastructure includes specialized TCM hospitals, while the majority of CHCs and village clinics also provide some form of traditional therapy. TCM development is a national strategy, with a government target for TCM services to cover all areas of medical care by 2030.

Health Sector Governance and Policy Implementation

Responsibility for governance of the healthcare sector is concentrated in the hands of two national agencies created in 2018, each of which is also represented at provincial and city level:

- **The National Health Commission (NHC)** is responsible for the regulation of medical and healthcare services. It oversees the government's healthcare reform program and plays a broad role in the development and implementation of healthcare policy.
- **The National Healthcare Security Administration (NHSA)** is responsible for oversight of the basic medical insurance (BMI) schemes. It formulates and administers health insurance policy,

including the pricing and reimbursement of drugs and medical services covered by the BMI schemes.

COVID-19 Pandemic

The government's response to the initial outbreak of COVID-19 in mainland China in February 2020 involved a strict 'zero-COVID' policy, with rigorous lockdown and quarantine measures imposed in affected areas for the following three years. By late 2022, this was becoming increasingly difficult to sustain, as outbreaks of highly transmissible viral variants disrupted economic activity and access to healthcare in major cities across the country.

In December 2022, the authorities announced an abrupt shift away from the zero-COVID strategy, triggering a huge wave of infections that overwhelmed the healthcare system and inflicted a heavy toll on China's population.

After subsiding through February and March 2023, COVID-19 case numbers rose again briefly in May and June, while the NHC reported a significant increase in respiratory infections across the country during November 2023. These included COVID-19 cases as well as spikes in the prevalence of influenza, mycoplasma pneumoniae and respiratory syncytial virus.

As of December 2024, the country had reported 99.4 million confirmed cases of COVID-19 infection to the WHO, and 122,389 associated deaths. Actual figures are likely to be much higher. Some independent analysts estimate that the virus was responsible for around 1.5 million excess deaths in China during the first wave following the end of the zero-COVID policy alone.

In terms of COVID-19 vaccine coverage, WHO data indicate that more over 1,284 million people in China had completed a primary (two-dose) vaccination course by February 2024, while 834 million had received at least one subsequent booster dose.

Health Policy

Health Policy Governance

National health policy governance was overhauled in 2018 with the creation of new agencies including the NHC, the NHSA and the National Medical Products Administration (NMPA) (see Regulatory Environment: Regulatory Agency). This removed previous overlaps in the remit of individual agencies, clarifying responsibility for individual policy areas.

While health policy is framed at national level, responsibility for its implementation is concentrated largely in the hands of provincial authorities. Differences in both local social and economic development and the capabilities and approach of provincial administrations mean that progress towards implementation of key policies often varies significantly at provincial and local level.

Healthcare Reform

Healthcare policy has been guided over the past 15 years by a comprehensive, long-term healthcare reform program, initiated in April 2009, which aimed to achieve universal access to affordable and equitable basic healthcare by 2020, and to narrow the gap between urban and rural populations. Medium-term policy goals have been set out in five-year plans, while longer-term policy directions, covering the period to 2030, have been mapped out by the State Council (see *Healthy China 2030*).

Key reforms have been piloted in the city of Sanming, in the south-eastern province of Fujian, with experience gained there informing the subsequent rollout of measures in 11 other provinces and cities and, in some cases, across the country as a whole.

Although significant progress has been achieved since 2009, there is still some way to go before long-term reform objectives can be realized in full. In the meantime, new health challenges have emerged, notably the COVID-19 pandemic. These have undermined progress towards some established goals and shaped both short- and long-term health policy. In particular, COVID-19 drove the implementation of measures designed to facilitate remote consultations. It also prompted the establishment of a new National Administration of Disease Prevention and Control (NADPC), signaling a shift towards national oversight of activity in the sector.

Healthcare Reform – Achievements and Outstanding Issues

Achievements	Outstanding Issues
Establishment of a universal basic medical insurance network covering around 95% of the population	Achieved, but affordability and access issues persist, while benefits available to individual patients vary.
Rollout of critical illness cover	Achieved, but health insurance coverage for serious diseases remains insufficient.
Reduction in the share of health spending accounted for by out-of-pocket payments	In progress, but further reductions are needed.
Establishment of a national essential drug system	Achieved, but implementation is relatively limited.
Strengthening healthcare provision at community level	In progress. Primary care is being strengthened but remains underdeveloped. Overcrowding is still a problem in major tertiary hospitals, while a fully functioning referral system has yet to be established.
Availability of a broader, more equitable range of services to rural and urban populations	In progress, but healthcare resources remain inadequate in parts of the country, especially in rural areas.
Reform of the public hospital system	In progress. Drug mark-ups have been eliminated, and hospital financing reforms are bedding in.

Source: IQVIA research

Health Status

By delivering improvements in the availability and affordability of healthcare services, the healthcare reform program was designed to deliver progressive improvements in key health indicators: to levels matching those in upper middle-income countries by 2020, and high-income countries by 2030. Targeted improvements in key health indicators, and in the treatment and control of key diseases, are outlined in both five-year plans and longer-term policy documents.

Life expectancy at birth exceeded the original of 77 years by 2020, and had reached 78.6 years by 2023. There have also been progressive reductions in infant, under-five and maternal mortality rates. While rates remain significantly higher in some rural areas, figures for 2023 show that the infant mortality rate had fallen to 4.5 per 1,000 live births. The under-five mortality rate fell to 6.2 per 1,000 and maternal mortality to 15.1 per 100,000 live births, on a par with advanced middle- to high-income countries worldwide.

National control and eradication programs have been established for infectious diseases such as HIV/AIDS, tuberculosis and viral hepatitis, while a national prevention and control plan for COVID-19 was introduced in 2020. The SARS-CoV-2 virus was reclassified from a Class A infection to Class B status in January 2023, however, signaling a shift from prevention to treatment.

Chronic non-communicable diseases (NCDs) pose a growing threat to both the health status of the patient population and healthcare finances. NCDs account for some 90% of all deaths in China and for 70% of the country's disease burden according to WHO data, with cardio/cerebrovascular disease, malignant tumors and respiratory diseases now the leading causes of mortality. Diabetes is also a growing problem.

National programs have been established to combat leading NCDs. These involve efforts to broaden screening initiatives, increase early diagnosis rates and improve the treatment of key diseases, including through better availability of reimbursed medicines. Demographic trends and lifestyle choices will further increase the incidence of NCDs as the population ages, however, and as urbanization continues. Efforts are being pursued to reduce the prevalence of key risk factors such as smoking and urban pollution, while provision of specialist care for the elderly, as well as reimbursement for outpatient treatment, are being strengthened.

Cancer is a particular focus, and in 2023, the government announced an extension of its existing national action plan for cancer prevention and treatment, to 2030. In line with objectives set under the *Healthy China 2030* plan, the plan targets improvements in screening and early diagnosis, an overall five-year survival rate of 46.6% and an early diagnosis rate of 55% for key tumor types by 2030.

Support for patients with rare diseases is also being stepped up, through investments in specialist provision and the inclusion of orphan drugs on the National Reimbursement Drug List (see Prescribing and Dispensing: National Reimbursement Drug List). A National Rare Diseases Registry has been established, while a collaborative network involving nearly 420 medical institutions nationwide is focused on the treatment of rare diseases.

Healthy China 2030

Long-term health policy goals were outlined in a *Healthy China 2030* blueprint unveiled in 2016. This called for healthcare considerations to play a central role in the development of future economic and social development policy, and for a shift in the focus of healthcare policy from treatment to health promotion and management. The blueprint established five overarching goals:

- Improvements in health status and life expectancy
- Effective control of health risk factors
- Improvements in the healthcare system and health service delivery
- Expansion of the health infrastructure
- Improvements in health system governance and oversight.

The blueprint set explicit targets for improvements in patient health indicators, healthcare provision and the exposure of patients to out-of-pocket healthcare costs. These were outlined in more detail, alongside policy initiatives designed to achieve specific goals, in a three-year guidance document and action plan (the *Healthy China 2030 Guidance Document and Action Plan*), released by the State Council in July 2019.

Healthy China 2030 Action Plan

Goal	Targets
Promotion of healthy lifestyles	Reduction in national smoking rates (to less than 20%), an increase in the proportion of the population engaging in regular exercise, and a continuous decline in the prevalence of adult obesity.
Interventions to support health at each stage of life	Focusing on improvements in maternal and child health (see Health Status), health promotion in schools, workplace health protection, and the promotion of healthy aging.
Prevention and control of major diseases	Focusing on cardiovascular and cerebrovascular disease, cancer, and chronic respiratory disease which together account for more than 80% of all deaths caused by chronic illness. Better management of diabetes is also on the agenda. Disease-specific targets for these conditions were set for 2022 and 2030, as follows: - Cardiovascular/cerebrovascular disease – a fall in the number of deaths to levels no higher than 209.7 per 100,000 population by 2022, and 190.7 per 100,000 population by 2030. - Cancer – five-year survival rate to reach at least 43.3% by 2022, rising to 46.6% by 2030. - Chronic respiratory disease – no more than 9.0 deaths per 100,000 population by 2022, falling to 8.1 per 100,00 by 2030. - Diabetes – standardized management rate to exceed 60% by 2022, rising to 70% by 2030.

Source: State Council, *Healthy China 2030 Guidance Document and Action Plan*, July 2019

14th Five-Year Plan (2021-2025)

Government priorities for economic and social development are set out in five-year plans. While the national plan provides high-level guidance, local and sector-specific plans refine priorities established in the national plan (also see Pharmaceutical Policy, below).

Progress towards medium-term health policy goals outlined in national five-year plans has been variable. While targeted improvements in key health indicators have generally been met or exceeded, some specific objectives have been missed, often where these required significant changes in the behavior of patients and/or providers. While measures outlined in the 13th Five-Year Plan (2016-2020) led to significant change in the medical insurance sector, progress towards a nationwide family doctor system and a functioning referral scheme has been more limited (see Primary Care).

The 14th Five-Year Plan, released during the first half of 2021, outlined the government's broad economic and social development priorities for the period to December 2025. Influenced in part by the COVID-19 pandemic, healthcare priorities include:

- **Building a strong public health system.** Reinforce disease prevention and control systems; improve early screening and comprehensive treatment of chronic diseases; and strengthen the national immunization program.
- **Deepening reform of the healthcare system.** Strengthen the public hospital sector through investment in hospital construction and establishment of a modern hospital management system; step up moves towards a hierarchical diagnosis and treatment system; improve grassroots healthcare provision; promote centralized procurement of drugs and health consumables; improve rapid review and approval mechanisms for innovative new drugs, vaccines and medical devices;

expand training of medical personnel to alleviate staff shortages (especially for general practitioners and pediatricians); encourage the provision and uptake of private medical services.

- **Improving the health insurance system.** Improve BMI financing, payment and planning; strengthen outpatient coverage; improve coverage for major diseases; encourage the development of commercial health insurance; and strengthen supervision of the insurance sector.
- **Promoting traditional Chinese medicine (TCM).** Strengthen the TCM system, leveraging its advantages in disease prevention, treatment, and rehabilitation.

The plan sets a number of explicit health-related targets, including a further one-year rise in life expectancy at birth and an increase in the doctor/population ratio (from 2.9 to 3.2 per 1,000). It also outlines concrete measures designed to help create a more hierarchical (tiered) public health infrastructure, including:

- Establishing 15 new regional public health centers to strengthen disease prevention; and upgrading/transforming 20 national major infectious disease prevention, control and treatment bases.
- Constructing national medical centers for the treatment of major diseases (including cardiovascular disease, respiratory disease and cancer), and establishing regional medical centers in underserved provinces.
- Developing county-level hospitals, involving upgrades to raise infrastructure and service standards at 500 county-level facilities to those of tertiary hospitals.

Medium-term health policy goals were fleshed out in the 14th Five-Year Plan for National Health, released in May 2022. This set out 21 explicit goals for the development of key health indicators, including a further one-year increase (to 78.9 years by the end of 2025) in average life expectancy at birth. More broadly, the plan targets:

- A shift in the focus of provision, from treatment towards disease prevention and management, and improvements to the public health system and infrastructure, including grassroots capability.
- An increase in health promotion and education initiatives and improvements in the prevention, diagnosis and management of major chronic conditions, including cardiovascular disease, respiratory disease and cancer, as well as improvements in mental health services.
- Improvements in the availability and quality of care, including through a hierarchical diagnosis and treatment system, with the aim of shifting a proportion of demand out of tertiary hospitals into county-level and grassroots facilities.
- Increased investment in provision for children and the elderly.

Pharmaceutical Policy

Improving access to effective, affordable medicines is a key health policy goal. The government is also keen to drive up pharmaceutical quality standards, and to limit costs associated with BMI drug coverage. In the longer term, it wants to see China emerge as a major force in the research and development of innovative medicines, and as a major exporter of both APIs and finished products (see Pharmaceutical Business Environment: Industry Structure; and Research and Development).

A 2017 State Council circular set out broad policy objectives covering major aspects of drug development, registration, manufacture and distribution, underscoring the government's desire to:

- Foster the manufacture and use of high-quality domestic generics while promoting the development of innovative drugs.
- Rationalize and consolidate drug distribution.
- Improve price transparency and supervision.
- Tighten regulation of promotional activity.

There have been a number of significant developments in the sector since then, including an overhaul of regulations governing new drug development and approval (see Regulatory Environment: Drug Registration), and the implementation of measures designed to drive up the quality of local generics (see Regulatory Environment: Quality Standards and Drug Safety; and Pharmaceutical Business Environment: Generics and Biosimilars). There have also been dramatic shifts in the number of drugs reimbursed by basic medical insurance schemes, and reductions in public hospital drug procurement prices (see Drug Procurement; as well as Prescribing and Dispensing: National Reimbursement Drug List; and Pricing: National Price Negotiations).

Under the current 14th Five-Year Plan, medium-term measures designed to achieve further improvements have been outlined in sector-specific five-year plans for the period to December 2025, covering:

- Drug safety and development (see Regulatory Environment: Drug Registration; and Quality Standards and Drug Safety)
- Pharmaceutical distribution (see Pharmaceutical Business Environment: Distribution)
- Pharmaceutical industry development. This plan, published in January 2022, envisages the development of an increasingly consolidated, innovation-driven industrial sector, capable of bringing first-in-class and best-in-class drugs to the global market (see also Pharmaceutical Business Environment: Industry Structure; and Research and Development).

Healthcare Financing and Expenditure

National healthcare spending increased consistently at double-digit rates from the launch of the healthcare reform program in 2009 until 2019, peaking at nearly 22% in 2011. While spending growth slowed into single digits in 2020, it continued to outstrip rates of economic expansion, driving up the proportion of GDP invested in healthcare from a pre-reform figure of 4.6% to a peak of 7.1% in 2020.

The reform program has also effected significant changes in the structure of national health expenditure since 2009. Since then, the share accounted for by out-of-pocket payments has declined progressively, while spending backed by social health insurance and government subsidies has risen. Out-of-pocket spending by patients, which accounted for approximately 40% of national health expenditure prior to the launch of the reform program, is now responsible for around 27% of the total.

Average per capita spending on health in 2023 was almost five times higher than the 2009 figure. Significant disparities persist between levels of spending in rural and urban areas, and between prosperous and poor regions of the country, however.

Impact of the COVID-19 Pandemic on Healthcare Spending

Government healthcare spending rose by 22% in 2020, reflecting measures to fund COVID treatment, subsidize corporate social health insurance contributions, and distribute additional funds to help

provincial authorities deal with the public health emergency. National health expenditure climbed by almost 10% as a result.

That trend was reversed in 2021, when government spending declined by almost 6%, with higher social and personal health expenditure responsible for an overall increase of 6.5% in national health spending. That was the smallest increase recorded since the government embarked on its healthcare reform program in 2009. A return to double-digit growth (+11.0%) was seen in 2022, but growth decelerated again in 2023, as government expenditure slowed sharply.

National Healthcare Expenditure, 2021-2023 (RMB billion)

	2021	2022	2023	Change 2023(%)	Share 2023 (%)
Total healthcare expenditure	7684	8533	9058	6.2%	100.0
Government expenditure	2068	2404	2415	0.4%	26.7
Social healthcare expenditure	3496	3835	4168	8.7%	46.0
Personal healthcare expenditure	2121	2294	2475	7.9%	27.3

Source: China Statistical Yearbook 2023

Pharmaceutical Expenditure

Pharmaceutical expenditure has historically been a major contributor to healthcare spending, accounting for almost 40% of the total in the early part of the last decade. While that figure was inflated by relatively low fees for medical services, pharmaceutical costs were traditionally amplified by the ability of hospitals to profit from the sale of dispensed drugs, and by relatively high prices for patent-expired brands and leading branded generics. Both these issues have been addressed by policymakers in recent years, through measures including:

- The elimination of hospital mark-ups on drugs
- The establishment and expansion of national tendering for multisource drugs.

These policies have driven a reduction in the share of total healthcare spending accounted for by pharmaceuticals. Patient exposure to pharmaceutical costs has also eased, reflecting recent expansion of the National Reimbursement Drug List (NRDL), which governs BMI drug coverage (see Prescribing and Dispensing: National Reimbursement Drug List; and Pricing and Reimbursement: Reimbursement).

Health Insurance

The healthcare reform program launched in 2009 aimed to achieve universal basic health insurance coverage by 2020. That target has broadly been achieved, with basic medical insurance (BMI) schemes now covering over 95% of the population.

Until 2016, the BMI system comprised three schemes, providing cover for, respectively:

- Urban employees – through the urban employee basic medical insurance (UEBMI) scheme
- Other urban residents – through the urban resident basic medical insurance (URBMI) scheme
- Rural residents – through the new rural co-operative medical scheme (NRCMS).

Since then, in a bid to harmonize coverage, the two schemes for urban and rural residents have been merged progressively at provincial level into a single program, known as the urban and rural resident basic medical insurance (URRBMI) scheme, which provides cover for almost one billion people.

Enrolment in Basic Medical Insurance Schemes, 2021-2023 (million)

Scheme	2021	2022	2023	Change 2023 (%)
UEBMI	354	362	371	2.4
URRBMI	1,009	984	963	-2.1
Total	1,363	1,346	1,334	-0.9

Source: China Health Statistics Yearbooks; Statistical Bulletin on National Medical Security Development, July 2024

At national level, the National Healthcare Security Administration (NHSA) has been responsible for oversight of BMI schemes since its creation in 2018. BMI funds are administered locally, however, with provincial and municipal fund managers responsible for setting contribution levels (percentage payroll deductions for the UEBMI; premium payments for residency-based schemes), and for decisions on the scope and level of reimbursement (see Patient Co-Payments, below). As a result, coverage is less comprehensive in poorer parts of the country, where BMI finances are weakest.

Benefits provided by the UEBMI are more generous than those available under the residence-based schemes. Accordingly, URRBMI affiliates are generally exposed to a higher proportion of their treatment costs, with higher per capita spending by UEBMI funds than by URRBMI schemes.

Overview of BMI Schemes

Scheme	Features
UEBMI	A mandatory scheme funded jointly by workers and their employers, providing reimbursement for inpatient services and a degree of outpatient drug coverage. The scheme features a pooled fund that finances inpatient care and outpatient bills for serious and certain chronic diseases, and personal health accounts used to pay for 'ordinary' (basic) outpatient services. Employees contribute 2-3% of their salary to personal medical savings accounts, while employers contribute a further 6-8%, of which – until recently – 70% was channeled into the pooled fund, with the remaining 30% paid into the employee's personal account. Reforms announced in 2021 will see employer contributions channeled entirely into the pooled fund, with additional revenues used to strengthen coverage of outpatient bills (see BMI reforms, below). Annual deductibles and coverage ceilings apply. Personal accounts can be used to cover expenses below the deductible but are rarely sufficient to cover costly treatment or fund chronic care. Nationally, contributions have consistently exceeded expenditure, although the scheme runs at a deficit in some economically weak provinces and cities.
URRBMI	A voluntary scheme for urban and rural residents, established through the merger of the URBMI and the former NRCMS. Financed by a mix of government subsidies and individual premiums. Coverage spans inpatient services and outpatient treatment for serious or chronic diseases. While the extent of outpatient cover still varies in accordance with local economic circumstances, the merger has been beneficial for rural residents, who previously had access to a more limited range of benefits.

Source: IQVIA research

Patient Co-Payments

Patient contributions to the cost of their care vary according to the scheme they are affiliated with and the province or city they live in. Coverage provided by the UEBMI is most comprehensive, with actual reimbursement of inpatient care costs averaging over 85% for affiliates of the employment-based scheme, compared with an average 70% for members of the URRBMI. Percentage contributions to outpatient care costs are higher, especially in poorer regions.

Drug co-payments and deductibles vary by scheme, by province/city, and by product. No contribution is needed if the prescribed drug is included in the national Essential Drugs List (EDL) and the patient is affiliated to a BMI scheme. Drugs included on List A of the National Reimbursement Drug List (NRDL) are also subsidized in full. The level of reimbursement for NRDL List B products is set by local authorities, however, and varies across the country. The amount is a percentage of the cost of the drug, regardless of whether it is a generic or an original brand. Co-payments are usually in the 20-30% range. UEBMI affiliates may use their personal medical accounts to cover co-payment costs.

Patients traditionally had to foot the entire bill for most outpatient drugs, although some municipalities and provinces provide a degree of outpatient drug coverage. In some cities, reimbursement is available at accredited retail pharmacies, with retailers receiving automatic payment of the percentage covered by the government and patients paying the co-payment. UEBMI coverage of outpatient medical expenses for treating some serious diseases has been rolled out, with provinces required to implement measures outlined by the State Council in April 2021 over a three-year period.

BMI Reforms

Policymakers are pursuing measures designed to broaden and deepen BMI coverage. This has seen:

- A progressive increase in the number of innovative drugs reimbursed by basic health insurance schemes (see Prescribing and Dispensing: National Reimbursement Drug List).
- The introduction of mechanisms enabling cross-provincial settlement of BMI claims.
- A shift in the structure of UEBMI funding, with a view to financing more comprehensive outpatient coverage.

BMI funds and providers began trialing the cross-provincial settlement of outpatient insurance claims in 2020. By the end of that year, around 48 million people (over 70% of whom were affiliates of residence-based schemes) had accessed treatment outside their home province. The initiative has broadened since then, with more than 110 million BMI affiliates accessing subsidized treatment outside their home province in 2022, including almost nine million who received inpatient care under the initiative.

Reform of the structure of UEBMI funding, announced by the State Council in April 2021, will:

- Allow affiliates to use their individual accounts to pay for some medical costs incurred by immediate family members (including out-of-pocket treatment expenses and the purchase of medicines at designated retail pharmacies).
- Expand UEBMI outpatient coverage, with employees able to claim reimbursement for 50% of covered outpatient costs and retirees eligible for higher reimbursement rates. Coverage expansion will be financed by shifting a proportion of employer contributions from personal medical accounts into the pooled fund.

The latter policy is designed to fund improvements in the reimbursement of outpatient costs, and to reduce spending by elderly and chronically ill affiliates, who incur high out-of-pocket costs for outpatient treatment. UEBMI mutual funds are expected to reimburse at least 50% of outpatient treatment bills, although guidance issued by the government does not reference explicit deductible thresholds, maximum reimbursement amounts or benefit coverage. These will be determined by local authorities in accordance with the financial health of their respective UEBMI funds.

The NHA calculated that the reforms would boost UEBMI mutual fund revenues by some RMB600 billion a year, without incurring additional costs to employers or affiliates. Provincial authorities were asked to implement the UEBMI reforms over a three-year period to 2024. Where finances permit, they have also been urged to broaden outpatient reimbursement coverage to more conditions over time.

Further reform of the health insurance system during the period of the 14th Five-Year Plan were fleshed out in the State Council's 14th Five-Year Plan for National Medical Security. This plan, which covers the period to 31 December 2025, targets the development of a stronger, more integrated, multi-level health insurance system, comprising the BMI schemes, critical illness insurance, commercial insurance, medical mutual assistance schemes, and long-term care insurance.

Objectives outlined in the plan include the implementation of measures to strengthen coverage, improve management and control, and regulate drug usage under the system.

Objectives of the 14th Five-Year Plan for National Medical Security

Objective	Targets
Strengthening coverage	Ensure that BMI affiliation rates remain in excess of 95%, and that coverage of hospitalization costs for UEBMI and URRBMI affiliates stabilizes at around 85% and 70%, respectively. Encourage commercial insurance, medical mutual assistance schemes and long-term care insurance. Improve outpatient benefits (notably for UEBMI affiliates with hypertension and diabetes). Move towards the standardization of BMI benefits, including drug coverage. Improve outpatient benefits. Improve serious illness coverage (including through financial relief for disadvantaged groups), medical assistance, and long-term care insurance.
Improving management and control	Improve the security and stability of BMI funds, through enhanced premium-collection systems and more effective budget management, including the use of flat-sum provider payment mechanisms. Establish supervision and inspection mechanisms. Improve management and service quality. Accelerate moves towards the digitization of health insurance.
Regulating drug usage	Improve supply and safety assurance capabilities for drugs, including through promotion of generic quality and consistency evaluation (GQCE) and a drug traceability system. Optimize adjustments to the NRDL. Harmonize the scope of drug reimbursement. Improve centralized procurement systems for drugs and medical consumables at national, provincial and cross-regional levels. Strengthen drug price control through the implementation of purchase price and transaction price monitoring mechanisms, and national drug price monitoring.

Source: IQVIA research

Critical Illness Insurance

Critical illness insurance was rolled out across the country from 2014 and the vast majority of the population is now covered. Initially covering 22 diseases, its scope was broadened in 2015. In most provinces, it now covers the cost of critical illnesses beyond a set threshold. Like the BMI schemes, critical illness insurance is managed at provincial/city level, and payment thresholds, reimbursement levels and patient contributions vary – as does the way implementation is managed by local authorities.

A small percentage of BMI fund revenues is generally allocated to purchase critical illness insurance policies from commercial insurance companies. The aim is for commercial policies to reduce the amount that patients have to pay for expenses incurred beyond the BMI reimbursement ceiling.

It was envisaged that reimbursement would initially cover 50% of costs, with that figure rising gradually over time. Subsidy levels were raised to 60% in 2019, though actual reimbursement depends on local budgets and is higher in parts of the country. Drug reimbursement is usually governed by the NRDL.

Some provinces (notably Zhejiang) have included coverage for rare diseases under the critical illness scheme. Where this is the case, provincial authorities have negotiated reductions in the price of orphan drugs to limit costs associated with such initiatives.

Long-term Care Insurance

A long-term care insurance scheme, aimed primarily at elderly patients, was launched in 2016. Piloted initially in 15 cities, it was trialed in around 50 cities in 2023, with some 170 million people participating in the scheme. Beneficiaries contribute RMB160 a year. The rollout has encouraged nursing homes and other long-term care facilities to be set up in the cities where the scheme is being piloted, although the quality and availability of care for frail elderly patients remains an issue.

Medical Financial Assistance Program

The Medical Financial Assistance Program, established nationally in 2008, provides supplemental financial support for disadvantaged individuals, either through direct medical assistance (accounting for around two-thirds of expenditure by the program) or financial help to purchase BMI policies. Administered locally, medical assistance schemes are subsidized by local and central government funds, and by private donations, varying in scale according to prevailing financial conditions. The program subsidized BMI participation for over 80 million low-income individuals in 2023, and provided financial assistance for outpatient and inpatient treatment to over 153 million.

City-Level Supplementary Medical Insurance Schemes

Pioneered by Shenzhen in 2015, city-level supplementary insurance schemes (*huiminbao*) have emerged as an option for patients to top up BMI cover, giving them subsidized access to products and services not covered by the BMI plans. While the schemes are publicly managed (by local NHSA offices), they are backed by commercial payers, working with third party administrators (TPAs).

The premise of *huiminbao* schemes is that, in return for a modest annual premium, individuals can access additional services at heavily subsidized rates – subject to an annual monetary cap, and usually after payment of a deductible. A particular attraction for patients is specialty drug formularies, listing costly drugs that do not feature on the NRDL.

While annual premiums, deductibles and coverage limits vary, patients generally have to pay between RMB50 and RMB150 for one year of cover. Where a deductible is payable, this is usually set at between RMB20,000 and RMB25,000, with caps on the value of annual benefits typically set at RMB2-3 million. After payment of the required deductible, patients can access high-cost drugs, which are typically subsidized by *huiminbao* schemes at rates of 50-60%.

While a number of commercial health insurers are involved in the sector, China Pacific Insurance Company (CPIC) has emerged as a leading player, with Ping An, Zhong An, and China Life also claiming significant early shares of the market. Pharmacy benefit management companies (PBMs) involved in the sector include MediTrust, MedBanks and Yuanxi Huibao.

More than 300 million people were reported to be enrolled in *huiminbao* schemes in 2023, according to data cited by the Asia Insurance Review. According to KPMG China, as of February 2024, 21 provinces/ regions and 142 cities across China were offering *huiminbao* products with effective dates for that year.

Growth in the sector has slowed, though, reflecting concerns about the sustainability of the model in its current form. Operated predominantly at city level, and offering coverage for individuals with pre-existing conditions, many *huiminbao* schemes have unfavorable risk profiles, leading to financial bail-outs. The schemes remain attractive to patients, however, and to drug manufacturers pursuing market access for innovative new products not included on the NRDL.

Private Health Insurance

The government views private health insurance as offering potential to complement public insurance plans and address shortfalls in coverage and reimbursement provided by BMI schemes. Critical-illness insurance and products covering high-end medical services or meeting the needs of elderly patients have been identified as priorities. Costs associated with the purchase of eligible commercial health insurance products for both individuals and companies have been tax deductible since 2017.

The private health insurance market expanded rapidly (albeit from a very low base) over the decade to 2020, with voluntary private plans estimated by the WHO to have provided around 9% of total healthcare spending in that year, up from only 2% in 2010. Private health insurance premiums, which had risen by over 50% in each of the two previous years, increased by only 10% in 2017, following a government crackdown on abuses in the insurance sector. However, they rose again by 24% and 30% in 2018 and 2019, respectively. A more pronounced slowdown was witnessed following the onset of the COVID-19 pandemic. The market showed signs of a recovery in 2024, with premium income rising by 6.1%.

While more sophisticated products have begun to emerge, disease-specific critical illness policies (CIPs) – often purchased in conjunction with life insurance cover – have traditionally dominated the sector, generating close to 50% of all commercial health insurance premium income in 2020. Outside the CIP market, beneficiaries may be covered either through individual or family policies, or by group plans taken out by their employer.

Short-term (usually one-year) medical insurance policies, typically featuring low premiums and high insured amounts, and often dubbed ‘million-dollar’ plans, have become increasingly popular. However, they offer no guarantee of renewal if patients fall sick and start to incur substantial healthcare bills. Longer-term, high-end plans, similar to the products that dominate the commercial health-insurance

sector in developed western economies, remain beyond the reach of most patients, and are largely the preserve of high-income individuals or high-ranking corporate executives.

The private health insurance market is dominated by leading domestic payers, including Ping An, PICC, China Life, New China Life, and China Pacific. Most foreign insurers involved in the market – including players such as AXA Life, Aviva, Generali, Prudential and Allianz Group – have levered entries through collaborations with local partners. More recently, China's leading internet companies have entered the sector, broadening the range of insurance products on offer. The removal of restrictions on investment by insurers in healthcare means leading players may now invest directly in providers and can partner with hospitals in areas such as data and claims management.

Hospital Sector

Public hospitals remain the dominant providers of care, but their numbers have been declining for several years now, while the private hospital sector has grown rapidly. Private facilities have outnumbered public hospitals since the middle of the past decade, and now account for over around 70% of the total. Public facilities still account for 69% of the country's hospital beds, however, and deal with over 80% of both inpatient admissions and hospital outpatient consultations.

Most public hospitals are categorized into one of three tiers, reflecting the range and complexity of services they offer, as follows:

- Tier 3 – larger institutions at national, provincial or city level providing high standard, specialized tertiary care services across regions, with advanced teaching and research responsibilities.
- Tier 2 – medium-sized facilities at city, county or district level, providing general secondary care services to multiple communities, with some teaching and research responsibilities.
- Tier 1 – smaller community, township or county facilities, providing primary care (including preventative) services, health maintenance and rehabilitation services in the community.

Number of Hospitals by Type and Ownership, 2021-2023

Type of Hospital	2021	2022	2023	Change 2023 (%)	Share 2023 (%)
Total hospitals	36,570	36,976	38,355	3.7	100.0
By ownership					
Public	11,804	11,746	11,772	0.2	30.7
Non-public	24,766	25,230	26,583	5.4	69.3
By categorization					
Tier III	3,275	3,523	3,855	9.4	10.1
Tier II	10,848	11,145	11,946	7.2	31.1
Tier I	12,649	12,815	13,252	3.4	34.6
Uncategorized	9,794	9,493	9,302	-2.0	24.3
By specialization					
General hospital	20,307	20,190	20,497	1.5	53.4
TCM hospital	4,630	4,779	5,053	5.7	13.2
Specialist hospital	9,699	10,000	10,581	5.8	27.6

Source: China Statistical Yearbooks; NHC Statistical Bulletin on Development of Healthcare in China 2023 (August 2024)

Patients have long preferred to seek treatment in large, specialist hospitals, where they perceive – often correctly – that the quality of care is superior. As a result, overcrowding is a major problem in these facilities, and bed utilization rates far outstrip those in lower-tier hospitals.

The workload facing public hospitals has increased significantly following the rollout of the BMI schemes. As well as exacerbating overcrowding, this has highlighted other problems, including variations in the cost and quality of care. The need to curb treatment costs has become more pressing for the government, which heavily subsidizes the residence-based BMI schemes (see Health Insurance).

Hospital Reform

Policymakers acknowledged that a substantive overhaul of the public hospital sector was needed when they embarked on the current healthcare reform program in 2009. Costs associated with providing public hospital care had been inflated by poor management and limited government subsidies, which accounted for less than 10% of public hospital funding. With allocations based on the number of beds filled and staffing levels relative to patient throughput, this had encouraged widespread overtreatment and the pursuit of other income streams – notably, profits on the sale of pharmaceuticals and medical devices dispensed on hospital premises. A raft of reforms has sought to address these issues.

Public Hospital Reforms – Key Goals and Progress

Measures	Details
Hospital financing	The 15% mark-up previously permitted on prescription drugs and diagnostic tests was eliminated nationally in 2017. Medical service fee hikes and an increase in government subsidies were envisaged to offset the impact on hospital finances. The elimination of mark-ups drove a progressive reduction in the proportion of public hospital income generated by drug sales, to 24% of inpatient costs and 38% of outpatient costs in 2022. Medical fee hikes and government funding have not fully offset the impact on overall hospital revenues, however, limiting physician salaries. Medical service pricing reforms are now being pursued in a bid to limit regional disparities and enable more dynamic price adjustments.
Drug budget controls	Capitated global budgets have been imposed on public hospitals in many cities, with hospitals expected to contribute to funding expenditure in excess of the global budget. Caps on the contribution of drugs to hospital revenues, designed to peg drug spending at 30% of total public hospital expenditure by 2020, had a significant impact on drug use in some public hospitals. Ceilings have since been relaxed.
Provider payments	BMI provider payment systems are being overhauled in a bid to encourage more cost-conscious approaches to treatment. This involves the progressive introduction of flat-sum reimbursement mechanisms in public hospitals. Diagnosis-related group (DRG)-based payments and ‘disease intervention packages’ (DIPs), which have been trialled at city level, are scheduled for rollout to all inpatient providers by 2025.
Tiered diagnosis and treatment	Another key objective is a tiered diagnosis and treatment system, featuring a functioning gatekeeper mechanism. Funding for county-level hospitals and municipal facilities has been ramped up, while the expansion of large public hospitals has been controlled. Alliances between hospitals (known as ‘medical treatment partnerships’) to filter expertise down to lower-level facilities are being encouraged. A 2019 State Council circular called for the establishment of integrated urban medical groupings in 100 cities, and of 500 county-level medical treatment partnerships. Tiered reimbursement rates, differential service charges and referral ratio controls are being used to encourage more patients to seek initial consultations in primary care or lower-tier hospitals.
Hospital management	State Council guidance issued in 2017 targets a ‘modern hospital management system’ emphasizing the non-profit nature of public hospitals. The introduction of a new merit-based evaluation system for hospital directors is planned. Hospitals will be handed greater autonomy, but supervision of the sector will be stepped up.

Source: IQVIA research

The COVID-19 pandemic, which highlighted shortcomings in the public hospital sector, prompted a renewed push for the implementation of existing reforms. A three-year action plan for the rollout of DRG/DIP hospital payment reforms, published by the NHSA in November 2021, envisaged the progressive rollout of flat-sum reimbursement models in all provinces during the period to 31 December 2024. The plan aimed to ensure that, by 2025, 70% of inpatient service costs funded by BMI schemes would be reimbursed via DRG and DIP models.

Early studies showed that, where they have been implemented, DIPs achieved reductions in both overall treatment costs and the proportion of those costs picked up by patients. They also had a positive impact on the efficient allocation of healthcare resources, with reduced average stay lengths

in tertiary hospitals and increased lengths of stay in secondary-level facilities. DRGs have been shown to produce similarly positive results, including more widespread adherence to clinical guidelines and reductions in the variance of treatment costs for specific interventions. These latter have been a particular issue under fee-for-service payment models.

Private Hospitals

While public institutions remain the dominant providers of hospital care, private hospitals are playing a growing role, complementing public healthcare services and helping to ease pressure on the public hospital infrastructure. Private hospital numbers and overall capacity in the sector have both risen rapidly. Nearly 70% of all hospitals were privately owned in 2023, although most are smaller facilities, with only 31% of hospital beds located in private facilities. The sector still handles only 18% of inpatient admissions and 16% of outpatient consultations, however, and has been hit hard by the COVID-19 pandemic, which led to a sharp decline in patient footfall, triggering a spate of bankruptcies.

The failure of private hospitals to meet a bigger share of demand is largely a reflection of their limited involvement in the BMI system. Healthcare reform plans envisage the eventual inclusion of private providers in the system, but BMI medical fees have limited appeal for profit-oriented players. In the meantime, demand for private hospital care is driven by dissatisfaction with standards of provision in overcrowded public hospitals among an increasingly affluent urban population. A growing number of patients are seeking treatment in private facilities, where waiting times are shorter, and where conditions and staff attitudes are better than those in most public hospitals. Demand in the sector has also benefited from increases in the number of people with private health insurance.

The government is actively encouraging expansion of the private hospital sector. In 2015 the State Council announced that the approval process for private hospitals would be streamlined, and that private hospitals would be integrated into the BMI system. In 2017 it called for further development of private healthcare services and said efforts should be made to attract foreign investment in the sector, promising to relax requirements for new entrants and to step up training for medical professionals.

Rules governing private and foreign investment in hospitals have been eased, although not as widely as anticipated. Foreign investment restrictions were lifted in the Shanghai Free Trade Zone in 2014, but restrictions on overseas participation in joint-venture hospitals continued to cap foreign holdings at 70%. Investors from Hong Kong, Taiwan and Macao, which are exempt from the restrictions, may set up 100% foreign-owned hospitals in cities above the prefecture level. A pilot program offering further liberalization was announced in September 2024 (see Key Developments and Trends: Healthcare Provision: Hospital Sector).

Local investors in the private hospital sector in recent years included Fosun Pharmaceuticals, property developers Vanke and Evergrande, and the Wanda conglomerate. Major foreign investors include Germany's Artemed Group, Columbia Pacific, IHH Healthcare and Raffles Medical Group. Notwithstanding problems caused by the COVID-19 pandemic, the sector remains an attractive long-term investment target, and continues to draw in new entrants. One of them is TikTok's Chinese owner, ByteDance, which acquired a chain of hospitals and specialist clinics through its purchase of Amcare Healthcare in August 2022.

Physician recruitment remains a challenge for private hospitals. While rules governing physician employment have been relaxed, allowing doctors to work at multiple sites, administrative challenges persist, and employer consent requirements remain in place. Medical practice mobility is also

hampered by the shortage of doctors in the public sector, where hospitals are reluctant to lose or share key staff. Relatively few doctors have so far been willing to practice at multiple sites.

Primary Care

Strengthening primary care provision is an established health policy goal, and initiatives designed to foster development of the sector have featured prominently in recent Five-Year Plans. The aim is to relieve pressure on top-tier public hospitals by shifting primary care provision into county hospitals, urban CHCs and rural township health centers, and to limit costs through more effective management of key diseases. The ultimate goal is the establishment of a ‘tiered’ or hierarchical approach to diagnosis and treatment featuring a family-doctor system, with primary care practitioners acting as gatekeepers to specialist and inpatient care (see Health Policy).

Early goals for primary care reform included establishing a functioning GP network and a progressive increase in the GP/patient ratio. Policymakers wanted to see GP numbers exceed 150,000 by 2015 and targeted a GP/patient ratio of 2-3 per 10,000 by 2020 – rising to five per 10,000 by 2030. In fact, GP numbers exceeded 150,000 by 2014 and topped 300,000 by 2018, while NHC data showed that the GP/patient ratio had reached 2.6 per 10,000 by 2019, and 4.0 per 10,000 by 2023. Both metrics vary significantly between provinces and individual cities, however. There are now around six GPs per 10,000 patients in some major cities and provinces, for example, while figures in some remote areas of the country are closer to one per 10,000.

CHCs and township health centers have traditionally been the main providers of grassroots inpatient care. Across the country as a whole, these facilities provide around 1.6 million inpatient beds – equivalent to almost 20% of total bed capacity. Bed utilization rates in both CHCs and township health centers are low, however, at around 50% and 57%, respectively, and far below rates in major tertiary and secondary hospitals (see Hospital Sector). Efforts to strengthen primary care provision have seen CHC numbers rise gradually in recent years, while there has been a significant increase in the number of grassroots clinics and infirmaries.

Primary Care Facilities in China 2021-2023

Type of Institution	2021	2022	2023	Change 2023 (%)
Total grassroots health institutions	977,790	979,768	1,016,238	-3.7
Community health centers/stations	36,160	36,448	37,177	2.0
Rural township health centers	34,943	33,917	33,753	-0.5
Village clinics	599,292	587,749	581,964	-1.0
Clinics/infirmaries	271,056	282,386	318,938	12.9

Source: NHC Statistical Bulletin on China's Health and Wellness 2021 (July 2023); NHC *Statistical Bulletin on Development of Healthcare in China 2023* (August 2024)

Family Doctor System

Since 2016, a family doctor system has been trialed in 200 cities piloting public hospital reforms. Public-hospital doctors have been encouraged to practice in grassroots medical facilities or to set up their own family-doctor practices, and efforts have been made to train general practitioners. Patients were asked to contract with a family doctor, with enrollment initially prioritizing vulnerable groups – including patients with chronic diseases and psychiatric disorders, seniors, pregnant women, children, and patients with disabilities – before being rolled out to the rest of the population.

On paper, family doctor networks are now established in more than 95% of cities. The best progress has been made in the country's most affluent cities, including Shanghai, Beijing, Tianjin, Hangzhou and Nanjing. Local governments there have allocated funds to support training and are encouraging doctors from tier 3 hospitals to visit CHCs on a weekly basis. Patients are incentivized to visit CHCs through the application of differentiated reimbursement rates. The system has a much more modest impact in areas where GPs are in short supply, however.

Drug Procurement

Tender System

Pharmaceuticals used in public hospitals have traditionally been purchased mainly through competitive tenders at provincial level, city level, or for groups of hospitals at local level. Since 2015, procurement must be conducted in accordance with national regulations requiring the purchase of all drugs through provincial procurement platforms. Provincial procurement agencies must aggregate demand estimates from hospitals, centralizing and consolidating all drug purchasing, with a view to increasing transparency, reducing kickbacks and improving payment and delivery. Procurement data must be released, to increase the transparency of the procurement process.

The regulations split drugs into five categories and require category 1 (multisource) products to account for 80% of the previous year's total procurement value by public hospitals – effectively ensuring that the vast majority of hospital drugs are purchased via tender. To be included in tenders, category 1 drugs must be produced by at least three manufacturers. The regulations recommend a 'double envelope' tender system, with no more than two winners in each group. Under the double envelope system, economic and technical as well as commercial criteria are taken into account. Where a bid price is significantly lower than other tendered prices, the risk to quality and supply is assessed. Preference is given to generics that meet international standards.

Since 2015, provinces have established online procurement platforms, which have increased the transparency of the tendering process and allow provinces to reference prices achieved elsewhere. Provinces reference prices either to the lowest price achieved in all provinces, the average price achieved in all provinces, or the lowest or average price in a selection of provinces.

Additional pressure on tender prices arises from:

- City-level price negotiations. These increase downward pressure on prices by aggregating volumes, and can drag down provincial prices where they are significantly lower.
- Group purchasing organizations. These have been established in several cities, and have begun to emerge in parts of the private hospital sector.

- Hospital-level price negotiations. These have been pursued in a bid to compensate for the abolition of hospital drug mark-ups, often involving requests for discounts or ‘commission’ payments in return for formulary listings.

2015 Regulations on the Categorization of Drugs for Procurement Purposes

Category	Procurement Requirements
Category 1	Essential and off-patent drugs (including generics and off-patent originators) with high clinical usage volume and high procurement value, produced by multiple manufacturers. Category 1 drugs must be purchased via tender through provincial drug procurement agencies.
Category 2	Certain patented drugs and drugs produced exclusively by one manufacturer. Category 2 drugs are subject to national price negotiations (see also Pricing: National Price Negotiations). Negotiated prices are made public and should not, in theory, be subject to any further negotiation at provincial or local level.
Category 3	Off-patent gynecology and pediatric drugs, first aid and emergency drugs, basic infusions, drugs with low clinical usage volume and common low-cost drugs. Category 3 drugs must be purchased directly by hospitals through a centralized online platform.
Category 4	Clinically indispensable drugs and drugs with low usage volume or in short supply. These are manufactured and supplied by producers designated through national tenders, with purchase prices negotiated for the relevant products.
Category 5	Narcotic and psychotropic drugs, drugs provided free of charge for the prevention and treatment of infectious and parasitic diseases, vaccines for the national vaccination program, contraceptives, and Chinese herbal medicines. Category 5 products are purchased in line with national regulations that applied prior to the introduction of the 2015 procurement guidelines.

Source: IQVIA research

National Volume-Based Procurement (VBP)

In November 2018, policymakers signaled the start of a new push to drive down procurement prices for multisource drugs, inviting manufacturers to bid for contracts involving the supply of 31 such products to hospitals in 11 of the country's biggest cities (Beijing, Shanghai, Guangzhou, Tianjin, Chongqing, Shenyang, Dalian, Xiamen, Shenzhen, Chengdu, and Xi'an). Winning bidders were offered a guaranteed share of the market in these cities (averaging 70% but ranging from 50% to 80% in individual cities). Public hospitals in the 11 cities were required to prioritize the purchase of products from successful bidders with effect from March 2019, until quantity commitments had been fulfilled.

Tenders were conducted in December 2018, resulting in consolidated procurement contracts for 25 of the 31 drugs. Contracts were awarded by a Joint Procurement Office (JPO) under a two-stage process. On average, procurement prices agreed in those contracts were 52% lower than the lowest price prevailing in the 11 cities during 2018. The approach has been expanded progressively since then:

- First, in September 2019, by the award of contracts for supply of the same 25 drugs to public hospitals nationwide.
- Subsequently, through the issue of invitations to tender for nationwide supply of a further 349 drugs in eight rounds of VBP conducted between 2020 and 2023

National VBP Activity

Round	Number of drugs	Aggregate sales of affected drugs (RMB billion) ²	Average price reduction (%)
December 2018 ¹	25	15.6	52
September 2019	25	55.2	25
January 2020	32	33.5	53
August 2020	55	58.3	53
February 2021	45	35.5	52
June 2021	61	55.0	56
November 2021 ³	16	17.4	48
July 2022	60	70.0	48
July 2023 ⁴	39	29.8	56
September 2023 ⁴	41	31.4	58

¹ VBP pilot, involving contracts for the supply of public hospitals in 11 major cities. ² Annual sales in the calendar year prior to VBP. ³ Involving contracts for the supply of six insulin product types. ⁴ Projected annual savings estimates by the government.

Source: IQVIA China analysis

In the 2018 VBP pilot, contracts were awarded to a single winning bidder. Since then, a substantial proportion of national contracts have been split between multiple (up to 10) winning bidders, with successful companies sharing up to 80% of the market.

Rules were tweaked further in 2022, principally to improve security of supply and prevent manipulative practices such as bid rigging. On average, more than five companies were selected for the supply of each drug subject to national tenders agreed in July 2022. Individual bidders had to specify both preferred supply areas and 'standby' areas where they would be able to supply the market if the main supplier selected for a particular region was unable to meet demand, or if quality issues disrupted the availability of a particular product.

For small-molecule drugs, national VBP tenders have only been issued for patent-expired molecules where at least two generic competitors to original brands have undergone generic quality consistency evaluation (GQCE). Insulin products, for which national VBP tender contracts were awarded in November 2021 (see below), were not subject to GQCE requirements, however.

The massive volumes guaranteed for winners of national VBP contracts have encouraged fierce competition, driving significant reductions in the price of affected drugs. The extent of those reductions, and the failure of higher volumes to offset cuts in the price of many products that have won national contracts, have been the subject of growing concern for manufacturers.

Multinationals have chosen not to compete widely for contracts on offer in recent VBP rounds, preferring to maintain existing prices while competing for remaining shares of the public hospital market, and ramping up efforts to generate sales through other channels, including private hospitals, retail pharmacies and the e-pharmacy market. Some well-established brands used in complex hospital settings have also retained significant shares of the public hospital market, despite losing out in VBP

tenders. Antibacterials used in ICU departments are a prominent example, with prescribers in that setting reluctant to switch away from trusted brands.

National VBP for Insulins

In what was widely perceived as a pilot to inform national VBP for other biologics, insulin manufacturers were invited in November 2021 to bid for two-year supply contracts, equivalent in volume terms to around 60% of national insulin usage. The tender split 42 insulin products into six groups (three for human insulins and three for modern insulin analogs). It pitted multinationals such as Novo Nordisk, Eli Lilly and Sanofi – which previously held around 70% of the Chinese insulins market – against eight domestic manufacturers, triggering price cuts that averaged 48%.

Acknowledging the more complex nature of insulins in comparison with small-molecule drugs, rules governing the award of supply contracts were modified, allowing for:

- The inclusion of all originators in supply contracts.
- The establishment of a price threshold, enabling all companies offering cuts of at least 40% to win a proportion of supply contracts.
- More liberal hospital procurement rules, enabling hospitals to purchase multiple products.

Bidders were categorized into A, B and C winners, with those offering the sharpest price cuts awarded the largest market shares. With local manufacturers generally outbidding multinationals, that meant most market-leading insulin brands faced both a significant price cut and the prospect of reductions in their volume sales.

Provincial VBP

Policymakers have actively encouraged the adoption of VBP at provincial level, while in January 2021 the State Council called for the establishment of multi-province purchasing alliances. The NHSA has indicated that responsibility for the renewal of contracts negotiated initially at national level will lie with provincial health authorities. Provincial and multi-province tenders have also been issued for other small-molecule drugs, including some that are not available in GQCE-approved generic form.

Multi-Province Procurement Activity in 2022

Leading Province	Number of Participants	Number of Drugs	Notable Features
Guangdong	10	276	Included growth hormone and blood products.
Guangdong	6	87	Focus on chronic disease treatments. Hospitals submitted purchase volumes after tenders.
Shanghai	3	47	Extension of chronic disease supply contracts to three years.
Liaoning	12	18	Including fourteen injectable drugs.
Jiangxi	16	15	Including drugs with generics that had not obtained GQCE certification.

Source: IQVIA China analysis

Selection, tendering criteria and the identity of participating provinces vary, making for an increasingly complex environment. Provincial health authorities in Guangdong and Shanghai have played a prominent role in the co-ordination of major multi-province tenders, some of which have levered price cuts averaging in excess of 50%. Pooling the purchasing power of multiple provinces helps alliances to lever more substantial discounts from bidders. Provincial authorities also boost their leverage by awarding substantial market shares to fewer winning bidders, with contracts often split between one or two suppliers, in comparison with the current average of around five for national VBP contracts.

E-Health

Digital technologies have been harnessed increasingly by a range of stakeholders in the healthcare system. The trend has been encouraged by regulators, who have acknowledged the potential of e-health to improve access to healthcare, increase levels of efficiency, and reduce pressure on major public hospitals. The government is also keen to harness healthcare big data as a strategic resource.

The use of digital technology to support remote consultations, electronic prescribing and online pharmacy surged in the wake of the initial COVID-19 outbreak. Regulators were happy to facilitate this activity, relaxing some telemedicine rules and expanding access to BMI reimbursement of digital health services.

E-Health Policy Development

The government's goal is to establish standardized data systems for electronic medical records, medical services, and health insurance information, with access to medical treatment facilitated through online appointments, online consultations, payment through mobile devices, and mutual recognition of examination and test results by medical institutions via electronic medical records. Healthcare big data will be harnessed to build information on medical conditions, medical services, and disease control and prevention.

Regulation of e-health – also referred to as “internet + medical services” or “internet plus” - has been stepped up progressively. Three regulations published in 2018 provide a legal basis for the administration of telemedicine and other online healthcare services. The NHC released *Detailed Rules for Regulation of Internet-based Diagnosis and Treatment (Trial)* in June 2022 to address loopholes, outlining detailed requirements for operators of internet diagnosis and treatment platforms. In parallel, provincial and municipal health authorities are imposing their own e-health rules.

Detailed rules governing online drug sales, which had been outlined in the 2019 revisions to the DAL, were published in September 2022 and came into force on 1 December 2022. The regulations apply to both online drug distributors and third-party e-commerce platforms.

E-Health Regulatory Overview

Scope	Requirement
Providers	Only licensed medical institutions can provide remote consultation and diagnosis. Physicians practicing telemedicine must generally have more than three years' experience and can only provide telemedicine services through the platforms or communication channels set by medical institutions. Physicians need to provide real-name authentication during online consultations.

Scope	Requirement
Prescribing	Physicians can issue online prescriptions for patients with common and chronic diseases. Regulations require initial face-to-face consultations before physicians practicing telemedicine may issue a prescription, however. Prescribing of special management drugs, such as narcotics or psychotropics is not allowed.
Drug Sales	Online sellers must be licensed and registered with local authorities, and must establish systems for quality and safety management, risk control, traceability, storage and delivery. Online retailers must also establish online pharmaceutical care systems, with licensed pharmacists employed to review prescriptions, prepare drugs and provide guidance on rational use. The online sale of vaccines, blood products, anaesthetics, psychoactive drugs and other controlled products is prohibited. For sales of other prescription drugs: • Online sales to individuals must be based on valid prescriptions and dispensed to named individuals. • Sellers must sign agreements with e-prescription providers, review scripts, and implement effective measures to avoid repeated use of paper prescriptions. • The prescription status of individual products must be made clear on sellers' websites, which must not depict prescription drug packaging or labelling on main webpages. • Online retailers must ensure the quality of distribution by any third-party service providers, in line with Good Supply Practice (GSP).
Reimbursement	Reimbursement was extended to limited online medical services during the pandemic, but further clarification is needed.

Source: IQVIA research; Chambers and Partners, *Digital Healthcare 2023 – China (June 2023)*; CMS, *Digital Health Apps and Telemedicine in China (May 2023)*

Mid-term E-Health Policy Goals under the 14th Five-Year Plan

Mid-term goals for the development of e-health during the period of the 14th Five-Year Plan were outlined in a sector-specific plan published in November 2022. Key targets include:

- The establishment by 2025 of a unified, interconnected national health information platform.
- The establishment of a national personal-health information index to support the development of a cross-provincial electronic medical records system.
- The establishment of an infectious disease monitoring information platform, providing early warnings of future potential health threats.
- A comprehensive upgrade of hospital information systems.
- Standardization of national health information systems.
- Development and expansion of internet health service provision, including telemedicine, with a view to improving and harmonizing patient access – especially in fields such as primary care, pediatrics and maternal health.
- Improvements in monitoring of drug use and pharmaceutical supply, including the establishment of an early warning system for potential drug shortages.

Private Investment in E-Health

Private investment in the sector has increased rapidly in recent years, and a multitude of mobile services – ranging from booking appointments and making payments, to online patient-doctor communication and consultation, as well as patient monitoring and disease management – are now marketed. Patients can consult leading doctors across the country, upload medical scans, and receive diagnoses and prescriptions electronically, while supplementary health insurance cover is available via mobile phones.

China's leading internet firms, Alibaba Group, Tencent and JD.com, as well as leading insurance companies such as Ping An, dominate the provision of online healthcare services. Leading e-health platforms include:

- Ping An Good Doctor: Provides online appointments, consultation and diagnosis services. Online medical platform, comprising an extensive network of hospitals, clinics, and pharmacies.
- WeDoctor: Operated by Tencent. Provides online appointment, diagnosis, prescription and payment services. Comprises a network of facilities, including internet hospitals, that provide online health services, including outpatient appointments, remote consultations, diagnoses, prescriptions, drug delivery and health management.
- Alibaba's AliHealth and JD.com's JD Health: Both provide online medical consultation services. Each has an extensive network of doctors.

Leading companies in the e-health sector have also expanded offline. Both Tencent's Doctorwork and AliHealth provide online-to-offline (O2O) medical care networks.

Investors in the sector are also expanding their interests in artificial intelligence (AI). Ping An has developed an AI-powered diagnosis and treatment system, while Tencent is building a medical AI portfolio, including its AI Medical Innovation System (AIMIS) Medical Image Cloud and an AIMIS Open Lab to help incubate medical AI applications.

Prescribing and Dispensing

Prescribing Influences

Hospital Prescribing

Driven by hospital financing mechanisms and patient expectations, overtreatment and irrational prescribing have traditionally been widespread in the sector. Since the launch of the 2009 reform program, a raft of government policies has sought to address the problem. Coupled with growing scrutiny of physician behavior, this has begun to have a significant impact on prescribing activity. Hospital physicians have been the main target of most measures designed to tackle overprescribing and to encourage the adoption of more rational, cost-conscious prescribing choices.

A National Prescription Guide has been in place since 2010, and by 2020, the NHC had drawn up clinical pathways for over 200 disease conditions, in a bid to improve care quality, reduce risks, increase efficiency and control costs. To date, poor compliance has limited the impact of these initiatives on key metrics such as hospital stay length and treatment costs, however.

Other developments have had a bigger impact on hospital prescribing, including public hospital financing reforms and drug budget controls; reimbursement claims controls; restrictions on antibiotic prescribing; and anti-corruption measures.

Key Impacts on Hospital Prescribing Trends

Measure	Aims and Impact
Public hospital financing reforms and drug budget controls	Hospital drug mark-ups were eliminated nationally in 2017 to reduce dependence on income from drug dispensing and discourage profit-driven prescribing. Caps on the hospital drug-to-revenue (DTR) ratio were imposed, cutting the share of hospital budgets allocated to medicines from around 50% to 30% or less. DTR caps were relaxed from 2019, with innovative new drugs added to the NRDL excluded from calculations. While the 30% DTR ratio is no longer enforced, it is often still loosely taken into consideration. Cost-containment measures that affect prescribing practice at hospital level include: • Quarterly or monthly value or volume caps on costly drugs, with prescribers required to switch to generics when ceilings are reached. • Caps on the value of individual prescriptions. • Limits on the duration of prescriptions (typically 1-2 weeks, although limits were relaxed following the onset of the COVID-19 pandemic). • First- and second-line prescribing restrictions. • Specialist-only prescribing restrictions for some high-cost drugs. • Ceilings on prescribing costs for inpatients. • Limits built into prescribing software, imposing daily caps on the value of drugs prescribed by individual physicians.
Reimbursement claims controls	Health insurance authorities have implemented sophisticated claims processing software, enabling closer scrutiny of prescribing and reimbursement claims. With hospitals obliged to absorb the cost of non-reimbursed claims, many have set up their own internal prescription monitoring initiatives.

Measure	Aims and Impact
Antibiotic prescribing regulations	2012 regulations to combat widespread overprescribing classified antibiotics into three categories ('restricted', 'unrestricted', or 'subject to special management') for prescribing purposes, and imposed tier-related limits on the range of antibiotics that may be stocked by public hospitals. A five-year national action plan to contain antimicrobial resistance, launched in 2016, outlined measures to tighten supervision of antibiotic production, sales and use, and called for the establishment of a national surveillance center to monitor antibiotic consumption.
Anti-corruption measures	Anti-corruption compliance requirements for providers and prescribers were published in 2013, while controls on medical representatives were tightened in 2017. Further controls on promotional activity – including a sales rep registration scheme – were implemented in 2020, while a new anti-corruption drive was launched in July 2023.

Source: IQVIA research

More recently, volume-based procurement (VBP) contracts have begun to exert a growing impact on prescribing choice in public hospitals, while the rollout of flat-sum reimbursement mechanisms for hospital inpatient activity will act as an additional driver of more cost-conscious prescribing.

Reassessing Hospital Prescriber Options

Rules governing hospital formulary listings – including limits on the number of drugs that may be stocked by public facilities – have constrained access to new drugs (see Hospital Formularies). They have also mitigated against the listing of more pediatric drugs and new treatments for age-related conditions. The NHC announced plans to relax existing formulary listing rules in July 2022, primarily to encourage more widespread listing of pediatric drugs. The move was in line with targets outlined in the 14th Five-Year Plan, which include improvements in healthcare provision for children.

Explicit regulations governing off-label drug use were introduced for the first time in March 2022. The law stipulates that, subject to patient consent, drugs may be used off-label when no other effective treatment is available. Off-label use must be backed up by scientific evidence, however.

Patient Finances

Despite progressive improvements in reimbursement coverage under the BMI schemes, patient finances still often affect prescribing decisions, especially where high-cost specialty drugs are concerned. Even when these products are listed on the NRDL, percentage-based co-payments are beyond the financial means of many low-income patients, who may be prescribed cheaper – often less effective – drugs as a result.

This is a particular issue for cancer drugs, with evidence of a correlation between patient incomes and the frequency with which newer, targeted oncology medicines are prescribed. Patient co-payments can also affect treatment options for chronic diseases, for which some cities have set annual reimbursement caps. These effectively translate into significantly higher patient co-payments.

CHC Prescribing

Prescribing in CHCs is dominated by the selection of fully reimbursed drugs on the EDL (see Essential Drug List). Policymakers originally sought to limit prescribing in CHCs completely to EDL products. While the use of essential drugs is still prioritized in CHCs, restrictions on prescribing outside the list have been relaxed over time, partly to encourage more widespread patient use of primary care facilities, but also to facilitate downward referral of patients from higher-tier hospitals to CHCs under the tiered diagnosis and treatment system.

Impact of the COVID-19 Pandemic on Prescribing

The COVID-19 pandemic drove significant shifts in prescribing activity, triggering a sharp increase in remote consulting and electronic prescribing, and a rise in the number of long-term scripts written for chronic patients. Some temporary prescribing rules implemented in 2020 were made more permanent in May 2021, when the NHC issued regulations permitting physicians to write initial scripts for periods of up to four weeks (up from one week), and subsequent scripts for up to 12 weeks. In-person consultations must occur before an initial long-term prescription is issued, however.

Essential Drug List

Establishing an essential drug system was part of the 2009 healthcare reforms. The first national Essential Drug List (EDL) was published in 2009. It has been updated (and expanded) twice since then, in 2012 and 2018. Management of the EDL is part of the NHC's remit. Essential drugs, which form a subset (List A) of products on the NRDL, are defined as those meeting basic healthcare demands, in suitable dosage forms, with reasonable prices and guaranteed supply. Their use is reimbursed in full by BMI funds.

The current version of the EDL was published in October 2018. While some older products were removed, the 2018 update involved a net increase of 165 in the number of drugs on the EDL, which rose to 685, more than double the figure of 307 on the initial 2009 list. Notable additions included around a dozen cancer drugs, more than 20 pediatric products and Gilead Sciences' direct-acting hepatitis C antiviral, Epclusa (sofosbuvir + velpatasvir). Another notable feature of the 2018 update was the inclusion of some drugs that had either yet to be listed in the NRDL, or which were on List B of the NRDL at the time.

2018 Expansion of the EDL

Product Type	Number of Active Ingredients	
	2012	2018
Western drugs/biologics	317	417
TCMs	203	268
Total	520	685

Source: IQVIA research

State Council Plans to Strengthen the Essential Drugs System

The State Council signaled its intention to strengthen the essential drugs system in a September 2018 circular, which advocated a more dynamic approach to updates of the EDL, and called for an increase

in the use of essential drugs. In November 2021, the NHC published draft measures for the administration of the EDL, in which it confirmed plans to adopt most of the measures outlined in the 2018 State Council Circular. In May 2022, the NHC's proposals were included in a draft NMPA regulation designed to implement Drug Administration Law (DAL) reforms. By the end of 2024, the EDL had still not been updated, however.

State Council 2018 Plans to Strengthen the Essential Drugs System

Measure	Aims and Impact
Management of the EDL	A more dynamic approach to management of the EDL, with full updates at least once every three years, and with new, more effective treatments added between updates where appropriate. Drugs to be selected for inclusion based on their cost-benefit profile as well as safety and efficacy. The range of drugs on the EDL should be sufficient to meet all major clinical needs, and should take into account the requirements of specific populations (notably children, but also the elderly).
Increased use of essential drugs	Provinces should set thresholds for the use of essential drugs by hospitals and health centers. Incentives for physicians to prioritize essential drugs should be introduced. Local governments should prioritize the use of essential drugs to manage chronic diseases.

Source: IQVIA research

National Reimbursement Drug List

The National Reimbursement Drug List (NRDL) provides a framework for the reimbursement of drugs prescribed to beneficiaries of the basic medical insurance schemes. NRDL listings are by international non-proprietary name (INN), and include details of indications, dosage forms, strengths and prices. The list is split into two separate schedules, with placement determining the level of reimbursement (see also Pricing and Reimbursement: Reimbursement).

- List A – Clinically essential drugs that are used widely and are available at low prices (reimbursed in full).
- List B – More expensive drugs deemed to be clinically essential (reimbursed partially).

Established in 2000, the NRDL was updated in 2004 and 2009. Manufacturers then had an eight-year wait before the next revision, which was finally announced in February 2017. By then, only 20% of 384 new drugs launched between 2003 and 2015 had been included in the NRDL, although some provinces included additional new drugs on provincial reimbursement drug lists (PRDLs) (see Harmonization of Provincial Lists, below).

Expansion of the NRDL

The creation of the National Healthcare Security Administration (NHSA) in 2018 paved the way for a more dynamic approach to updating the NRDL. The new agency, which oversaw an update of the list in 2019, indicated in July 2020 that it would be revised on an annual basis in future. It has implemented a range of measures designed to mitigate the impact of a more expansionary approach to NRDL listing on BMI finances, including:

- Volume-based procurement (VBP) of more multisource List A drugs.
- Negotiation of acceptable prices for costly drugs identified as candidates for inclusion in List B.
- A more proactive approach to de-listing of older drugs.

Between 2017 and 2023, over 550 drugs were admitted to List B, while some 200 products deemed to be of limited clinical value were removed from the NRDL.

New Drugs Added to the NRDL, 2017-2023

Year	Number of Drugs Added to the List
2017	36
2018	17
2019	70
2020	119
2021	74
2022	111
2023	126
Total	553

Source: IQVIA research

Harmonization of Provincial Lists

Until 2019, up to 15% variance was permitted between the content of provincial reimbursement drug lists (PRDLs) and the NRDL, to reflect local economic conditions and patient needs. In reality, provincial listing policies had resulted in the emergence of more substantial variations in coverage.

Some of the country's least prosperous provinces struggled to fund NRDL coverage requirements, delaying PRDL updates in the wake of NRDL expansion and imposing substantive patient co-payments for costly List B drugs. By contrast, some more wealthy provinces had pursued the progressive expansion of their PRDLs in the absence of more frequent NRDL updates, and had imposed lower co-payments for high-cost drugs.

In a bid to harmonize patient access to subsidized medicines, the NHSA announced in July 2019 that provincial reimbursement lists would in future have to mirror fully the NRDL. While local authorities were given three years to comply, some adjusted their reimbursement lists immediately after publication of the new NRDL in November 2019, and subsequent updates have been implemented within months by most provinces.

Impact of the NRDL on Prescribing

NRDL listing has a positive impact on demand for most listed products, which are being prescribed more widely following inclusion on the national list. This is reflected in the performance of some recently listed drugs, with the impact of price cuts required to secure NRDL listings outweighed by subsequent volume increases, resulting in sometimes dramatic revenue gains. Hospital formulary

listings, however, can still act as a barrier to the prescription of costly NRDL drugs (see Hospital Formularies).

Demand for some newly listed drugs was blunted by the COVID-19 pandemic and the repeated localized lockdowns implemented to limit its spread. The removal of the zero-COVID restrictions in December 2022 laid the foundations for a return to more normal patterns of patient demand.

Hospital Formularies

Gaining access to public hospital formularies is essential for product usage in individual facilities, even where drugs are included on the NRDL and have won tenders. The process can be lengthy, however, and is lacking in transparency.

Formulary committees (*yaoshi weiyuanhui*) usually include the hospital director, the chief pharmacist, heads of department, specialists and hospital administrators. Most meet on a quarterly basis, but gaps between meetings in some facilities may be considerably longer. Some hospitals, especially those in major cities, only consider new formulary listings once a year. In recent years, measures have been implemented encourage more frequent meetings to assure inclusion of newly NRDL-listed drugs (see NRDL Listing as a Driver of Formulary Access, below)

Listing decisions take price and reimbursement status into consideration, as well as efficacy and safety. The listing process is not transparent, however, and committee membership is prone to regular changes in an effort to prevent bribery. As a result, identifying key hospital contacts can be difficult for drug sponsors.

Most hospitals have traditionally listed two brands per molecule (usually the off-patent originator or a leading domestic branded generic, and a cheaper local generic), making it difficult, if not impossible, for a third brand to gain access to the formulary. Hospitals switch between the brand and the generic or remove the brand from the formulary completely when budgets are under particular pressure.

Since the implementation of VBP, off-patent originators (OPOs) and other brands that do not win VBP contracts have found it increasingly difficult to maintain hospital formulary listings. Most public facilities now list one or two VBP winners, with fewer hospitals willing to maintain non-VBP brands on their formularies.

NRDL Listing as a Driver of Formulary Access

The number of innovative new drugs added to List B of the NRDL has risen sharply since 2017 (see National Reimbursement Drug List). The upside of NRDL listing for manufacturers – who had invariably given considerable ground on price – was initially limited, reflecting the time taken to update provincial and hospital formularies, and a reluctance on the part of hospitals to prescribe costly drugs. Regulators have since implemented measures designed to maintain interest in the pursuit of NRDL listings, including:

- A mandate for provincial authorities and hospitals to update their formularies swiftly in the wake of NRDL updates.
- The exclusion of drugs in List B of the NRDL from hospital drug-to-revenue (DTR) ratio calculations.
- The relaxation of caps on the contribution of drugs to hospital revenues.

Many hospitals now hold special meetings in the wake of NRDL updates, at which discussions focus purely on the inclusion of new List B drugs on the hospital formulary. While this means formulary listing has become easier for innovative new drugs added to the NRDL, it remains a challenge for newly listed products where a cheaper therapeutic alternative is already listed. Caps on the size of hospital formularies have also posed a growing challenge as the number of new drugs included on the NRDL has increased.

As a result, while manufacturers acknowledge that hospital formulary access has improved, it may still take two to three years for a new drug to obtain listings in all relevant facilities. A June 2023 LEK Consulting analysis, *Hospital Priorities 2022*, showed that only 38% of NRDL drugs were listed automatically by hospitals in 2022. It found that automatic listings were particularly difficult in tier 3 hospitals, and that prescription limits restricted access to some costly NRDL drugs. Acknowledging the impact of hospital formulary management on patient access to reimbursed medicines, regulators have established a 'dual-channel' mechanism, enabling the reimbursement of NRDL drugs with nationally negotiated prices in designated retail pharmacies (see Dispensing).

Formulary inclusion for products that have not achieved reimbursement listing depends on whether a drug is deemed necessary to treat a serious disease, as well as on individual hospital and physician support for the product in question. Few drugs without NRDL listings gain widespread access to public hospital formularies.

Dispensing

Hospitals dominate dispensing activity, generating over 80% of drug sales in 2024, according to IQVIA data. Dispensing through retail pharmacies has been held back by the hospital-led nature of healthcare provision, and by lack of reimbursement for drugs dispensed outside hospitals. Dispensing through retail channels is generally highest in provinces or cities where reimbursement under the BMI schemes is most limited.

Patients purchasing drugs in retail pharmacies must in theory present a prescription for any product not classified as OTC. In reality, the situation varies: while retail pharmacies in major cities tend to demand a physician's script before dispensing prescription-only medicines, regulations are enforced less rigorously in many rural areas.

Government policy favors the progressive separation of prescribing and dispensing (SPD) functions. Measures designed to encourage more widespread dispensing outside the hospital setting include:

- The abolition of hospital drug mark-ups in 2017, and the imposition - no longer enforced, but often still taken into consideration - of drug-to-revenue thresholds as a means of controlling hospital drug budgets. This has encouraged the release of more prescriptions and the emergence of direct-to-patient (DTP) pharmacies focused on the sale of specialty medicines.
- The introduction in May 2021 of a 'dual-channel' policy, aimed at enabling patients to access reimbursement for some NRDL drugs dispensed in designated retail pharmacies.
- The implementation in December 2022 of explicit national rules governing online prescription drug sales (see Healthcare Provision: E-Health).

The COVID-19 pandemic triggered further shifts in dispensing, reflecting sharp increases in remote prescribing and online drug purchasing.

The Dual-Channel Policy

In policy guidance issued in May 2021, the NHSA and the NHC confirmed that the dual-channel initiative would be applicable to NRDL drugs with negotiated prices which are of high clinical value, meet urgent clinical needs and have limited therapeutic alternatives. To ensure the success of the initiative, the guidance called for:

- Provincial healthcare security administrations to draw up a dual-channel drug list (DCDL), based on an evaluation of local clinical needs.
- Efforts by provincial authorities to increase the number of qualified designated retail pharmacies, encourage stocking of DCDL drugs by designated retail pharmacies, and ensure that DCDL products were reimbursed by hospitals and pharmacies at the same rate.

Early efforts to establish dual-channel reimbursement for NRDL drugs have encountered a number of barriers. These include a lack of financial incentives for designated retail pharmacies to stock affected products, and issues around hospital-prescribing software, which can prevent doctors from issuing scripts for drugs that do not feature on the hospital formulary.

Pricing and Reimbursement

Pricing System

Pricing policy was overhauled in 2015. Before then, maximum retail prices were set at national level for all fully reimbursed drugs (those on the EDL and List A of the NRDL) and were subject to periodic reviews and cuts. Provincial pricing authorities were responsible for setting the price of drugs on List B of the NRDL and of any other products on their respective provincial EDLs. 'Independent pricing' was available for off-patent originators and originator TCMs, allowing premium prices for those products on the basis of superior efficacy, quality or safety over generic alternatives. All non-reimbursed drugs were free from price control, although manufacturers of unsubsidized patented drugs had to register prices with regulators prior to launch.

The relaxation of price controls in 2015 prompted a shift to a largely market-based approach, although different pricing principles apply to particular product types. Significantly, this saw the abolition of the 'independent pricing' category. As a result, patent-expired originators have been forced increasingly to compete directly on price with local generics for procurement contracts (see Healthcare Provision: Drug Procurement).

Companies must file prices with regulators at launch. For imported products, information on prices in other countries must be provided. Public hospitals cannot begin purchasing new drugs until they have been listed on provincial online procurement platforms.

Pricing Regulations by Product Type

Product Type	Approach
Multisource drugs reimbursed under the BMI schemes	Held in check by provincial tendering and volume-based procurement (VBP) at national and provincial level. May eventually be subject to explicit reference pricing (see Payment Standards).
Patented drugs and other single-source products on the NRDL	While list prices are free from regulatory control, actual prices of high-cost patented or single-source drugs on the NRDL are determined via negotiations between the NHSA and manufacturers. When initial NRDL listing agreements expire, prices are either renegotiated, or are adjusted in line with algorithms determined via reference to their expected and actual impact on BMI budgets (see National Price Negotiations).
Non-reimbursed blood products and those covered by national disease control programs	Prices are determined via tendering or procurement negotiations.
Narcotic analgesics and class I psychotropic drugs	Subject to price controls, with ex-factory prices set by the authorities. Maximum distribution margins for these products, previously set by regional authorities, were phased out in 2020.
All others	Manufacturers are free to set their own prices.

Source: NDRC et al, *Opinions on Promoting Drug Price Reform*, May 2015; IQVIA research

Responsibility for pricing policy, which was previously split between several government agencies, has been concentrated in the hands of the National Healthcare Security Administration (NHSA) since its establishment in 2018. The NHSA has a broad remit spanning pricing and reimbursement policy, price regulation and drug procurement. While it has pursued further rapid expansion of the NRDL, the

NHSA has implemented a raft of measures designed to limit the impact of a more reimbursement expansionist strategy on BMI finances.

Costs associated with the reimbursement of drugs for BMI affiliates have risen significantly over the 15 years, reflecting expanded enrolment, the merger of the urban and rural residents' schemes and, since 2017, expansion of the NRDL, which includes a rapidly growing number of high-cost innovative new drugs. Coupled with concerns about the financial burden imposed by medicines on patients, this has driven a more aggressive approach to drug price control.

Key Measures to Constrain Drug Prices

Constraint	Impact
Centralized volume-based procurement (VBP)	National volume-based tenders for the procurement a growing number of multisource drugs available in bioequivalence-approved generic form encourage aggressive bidding at low prices. Provincial and multi-provincial VBP contracts are also increasingly prevalent and have included the purchase of some generics without bioequivalence certification (see Healthcare Provision: Drug Procurement).
National price negotiations	National price negotiations are used to lever sharp reductions in the price of innovative new drugs in return for inclusion on the NRDL (see National Price Negotiations).
Hospital drug procurement mechanisms	Tender-based procurement mechanisms employed at provincial, city, GPO and individual hospital level ensure that drug prices in hospitals are kept in check (see Healthcare Provision: Drug Procurement).
Price transparency and monitoring	The State Council has called on regulators to step up drug price monitoring, and for enhanced price transparency by linking ex-manufacturer drug price information to a centralized procurement platform.
Tariff and tax cuts	Import tariffs and VAT applied to more than 100 imported cancer drugs were reduced in 2018, with cuts (averaging approximately 5%) in the price of more than a dozen high-profile brands following talks with the NHSA. Import tariffs for some APIs were reduced or abolished at the beginning of 2019.

Source: IQVIA research

NRDL price negotiations and the progressive expansion of national VBP have driven substantial reductions in the prices of both new drugs reimbursed by BMI schemes and older products available in bioequivalent-certified generic form. This has led to concerns about the impact of current policies on the sustainability of the domestic pharmaceutical industry, and on the willingness of foreign companies to bring innovative new drugs to market in China. Acknowledging those concerns, the NHSA has begun exploring measures to facilitate more rapid market access for new drugs.

National Price Negotiations

Keen to broaden access for BMI scheme affiliates to innovative new drugs – but determined also to limit the impact of more widespread reimbursement on BMI finances – regulators piloted national price negotiations in 2016, paving the way for the inclusion of three innovative medicines in List B of the NRDL in February 2017 (see Reimbursement). Based on experience from that pilot, selection and negotiating procedures have since been refined progressively. Simultaneously, measures have been implemented to enhance the attraction of NRDL listings to manufacturers. These include:

- Regular, annual updates of the NRDL, offering rapid market access for innovative new drugs.
- The possibility for manufacturers to apply directly for NRDL listings, and to make the case for the inclusion of their products.
- A mandate for full compliance by all provinces with the NRDL, and for hospitals to update their formularies immediately after the publication of NRDL revisions (see Prescribing and Dispensing: National Reimbursement Drug List).

Negotiation Procedures

Manufacturers whose products are on the shortlist of candidates for inclusion on the NRDL are invited to negotiate reimbursement prices for costly innovative new drugs on that list. The nature and extent of negotiations – which were initially inflexible and lacking in transparency – has been refined progressively.

Before negotiations begin, representatives from NHTA expert committees conduct a cost-effectiveness analysis (CEA) and a budget impact analysis (BIA). Regulators also take account of:

- International prices, benchmarking prices for the drug in a basket of 12 other markets (Japan, France, Germany, Italy, Spain, the UK, Canada, South Korea, the US, Australia, Turkey, and Taiwan)
- Local comparator prices, including the relative cost of alternative therapies already on the NRDL.

Based on these analyses, the NHTA arrives at an ‘envelope price’ for the drug. Previously, where prices offered by manufacturers were more than 15% above the envelope price, negotiations were terminated immediately. Since 2022, however, companies have been given the chance to submit a revised second offer.

NHTA Approach to Price Negotiations

Following its establishment in 2018, the NHTA’s initial focus was on securing the lowest possible price for drugs admitted to the NRDL. While this mitigated the impact of listings on BMI budgets, it also limited the number of innovative new drugs admitted to reimbursement. Strategies employed by the NHTA to drive down prices included:

- **Candidate choice:** The preference was for drugs with competitors nearing the market and original brands approaching patent expiry.
- **International reference pricing:** Regulators referenced prices in a basket of 12 countries during the calculation of envelope prices. The inclusion of low-priced markets in that basket limited envelope prices, with Turkish prices emerging as a *de facto* benchmark for patented drugs seeking access to the NRDL.
- **Competitive negotiations:** Employed in 2019, when regulators announced they would list only two of six direct-acting antiviral drugs for the treatment of hepatitis C, excluding other drugs in the class from reimbursement for at least two years.
- **Non-disclosure clauses:** Enabling manufacturers to preserve list prices, which encouraged them to offer more substantial discounts.

- **Re-negotiation of NRDL listings:** Employed for the first time in 2019, prompting a further average decline of more than 26% in the price of 27 drugs added to the NRDL in 2017 (see Renegotiation of NRDL Listings).

The NHTA's increasingly aggressive approach to price negotiations drew criticism from both patient advocates, who said it was denying them access to life-changing or life-transforming drugs, and from manufacturers, which warned that it posed a threat to both foreign investment and local industry innovation.

Acknowledging those criticisms, regulators have implemented a number of changes to procedures governing the expert review of NRDL candidates, and to rules for both establishing envelope prices and determining the outcome of price negotiations. Notable changes introduced in 2022 included more open, flexible price negotiation procedures for innovative drugs, the introduction of competitive bidding for multisource drugs, and the establishment of an algorithm to determine price adjustments for products already on the NRDL (see Renegotiation of NRDL Listings).

Further significant changes, applied in 2023, included value ratings for new drugs, the application of explicit ICER thresholds to some innovative products, and adjustments to the algorithm used to determine price adjustments for products already on the NRDL (see Health Technology Assessment).

Outcome of Negotiations

National price negotiations have paved the way for the inclusion of more than 600 new drugs on the NRDL since the approach was first piloted in 2016. Since 2019, this has become an annual process, during which negotiations on shortlisted candidates are conducted towards the end of the year. Where these are successful, new drugs are included on an updated version of the NRDL that comes into effect at the beginning of the following year.

The number of NRDL listing recommendations or applications received by the NHTA has risen progressively in recent years. A substantial proportion of drugs on the longlist do not progress to the price negotiation stage, however, while not all negotiations are successful.

National Price Negotiation Rounds 2016-2022

Round	NRDL Listings	Details
2016	3	Pilot, in which 30 companies were invited to enter into national price negotiations. Agreements were reached on just three products (including two multinational brands). Discounts of between 53% and 67% were secured for those drugs, which were added to List B of the NRDL in February 2017.
2017	36	Refined selection and negotiation procedures led to the agreement of acceptable prices for 36 of 44 selected drugs, which were added to List B of the NRDL in July 2017. Multinational brands accounted for 31 of the 36 products, which included 15 patented cancer drugs. Negotiated discounts averaged 44%, while the maximum discount was 70%.
2018	17	Negotiations with manufacturers of 18 cancer drugs, led by the newly created NHTA, secured discounts averaging almost 57% on 17 products, which were added to List B of the NRDL in October 2018. Ten of the 17 products added to the NRDL had been approved in China since the beginning of 2017.
2019	70	Acceptable prices were negotiated for 70 of 119 high-cost drugs identified by regulators as candidates for inclusion in the NRDL, of which 47 were multinational brands. Discounts averaged 60.7% but reached 85% for

Round	NRDL Listings	Details
		hepatitis C drugs, which were the subject of what amounted to a competitive bidding procedure.
2020	119	Acceptable prices were negotiated for 119 of 162 candidates (including 96 products with exclusivity), in respect of which price cuts averaged 50.6%. The share of listings granted to local companies rose from 35% to 44%, reflecting their willingness to cut prices to lower levels than multinationals.
2021	74	74 (63%) of 117 candidates that entered into negotiations were successful. Manufacturers of those products agreed to price cuts averaging 61.7% in return for their inclusion on the NRDL. Forty-six drugs added to the NRDL in January 2022 had been approved by Chinese regulators within the previous two years.
2022	111	NRDL listings for 111 drugs were approved by the NHSA in January 2023, with manufacturers agreeing to cut prices by an average of 60.1%. Twenty-three of those drugs had been approved by regulators since the beginning of 2022. For the first time, 17 drugs were listed following competitive bidding.
2023	126	NRDL listings for 126 of 148 shortlisted drugs were approved in December 2023, triggering their inclusion in the NRDL from 1 January 2024. 121 drugs were listed following successful price negotiations, which triggered price cuts averaging 61.7%. Five drugs were listed following competitive bidding.
2024	91	NRDL listings for 91 out of 117 shortlisted drugs were completed in November 2024, with 89 products included on the list following negotiation/bidding procedures that reduced prices by 63% on average. Two products were incorporated directly into the NRDL, after being selected for national centralized procurement. The listings took effect on 1 January 2025.

Source: IQVIA research

Renegotiation of NRDL Listings

Negotiated NRDL listings are valid for an initial period of two years, after which listings are reviewed, and prices may be adjusted. In 2019, the NHSA sought to renegotiate the price of 31 drugs that had been added to the NRDL in 2017. This led to a further average decline of 26% in the price of 27 drugs. Manufacturers of four other products refused to accept additional cuts, prompting their removal from the NRDL.

Demands by regulators for additional price cuts were particularly severe when manufacturers were pursuing reimbursement of new indications for NRDL-listed drugs, and where new competitors to a listed drug had been launched. With more manufacturers threatening to remove products from reimbursement, rather than accept further substantial price cuts, the NHSA's aggressive approach to renegotiation of initial listings raised the prospect of delisting for a growing number of innovative medicines.

To prevent that from happening, regulators introduced a new approach to NRDL listing reviews in 2022. This involved the use of an explicit algorithm to determine:

- Whether or not the price of a recently listed drug should be renegotiated.
- For drugs not subject to renegotiation, whether, and by how much, their price should be reduced.

Under the new rules, drugs scheduled for renewal of NRDL listings are split into three groups:

- Group 1 – Drugs moved permanently into List B.

- Group 2 – Drugs subject to simplified renewal procedures, with price adjustments determined via reference to an explicit algorithm.
- Group 3 – Drugs subject to price re-negotiations.

Products in Group 2 are split into two subgroups: those for which no additional indications have been reimbursed; and those with newly reimbursed indications.

Simplified Renewal Pricing Rules (2022)

Category	Price Adjustment Rules
No indication expansion	Price adjustments are calculated in accordance with actual budget impact figures and the ratio between actual and agreed budget impact figures. Where BMI spending on a drug is 10% or less above anticipated levels, prices are not adjusted. Where BMI spending is 10-200% higher than anticipated, prices are adjusted downwards. Cuts may vary: from 5% for drugs with a modest overall budget impact and sales that have not exceeded anticipated levels by more than 40%; to 25% for drugs with a high overall budget impact (more than RMB4 billion) and sales that have exceeded anticipated levels by 70-100%. Where spending is more than two-times higher than anticipated, the price of a drug must be renegotiated, regardless of its actual impact on BMI budgets. For drugs that have been on the NRDL for more than four years, the magnitude of price cuts is halved.
Indication expansion	Prices are adjusted in accordance with the expected additional budget impact resulting from indication expansions over the next two years. Where reimbursement spending is expected to rise by 10% or less, no price cut is required. Where reimbursement spending is expected to rise by 11-100%, prices are adjusted downwards by between 5% and 25%, depending on the extent of the anticipated spending increase and the value of BMI reimbursement spending on the drug. Where the reimbursement of new indications is expected to increase BMI spending on the drug by more than 100%, the price must be renegotiated. For drugs that have been on the NRDL for more than four years, the magnitude of price cuts is halved.

Source: IQVIA research

In a further change, announce in the run-up to the 2023 round of negotiations, drugs listed via negotiation that have been on the NRDL for more than eight years will be included automatically in the regular List B, with no need for further price revisions.

Reimbursement

Drug reimbursement available to BMI affiliates is determined by the NRDL, which was established in 2000. Historically, updates to the list were sporadic, while provinces managed their own drug reimbursement lists, which varied widely in scope. The NRDL has been updated at least once a year since 2017. With provincial lists now also required to mirror the national version, NRDL listings determine BMI affiliate access to pharmaceutical reimbursement.

Listings are by international non-proprietary name (INN), and include details of indications, dosage forms, strengths, and prices. The list is split into two separate schedules, with placement determining the level at which a drug is reimbursed. Products contained in List A must be reimbursed in full. Those

in List B, which comprises more expensive products, are subject to partial reimbursement – generally at rates of between 70% and 90%, depending on the state of BMI finances in a particular city or province.

With patients having to cover the remainder of List B product costs, co-payments can pose a significant challenge, especially with more expensive innovative new drugs. In an effort to limit out-of-pocket spending on these products – and to mitigate the impact of their inclusion in the NRDL on BMI finances – regulators have pursued an aggressive approach to price negotiations, which ultimately determine whether or not single-source products are added to the reimbursement list (see National Price Negotiations).

Structure of the NRDL

List	Conditions
List A	Products deemed to be clinically essential, which are widely used, of proven efficacy, and available at low prices. Full reimbursement of products on List A is mandatory for all provincial and municipal BMI schemes.
List B	More expensive drugs regarded as clinically essential. List B products are partially reimbursed, with patients making percentage-based co-payments. Provincial lists have to mirror the NRDL exactly. Provinces retain the freedom to set the level of reimbursement for List B drugs, however, and subsidy levels vary, reflecting the relative state of BMI finances in individual provinces or cities.

Source: IQVIA research

NRDL Selection Process

Historically, NRDL listings reflected the outcome of deliberations by government-appointed expert panels. The selection process has been refined progressively over time, however, with manufacturers now permitted to apply directly for inclusion of drugs on the list. Candidates for inclusion must meet at certain eligibility criteria, which are defined for each update, and usually include:

- Recently approved drugs authorized in the previous five years containing active ingredients new to China.
- Drugs with significant changes in indication or function.
- Drugs included in the latest (2018) version of the Essential Drugs List.
- Drugs included in the list of priority pediatric products.
- New generics.
- Drugs indicated in the treatment of rare diseases.

The selection procedure has evolved with each update since 2017. The NHSA conducts an initial review of applications, after which candidates are assessed by two expert groups – a ‘comprehensive group’ comprising pharmacists, hospital directors and health economics/outcomes research (HEOR) experts; and a ‘clinical group’ comprising heads of medical associations.

The review process involves an integrated, joint review by these two expert groups, during which products are scored against a range of criteria, including efficacy, safety, innovation and economic

impact. Scores determine whether or not a drug is included on an eventual shortlist for national price negotiations. As well as generating a shortlist of candidates for inclusion in the NRDL, expert groups are responsible for the choice of comparator against which new drugs are benchmarked.

Routes to NRDL Listing

Costly patented drugs on the NHTSA shortlist undergo price negotiations between the Administration and manufacturers. Where these lead to agreement on a price acceptable to the NHTSA, the drug is added to List B of the NRDL. Negotiated listing decisions are valid for an initial period of two years, after which the terms of inclusion on List B of the NRDL are re-negotiated, often triggering further price cuts (see National Price Negotiations).

Multisource drugs may gain access to the NRDL via one of two routes:

- Direct access – where the manufacturer of a drug wins a national VBP supply contract (see Healthcare Provision: Drug Procurement), the product is included on the NRDL.
- Competitive bidding – where several companies are pursuing access to the NRDL for products based on the same active ingredient, applicants are invited to participate in a competitive bidding procedure. This determines whether, and at what price, the drug will be included on the NRDL (see Competitive Bidding, below).

The selection process takes place over a period of several months. In recent years, price negotiations for shortlisted drugs have been conducted in the final quarter of the calendar year, with reimbursement of drugs added to the NRDL commencing in January of the following year.

Competitive Bidding

The competitive bidding procedure has been used to determine NRDL access for multisource drugs since 2022. Following the calculation by NHTSA experts of an envelope price, multiple manufacturers are invited to submit price offers. Where all offers are above the envelope price, the drug will not be listed. Where one or more offers are equivalent to or below the envelope price, the drug is added to List B of the NRDL.

Where the process results in drug listing, the lowest price offered becomes the BMI reimbursement payment standard for the molecule (see Payment Standards). The payment standard may not fall below 70% of the envelope price. Where a drug is admitted to the NRDL on this basis, versions with prices higher than the payment standard are subject to co-payments, with patients covering the difference. Companies that participated in the NRDL bidding process may not sell their product at prices higher than their bid offer during the validity period (provisionally set at two years).

Recent NRDL Updates

Patient access to reimbursement for innovative new drugs has broadened progressively since the introduction of national price negotiations as a route to NRDL listing for these products in 2017. Since then, more than 550 new drugs have been included in the national reimbursement list on the basis of price negotiations or competitive bidding. With fewer than 200 products removed from the list during that period, the NRDL has expanded progressively. Following the most recent update, which took effect on 1 January 2025, the list included 3,159 products, of which 1,765 were western medicines and 1,394 TCMs.

Expansion of the NRDL (2017-2023)

Year	Details
2017	February – The first substantial update of the NRDL since 2009 saw 339 drugs approved for reimbursement. Viread (tenofovir), Iressa (gefitinib) and Conmana (icotinib) were added to List B following the successful completion of national price negotiations for the three products in 2016. July – A further 36 high-cost drugs (including 15 cancer treatments) were added to List B, following national price negotiations.
2018	October – A further 17 high-cost cancer drugs were included in List B, following national price negotiations between manufacturers and the newly established NHSA.
2019	August – In the first comprehensive update presided over by the NHSA, 148 drugs were added to the NRDL, including 36 products indicated for the treatment of chronic NCDs, five rare disease treatments and 38 paediatric drugs. In the first major round of de-listings, 150 drugs were removed from the NRDL. November – Following a further round of national price negotiations, 70 more high-cost drugs were added to List B. Negotiations on 58 other candidates for inclusion were unsuccessful.
2020	December – A further 119 innovative new drugs were admitted to List B, including 96 drugs with exclusivity. Price negotiations for a further 43 candidates were unsuccessful. Twenty-nine drugs were removed from the NRDL.
2021	December – A further 74 products were added to List B, 67 of which were branded drugs with no generic equivalent. Eleven drugs deemed to be of limited clinical value were removed from the NRDL.
2023	January – The 2022 update was delayed by disruption associated with the COVID-19 pandemic. An updated version of the NRDL, containing 111 products included in List B for the first time, was eventually published in January 2023. Reimbursement of those products commenced in March 2023. Negotiations on a further 22 shortlisted candidates were unsuccessful. December – The 2023 update, which was published in mid-December, saw a record 126 new drugs included on List B of the NRDL, of which 121 were admitted to reimbursement following the successful conclusion of national price negotiations. Only 22 of 148 products on the NHSA shortlist did not feature on the updated NRDL, equating to a negotiation success rate of 85%.
2024	November – The 2024 update added 91 new drugs to the list, including 89 that were listed following successful negotiation/bidding procedures. A total of 117 drugs not previously listed on the NRDL were shortlisted to enter negotiations, of which 89 were listed, giving a success rate of 76%.

Source: IQVIA research

Reimbursement Beyond the NRDL

Supplementary and private health insurance plans can provide subsidized access to drugs not listed on the NRDL. Alternatively, they may help to limit patient contributions to the cost of expensive products reimbursed partially by BMI schemes.

While critical illness plans offering lump-sum payments following diagnosis have traditionally dominated private payer offerings, other, more sophisticated options have begun to emerge. These include city-level schemes (*huiminbao*) that provide supplemental coverage for high-cost specialty medicines; so-called ‘million dollar’ plans that offer broader coverage in return for higher premiums and deductibles; and specialty drug plans (see Healthcare Provision: Health Insurance).

These plans have had a limited impact on sales to date, with manufacturers pursuing inclusion on supplementary or private health plan formularies primarily in a bid to generate patient and prescriber exposure to new drugs, rather than as a source of significant revenues. In the long term, however, these sectors may emerge as drivers of more substantial prescription volumes for drugs outside the NRDL.

Payment Standards

Introducing the government's 2015 pricing reforms, policymakers signaled their longer-term plans to establish a reference pricing mechanism for drugs reimbursed by BMI schemes. This was understood to involve a single reimbursement price (or reimbursement 'payment standard') for multisource drugs, with patients having to pay the difference between the reimbursement threshold and the price of more costly versions of a drug.

The State Council reiterated calls to establish BMI reimbursement payment standards in 2018, while the NHSA confirmed in 2019 that it planned to implement the proposals. It began to do so in 2022, when rules governing that year's NRDL included a new competitive bidding procedure to determine whether or not multisource drugs would be admitted to reimbursement.

Under the new rules, expert NHSA committees calculate an envelope price for multisource drugs with one or more applications for admission to the NRDL. Listing decisions and payment standards for those drugs are determined as follows:

- Where no manufacturer submits a price offer equal to or below the envelope price, the drug is not listed on the NRDL.
- Where one or more manufacturers submits a price offer equal to or below the envelope price, the drug is added to the NRDL.

The lowest bid acts as the reimbursement payment standard for drugs admitted to the NRDL – subject to a floor price equivalent to 70% of the envelope price. BMI affiliates prescribed more expensive versions of the drug must pay the difference between the payment standard and the price of the dispensed product. Manufacturers of products that participated in the bidding procedure may not subsequently increase their prices during the two-year provisional validity period.

If the multisource drug has multiple indications, the most commonly used indications clinically are selected for bidding. Once bidding is completed, the payment standard for other indications is then determined in accordance with the NHSA's price difference rule.

During the 2022 NRDL update, 40 non-exclusive products passed the NHSA's formality review, 17 of which achieved reimbursement listing via competitive bidding. The average price cut under the procedure was 58%. Pfizer was the only multinational to win access via the procedure for its Ibrance (palbociclib). A further five drugs were admitted to the NRDL as a result of competitive bidding procedures in the 2023 update. Japanese company Taiho was the only foreign company to submit a successful bid (for its trifluridine/tipiracil cancer drug, Lonsurf).

Health Technology Assessment

Policymakers have worked to build local HTA expertise through collaborations with overseas agencies, and to enhance the quality and availability of local cost-effectiveness data through the development of integrated health information platforms.

A China Health Policy and Technology Assessment Network – established in 2016 and recognized by the WHO as an official HTA collaborating center in 2017 – oversaw the conduct of early Chinese HTA studies. These included economic evaluations of antiviral hepatitis B and C therapies and budget-impact analyses for non-small cell lung cancer drugs. The submission of pharmacoeconomic and budget impact data became a mandatory part of the national price-negotiation process in 2018, while a new National Center for the Evaluation of Medicines and Health Technologies was established by the NHC in October of that year.

Updated guidance on the use of pharmacoeconomic analysis in pricing and reimbursement assessments was published in August 2020. The guidance states that, for new drugs, clinical efficacy data from a randomized controlled trial are preferred, if available and applicable. For drugs that have been on the market for some time, effectiveness data from a real-world study should be used where it is not possible to supply new efficacy data, or if data are not applicable.

The NHSA piloted a new ‘value rating’ system during the 2023 NRDL update, allowing innovative new drugs recognized as responding to unmet medical need or offering significant improvements over current therapy the chance to obtain premium pricing. Under the new system, drugs were assigned to one of four categories at the review stage, with the assigned value-based classification determining their price in relation to reference drugs (see Key Developments and Trends: Pricing and Reimbursement: Health Technology Assessments).

ICER thresholds were applied to breakthrough therapies: up to 1.5-times GDP per capita, ranging from 0.5-times to 0.8-times for chronic disease treatments, 0.8-times to 1.0-times for oncology drugs, and up to 1.5-times for rare disease therapies. These are subject to an annual price cap of around RMB500,000. For drugs offering significant clinical improvements over, or equivalence to, current therapies, the ICER thresholds are 0.5-0.8-times GDP per capita.

Rapid HTA reviews are now conducted for all drugs. More detailed assessments apply to drugs identified as candidates for inclusion in the NRDL. For the latter, a two-step process involves:

- Step one – Manufacturers submit data on price, annual expenditure, efficacy, safety and innovation. Expert advisory groups assign value scores to individual drugs and recommend whether they should be added to the NRDL, subject to price negotiations, or rejected for reimbursement.
- Step two – Manufacturers of drugs recommended for price negotiation submit more detailed dossiers, including cost-effectiveness and budget impact analyses. Two health economic experts then undertake an independent assessment of the dossier and recommend a ‘base level’ negotiation price for the drug.

Although procedures have been refined over time, the potential impact of a reimbursement listing on BMI budgets remains a key factor determining the outcome of price negotiations. A number of challenges to the effective use of HTA remain. These include the availability and quality of local data, a shortage of HTA expertise, and the need for improvements in the transparency of HTA procedures.

Discounts and Margins

Consumer drug prices in China have traditionally been inflated by tariffs, value-added tax, complex distribution channels and, until 2017, the imposition of substantial hospital mark-ups. With products often handled by multiple distributors before reaching hospitals, and with hospital mark-ups set at 15%, overall mark-ups averaged 60-75% for multinational brands and 20-70% for drugs manufactured by local companies.

Prior to the relaxation of price controls in 2015, the system featured regressive ceilings on maximum mark-ups applied to the ex-factory price of reimbursed drugs, covering both wholesalers and retailers. While the ceilings disappeared following the relaxation of price controls, margin structures remained largely unchanged.

The gap between ex-manufacturer and public prices narrowed with the roll-out of the 'two-invoice' policy from 2017, truncating what was traditionally a lengthy distribution chain (see Pharmaceutical Business Environment: Distribution). The abolition of hospital mark-ups in 2017 was also designed to reduce the gap between ex-manufacturer and consumer drug prices, although hospitals have continued to push for discounts.

Regulatory Environment

Regulatory Agency

A major institutional restructuring exercise, designed to eliminate duplication of responsibilities, streamline regulation, and improve co-ordination between individual institutions, saw the establishment in 2018 of a new 'super-agency' known as the State Administration for Market Regulation (SAMR). The SAMR, which reports directly to the State Council, bears ultimate responsibility for regulation of the healthcare and life science industries, among other sectors.

The institutional overhaul also saw the abolition, merger or restructuring of around a dozen existing agencies – including the China Food and Drug Administration (CFDA), which was replaced by a new National Medicinal Products Administration (NMPA). While the CFDA's remit included food products, the NMPA's focus is narrower. Reporting directly to the SAMR, the NMPA is responsible for oversight of the drug, medical device and cosmetics sectors, from development through to regulatory approval, production, distribution and consumption.

Drug Registration

Drug registration, which is the responsibility of the NMPA and its Center for Drug Evaluation (CDE), is governed by the Drug Registration Regulation (DRR). Implemented in 2007, the regulation has been amended significantly since 2015 as part of a drive to cut lengthy approval times and encourage applications for the registration of more new generics and innovative medicines. Coupled with an increase in regulatory resources, this has driven a dramatic reduction in average approval times and encouraged a sharp rise in the number of new drugs being submitted for review. Further change is being driven by China's accession (in 2017) to the International Council for Harmonization (ICH).

Developments since 2015 have included:

- **Revisions to the drug classification system.** Changes implemented in 2016 saw drugs split into five categories for registration purposes, with classification determining the range of data required to support applications and review timelines. The definition of 'new drugs' became highly restrictive, applying only to innovative new chemical entities (NCEs) (class 1) or improved formulations/new indications (class 2) that had never been marketed globally. Two categories of generics were also recognized, including those approved elsewhere but not in China (class 3) and those already approved in China (class 4). Imported drugs – whether innovative or generic – were grouped in a single category (class 5).
- **A major increase in regulatory resources.** In 2015, only 120 CDE staff were responsible for technical reviews of NDA submissions. By 2020, that figure had risen to 1,000, and a further significant increase in reviewer numbers – to 2,000 – is being pursued.
- **Tighter regulation of generics.** New generics are now approved only if their quality and efficacy are consistent with that of the originator. New generic applications must contain pharmacokinetic and Phase III clinical trial data, while subsequent generic submissions must be supported by bioequivalence studies.
- **The establishment of priority review and approval pathways.** The NMPA has begun to grant trial waivers and conduct abbreviated reviews for some innovative new drugs already approved by

regulators in the US, the EU and Japan. The agency has published lists of priority drugs, inviting manufacturers to apply for abbreviated.

Revised Drug Registration Regulation

Amendments to the Drug Administration Law (DAL) are driving more fundamental reform of the DRR. A revised version of the regulation came into effect on 1 July 2020, paving the way for the formal introduction of a marketing authorization holder (MAH) system and ushering in changes to rules governing new drug applications, clinical trials, variations and renewals.

The revised regulation has prompted a further change in the classification system used to determine the information sponsors must provide to support applications for clinical trials and marketing authorizations. It recognizes three registration categories for each of three drug types (small molecule drugs, biologics, and TCMs). For western medicines, categories are as follows:

- **Small molecule products:** Innovative new drugs; improved new drugs; and generics.
- **Biologics:** Innovative biologics; improved new biologics; and 'marketed biological products' (with the latter category including biosimilars).

The revised DRR clarifies timelines for the review of approval applications submitted to the NMPA. These set a target of 200 working days for the completion of normal review procedures. The new DRR also establishes timelines and qualification criteria for expedited review pathways. Priority reviews must be completed within 130 working days, while a reduced time limit of 70 working days applies to drugs that have already been approved outside China and are urgently needed for the treatment of rare diseases.

Priority Review and Approval Pathways

Program	Applicable Drug Types
Breakthrough therapy program	Innovative or improved drugs indicated for the prevention or treatment of diseases that pose a serious threat to life or have a significant impact on quality of life – where no effective treatment exists, or where evidence shows the drug offers significant clinical advantages over existing therapy.
Conditional approval program	Drugs that treat life-threatening conditions for which no effective treatment exists, where early trial data indicate efficacy and potential clinical value. Drugs required urgently for public health where trial data indicate efficacy and potential clinical value. Vaccines required urgently for major public health emergencies.
Priority review program	Urgently needed drugs in short supply, and innovative or improved new drugs for the prevention and treatment of serious infectious and rare diseases. New paediatric drugs or dosage forms. Urgently required and innovative vaccines. Drugs approved under breakthrough or conditional approval programs.
Special approval program	Drugs required in the prevention and control of pending or existing public health emergencies.

Source: IQVIA research

Registration certificates issued for products authorized under the conditional approval pathway note the period for which such approvals are valid, and detail post-marketing surveillance measures required of MAHs. Failure to comply with those requirements may see conditional approvals revoked.

Amid growing concerns that manufacturers were abusing the conditional-approval pathway, proposed amendments to existing rules were published by the NMPA in August 2023. These would impose much stricter qualification criteria for conditional approval status, as follows:

- Where a conditional approval had been granted, applications for conditional approval of products in the same class, with the same target indication and on the basis of similar clinical studies, would be refused.
- Applicants would have to begin confirmatory trials before submitting applications for conditional approval. Those trials would have to be completed within four years, during which period applicants would need to provide the CDE with annual progress updates. Where those conditions were not met, the sale of a drug commercialized on the basis of a conditional approval could potentially be suspended.

Biosimilars

Copies of original biologics reached the Chinese market more than 20 years ago. Few had been tested in head-to-head trials with originators, however, and domestic biologics launched prior to 2019 are referred to as ‘alternative biologics’. The country’s first ‘true’ biosimilar, reviewed in line with regulations published in 2015, was granted approval by the NMPA in February 2019.

Modelled on the EU’s biosimilar guidelines, the 2015 regulations outline the principles for technical review of biosimilar applications, comparability criteria, and conditions for the extrapolation of indications. They do not create a separate, abbreviated pathway for the approval of biosimilars, however. This means that, while data requirements differ, applicants have to go through the same approval process as for innovative biologics.

Key Features of Biosimilar Guidance

Feature	Details
Definition	Biosimilars are defined as therapeutic biological products which are similar to a reference product already approved in China or elsewhere in terms of quality, safety and efficacy.
Reference products	The reference product cannot be another biosimilar and must be approved in China when comparative clinical studies are conducted. Originator biologics registered outside China are not accepted as reference products.
Characterization	A biosimilar must have the same amino acid sequence as the reference product.
Biosimilarity	Biosimilarity is proven through analytical studies to demonstrate similarities with the reference product in terms of physical and chemical properties, bioactivity, and purity; by animal studies (pharmacokinetic / pharmacodynamic studies, and toxicological assessment); and by human clinical studies.

Source: IQVIA research

Clinical Trials

Historically, onerous regulatory requirements and poorly administered approval procedures were a significant constraint on clinical trial activity, contributing to China’s lengthy drug lag. A major overhaul

of clinical trial requirements and regulations has been seen since 2015, however. The changes, which have been particularly beneficial for multinationals, have been a significant contributing factor to recent increases in the number of submissions filed for the registration of innovative new drugs in China.

Further improvements in the regulatory framework governing clinical trials were ratified in July 2020, when the new DRR came into effect, while other changes – including measures that would expedite ethics review procedures – are being pursued as part of DAL reforms. China’s accession to the ICH is also driving alignment of local regulations with international standards.

Clinical Trial Regulations

Measure	Details
Acceptance of foreign trial data	Data from multi-regional clinical trials (MRCTs) are accepted for new drugs, subject to the inclusion of data proving that ethnic factors do not affect safety or efficacy in Chinese patients. Pre-submission meetings with the CDE have been introduced to ensure that foreign data and documentation meet local regulatory requirements.
Simplification and streamlining of clinical trial approval procedures	A trial approval now relates to all phases of clinical activity (sponsors were previously required to seek separate approval for each phase). Phase I MRCTs of imported drugs may now be conducted in China without prior registration in overseas markets or prior entry into either Phase II or III clinical trials abroad. The CTA approval process has been streamlined. Sponsors may begin a trial if the CDE does not query or reject a CTA filing within 60 days. Bioequivalence studies must be filed, but do not require explicit approval.
Priority review and approval of clinical trial applications	Trial applications for new drugs offering clinical advantages over existing treatments for certain diseases (including hepatitis, cancer, HIV, tuberculosis and rare diseases, as well as pediatric and geriatric drugs) may qualify for priority review and approval. This applies to both innovative drugs and new generics.
Local trial waivers	Local trial waivers may be issued for priority drugs, subject to the submission and review of all foreign trial data. Having reviewed foreign data, regulators may reach one of three decisions on an application: - Full acceptance – amounting to a local trial waiver. - Partial acceptance – where there is uncertainty regarding ethnic impacts on safety and efficacy (conditional approval may be granted, subject to the post-approval collection of additional safety and efficacy data); - Non-acceptance – where data are deemed incomplete or inadequate, local trials are required.
Relaxation of clinical trial site certification requirements	Any facility with the necessary infrastructure may now become a clinical trial site, subject to registration with the NMPA.

Source: IQVIA research

More recently, measures to implement DAL reforms have included provisions that will have a positive impact on clinical trial sponsors. Among other things, these promise to streamline ethics review procedures for drugs that are part of MRCTs – principally by enabling mutual recognition of ethics reviews by participating trial sites, which have been delayed by six to eight weeks on average by existing procedures.

In November 2022, the CDE published draft guidance on the review process for clinical trial protocols. The guidelines emphasize the need to include complete trial protocols in submissions. They also underline the importance of consultation with regulators prior to the start of pivotal Phase III trials.

Improvements in the regulatory framework governing clinical research have led to a sharp increase in the number of trials being conducted in China, and in the number of CTAs being filed with the NMPA. This has increased competition for trial sites, however, driving up clinical trial costs.

Regulators have also become increasingly unhappy with the number of trials being conducted in specific, relatively narrow therapy areas, branding this a waste of industry resources. In 2021, the CDE issued new draft guidance to discourage development of ‘me-too’ drug candidates by multiple companies. Focused specifically on oncology, the guidance requires sponsors to use existing best-in-class therapies in control groups for randomized controlled clinical trials of these products.

Regulations Governing Data Protection and Human Genetic Material

Regulations governing data protection and human genetic material (HGM) have emerged as potential challenges for sponsors, especially where Chinese studies are part of MRCTs. Under regulations governing the export of human genetic resources (HGR), published in 2019, foreign companies had to apply for permission to collect and transfer samples and associated data, posing challenges for sponsors conducting MRCTs with Chinese trial sites.

New draft rules governing the management of HGM have since been issued in a bid to resolve issues raised by the 2019 regulations. Trial sponsors, which already had to comply with new rules outlined in the Internet Security Law and a new Data Security Law, were now also subject to provisions in a Personal Information Protection Law (PIPL), which took effect in November 2021. The PIPL specifies rules governing the collection, storage and use of personal data. It requires foreign companies to establish designated representatives in China to handle personal-information protection issues.

The PIPL's scope reaches beyond China, imposing exacting requirements for the export of personal data. Subsequently, restrictions on personal data exports from China were generally eased by the Cyberspace Administration of China's (CAC) final rules on *Regulating and Facilitating Cross-border Data Flows*, although the precise implications for clinical-trial data remain unclear, while HGM data remain subject to strict export controls.

Drug Administration Law

The Drug Administration Law (DAL) is the principal piece of legislation governing the pharmaceutical industry in China. Amendments to the DAL since 2015 have provided a framework for reform of rules governing all stages in the development and commercialization of medicines. These have affected a broad range of stakeholders, including distributors and retailers as well as pharmaceutical manufacturers. Major changes have included:

- An overhaul of the regulatory framework governing clinical research (see Clinical Trials).
- The implementation of a significantly revised DRR (see Drug Registration).
- The publication of explicit rules governing the online sale of prescription medicines (see Healthcare Provision: E-Health).

The ratification of amendments to the DAL in December 2019 paved the way for the implementation of a revised DRR, marking a switch away from a manufacturer-based authorization regime – which required applicants to have their own local manufacturing facility – to a marketing authorization holder (MAH) system.

The new regime enables both domestic and foreign companies to obtain marketing authorizations, regardless of whether they have local manufacturing capabilities. The NMPA has subsequently published guidance on specific aspects of the reforms, including regulatory aspects of contract manufacturing agreements and rules governing product recalls.

Draft regulations published by the NMPA in May 2022 will codify existing DAL reforms and lead to additional changes. Key areas covered by the draft include:

- New drugs – confirming the strict definition of ‘new drugs’, which affects registration procedures and whether individual products qualify for exclusivity periods and patent-term extension (see Intellectual Property).
- Regulatory data protection and market exclusivity – outlining exclusivity periods for new pediatric drugs, rare disease treatments and first generics; and acknowledging the concept of patent linkage and regulatory data exclusivity (see Intellectual Property).
- Innovation – reiterating calls for true innovation rather than duplicative investment in me-too products.
- Administrative requirements – confirming that a change of investigational new drug (IND) sponsor will require CDE approval.

Quality Standards and Drug Safety

The imposition and enforcement of tighter quality standards governing drug development, manufacture and distribution is a key health policy goal. As well as addressing patient safety concerns, improvements in the quality of locally manufactured drugs are being pursued in a bid to:

- Drive down the price of off-patent originator products, delivering savings for patients and payers.
- Provide a platform for the emergence of China as a more important exporter of finished pharmaceuticals.
- Drive consolidation of the domestic industry, encouraging the emergence of larger, stronger players capable of competing effectively for market shares both at home and abroad.

Compliance with a range of quality standards has been required for companies engaged in the clinical development, manufacture, and distribution of drugs since 2015, while bioequivalence certification requirements are being applied progressively to generics.

Key Quality Standards

Standards	Details
Good Clinical Practice (GCP)	China's current GCP guideline was introduced in 2003. Requirements have historically been poorly policed, however, and data falsification scandals have made headlines. In 2017 the Supreme Court recognized clinical data fabrication as a serious crime, punishable in some circumstances by death. China has now adopted ICH guidance on clinical safety data management.
Good Manufacturing Practice (GMP)	Compliance with GMP has been mandatory for the manufacture of sterile products since December 2013, and for all other products since December 2015. Companies that failed to meet those deadlines were prohibited from filing new-drug registration applications and faced closure of their plants, but were allowed to transfer operations to GMP-certified manufacturers.
Bioequivalence (BE)	All oral solid-dose generics on the EDL approved before October 2007 must complete GQCE testing. A five-year State Council drug-safety program, adopted in 2016, required quality consistency evaluation certification for 289 generic drugs on the EDL by 2020, although the deadline was subsequently extended.
Good Supply Practice (GSP)	Full compliance with revised GSP standards has been mandatory since 2015. Rules require distributors to operate quality control systems and use electronic drug monitoring systems. The 2015 revisions added requirements for information management, cold chain compliance, transportation and sales records. While certification requirements were abolished under the new DAL, monitoring of the sector is being stepped up.

Source: IQVIA research

Failure to meet required standards can result in fines, the suspension of trading licenses, the cancellation of drug registrations or the closure of manufacturing plants. Coupled with China's accession to the ICH in 2017, this has begun to shift oversight from certification schemes to inspection-based policing of quality standards. The change of approach was cemented by amendments to the DAL ratified in 2019. Breaches have nevertheless been reported.

IND and NDA application numbers have risen dramatically in recent years, stretching the NMPA's capacity for GMP and GCP inspection. To optimize the use of available resources, the CDE introduced a risk-based approach to on-site inspections at the beginning of 2022. The system categorized inspections as either low-, medium- or high-risk, according to the type of product under review. Innovative small-molecule drugs, reformulated new drugs, and biologics are all classed as high risk. Regulators are also policing quality standards for imported drugs more closely.

14th Five-Year Plan for Drug Safety

The 14th Five-Year Plan for National Drug Safety and High-Quality Development – published in December 2021, and spanning the period to December 2025 – calls for improvement in a number of areas. Key goals include:

- Optimizing the regulatory environment to support high-quality development of the pharmaceutical industry.
- Obtaining WHO certification for vaccines regulation.
- Pursuing new measures to encourage development and innovation in the TCM sector.

- Improving the expertise of regulatory reviewers and inspectors and an increase in sampling activity for products covered by basic medical insurance (BMI), imported drugs, pediatric medicines and TCMs.
- Establishing a pharmacovigilance system spanning the entire lifecycle of drugs.
- Establishing a digitized version of the national pharmacopoeia.

Quality Consistency Evaluation of Generics

The State Council called for the introduction generic quality consistency evaluations (GQCEs) in August 2015. A CFDA circular issued later that year required all oral solid-dose generics on the EDL approved before October 2007 to complete GQCE testing by the end of 2018, or risk cancellation of their registrations. The deadline was subsequently extended.

GQCE involves comparative testing of generics against originator drugs. *In vitro* dissolution testing is the primary method. *In vivo* testing of some drugs is required, although regulators have waived or reduced *in vivo* testing requirements for certain drugs. Nevertheless, early progress was slow, reflecting a combination of the cost implications for generic manufacturers and the limited number of institutions equipped to carry out bioequivalence testing, as well as a high failure rate.

The GQCE policy states that, when the first generic version of a drug has been certified as compliant, all other versions of the product must undergo GQCE registration within three years, or face removal from the market. In practice, regulators may extend that deadline by up to five more years where the removal of non-certified products would risk jeopardizing the supply of a drug.

GQCE-certified generics are awarded the majority of national VBP contracts (see Healthcare Provision: Drug Procurement). Manufacturers have been forced to slash their prices in order to win VBP contracts, however, raising concerns that local companies will struggle to recoup substantial investments in bioequivalence certification.

Incentives Attached to GQCE Certification

Incentive	Details
Interchangeability	GQCE-certified generics are deemed interchangeable with the originator.
Reimbursement	Bioequivalent generics will be prioritized for reimbursement under the BMI schemes.
Procurement	Bioequivalent generics will be prioritized in hospital procurement. Only GQCE-certified generics can bid for national public hospital supply contracts under the VBP system.
Labeling	Manufacturers of GQCE-certified generics may advertise the status of their products on drug labels and package inserts.

Source: IQVIA research

Most early national VBP contracts were for the supply of oral drugs. Regulators issued guidance on GQCE certification requirements for injectables in May 2020, and national VBP contracts have been awarded for a growing number of injectable drugs since then.

Intellectual Property

First enacted in 1985, China's Patent Law has been the subject of four major amendments since then. The most recent changes, which were ratified in 2020 and which, on paper at least, came into effect in June 2021, saw the establishment of a patent linkage mechanism and patent term extension provisions (see Patent Linkage below).

Enforcement of existing IP rules and the implementation of reforms is overseen by the China National Intellectual Property Administration (CNIPA), which was established as part of an institutional restructuring program undertaken in 2018. It reports directly to the new State Administration for Market Regulation (SAMR).

Intellectual Property Protection Rules

Rule	Details
Patents	Although pharmaceutical products have been patentable since 1992, until 2008 patents were only issued for innovations that were new worldwide. The patent term is 20 years, calculated from the date on which an application is filed. Patentability criteria preclude the grant of patents for 'specific therapeutic methods', ruling out patents for new dosage regimens, treatment of new patient subgroups or new routes of administration. Reforms implemented in 2021 allow for the restoration of patent terms eroded by regulatory delays.
Data protection	Current rules offer six-year periods of protection against the unfair commercial use of data submitted in support of products containing a new chemical ingredient. Protection only applies to drugs launched first in China, however, and data protection is not available for biologics. New provisions promise broader protection.
Market exclusivity	New drugs (categories 1 and 2) benefit from safety monitoring periods that provide <i>de facto</i> market exclusivity for up to five years. Exclusivity periods will be granted more widely under new rules.
Generic registration	While the NMPA may accept applications for the registration of generics up to two years before the expiry of an originator patent, marketing authorizations may only be issued when relevant patents have expired. In the absence of a patent linkage mechanism, generic versions of patented drugs have historically been approved on a regular basis. Reforms implemented in 2021 establish an explicit patent linkage mechanism, allowing for stays on the approval of generics which originators claim would infringe their IP. The reforms encourage generic challenges to patents by awarding exclusivity periods for applicants that successfully contest a patent.

Source: IQVIA research

Patent Linkage

New patent linkage rules ratified in 2020 loosely resemble the US 'Hatch-Waxman' system, providing both incentives for generic developers to challenge originator patents at an early stage, and a mechanism for the timely resolution of patent infringement disputes. Unlike the US system, however, the rules apply equally to both small molecule drugs and biologics.

In June 2021, the NMPA launched an online equivalent to the US FDA's Electronic 'Orange Book', containing a list of patents covering both small molecules and biologics. Companies seeking approval for generic or biosimilar versions of those drugs must certify that either:

- There is no patent for the originator on the NMPA registration platform.
- The patent has been terminated, declared invalid, or the generic applicant has been granted a patent license by the holder.
- The generic applicant agrees that its drug will not be marketed before the listed patent expires.
- The patent should be declared invalid, or the generic falls outside the scope of the patent.

Where originators believe the generic or biosimilar would infringe their intellectual property, they may seek either:

- An administrative adjudication by the CNIPA – quicker (delivered in theory within 45 days) but offering no preliminary injunction.
- A civil action in the Beijing IP court – more difficult to initiate, and slower, but providing for a preliminary injunction.

To encourage generic and biosimilar developers to pursue early market entries, the new rules offer 12-month exclusivity periods to companies that are first to invalidate an originator patent and market a follow-on product. To discourage frivolous legal challenges by patent holders, they may be liable for compensatory damages where a court subsequently finds that the patent holder knew – or should have known – that a generic or biosimilar did not breach relevant patents.

Patent Term Extension

The second major change to the Patent Law triggered by the ratification of reforms in 2020 involves the introduction of patent-term extension (PTE) provisions, where effective protection terms have been eroded by regulatory delays. The 2020 rules require patent holders to apply for these extensions not more than three months after a marketing authorization has been issued. Extensions may be granted for a maximum of five years, while total effective patent terms, including extensions, may not exceed 14 years.

Since the CNIPA said the provisions would not be applied retroactively, no drug approved in China before 1 June 2021 qualified for PTE. Originators may also file for patent term adjustments (PTAs) where significant time has elapsed between the filing and grant of a patent request. Again, the rules require applicants to request a PTA, while – unless the courts subsequently rule otherwise – the new provisions will not be retroactive.

The fears of originator companies that the CNIPA would take a narrow view of ‘new’, innovative’ and ‘improved’ drugs eligible for extended patent terms were confirmed when the Administration issued the final version of its revised *Implementing Rules for the Patent Law and the Guidelines for Patent Examination*, effective from 20 January 2024. The implementing rules exclude most multinational brands currently on the Chinese market from applying for PTEs and make any future applications contingent on drugs being filed for approval in China before they are granted marketing authorizations anywhere else in the world.

Pharmaceutical Business Environment

Industry Structure

China's pharmaceutical industry is highly fragmented, with around 4,000 local companies operating in the market. It is an important sector, accounting for around 4% of the country's total industrial output, according to the Ministry of Industry and Information Technology. As such, it is the subject of sector-specific five-year development plans drawn up by the government.

Industry dynamics have shifted significantly in recent years, reflecting the impact of substantive regulatory reforms and the establishment of new pricing, reimbursement and procurement rules. These have driven a sharp increase in the number of novel drugs brought to market in China. They have also meant a dramatic rise in BMI drug coverage. To limit the impact of more widespread reimbursement on BMI finances, regulators have:

- Negotiated sharp cuts in the prices of newly listed drugs.
- Issued national tenders for the supply of multisource drugs, leading to equally sharp cuts in hospital procurement prices for those products.

These developments have affected the structure of the Chinese market. They have also begun to shape commercial strategies employed by both local companies and foreign investors.

Local Industry

Consolidation of the local industry has been actively encouraged by the government, which is keen to improve the quality of domestic generics and encourage local innovation. In its latest development plan for the sector, covering the 14th Five-Year Plan period to December 2025, the government has prioritized innovation-driven expansion of the local industry, the creation of 'homegrown' multinational brands, and the emergence of domestic companies with global research and manufacturing capabilities that generate a substantial proportion of their revenues overseas. The plan sets a number of explicit targets, including:

- A further increase in domestic market shares generated by the country's 100 biggest pharmaceutical companies.
- A sector-wide increase in operating income (CAGR of at least 8%).
- An increase in R&D spending by the industry (CAGR of 10% a year).

The domestic industry is polarizing gradually between, on the one hand, large local players that have obtained international GMP certification and registrations in western markets; and, on the other, small, low-cost generics companies focused exclusively on supply to the Chinese market. GQCE certification requirements and the preferential purchase of bioequivalent generics for use in public hospitals has begun to make life increasingly difficult for the latter segment.

Mergers, acquisitions and collaborations are a growing feature of the local industry, with leading players pursuing aggressive expansion strategies in a bid to consolidate their footprint. Leading domestic manufacturers are also investing more heavily in research and development activities, and a growing number are entering into co-development partnerships. This has begun to bear fruit, with domestic manufacturers securing NMPA approval to market an increasing number of NCEs.

Foreign Investment

China has emerged as a significant contributor to global revenues generated by most multinationals, six of which now rank among the ten leading companies in combined retail and hospital channels audited by IQVIA (see Appendix A).

Off-patent brands were traditionally the main contributors to revenues posted by most multinationals in China. The picture has changed rapidly in recent years, however, reflecting the combined impact of regulatory, pricing, reimbursement and procurement reforms. These factors have seen a growing number of off-patent originators relinquish market share to local generics that have won national VBP contracts. At the same time, they have driven a rapid increase in the number of innovative new drugs granted approval by the NMPA and admitted to the NRDL – albeit at sharply reduced prices.

These trends have prompted multinationals to re-assess their strategic approach to the Chinese market, scaling back support for some patent-expired brands while investing more resources in the development, registration and commercialization of innovative new drugs.

Although a deterioration in political relations between China and the west may affect some future investment decisions, it appears unlikely to result in a significant reduction in overall levels of investment by foreign pharmaceutical companies in China, not least due to the sheer scale of the potential commercial opportunities the country continues to offer.

Research and Development

In line with the government's desire to see China emerge as a hub for international pharmaceutical product development, the sector has been accorded priority status in recent Five-Year Plans. The government's 'Made in China 2025' strategy, unveiled in 2015, set a series of five- and ten-year targets for the local industry in terms of drug innovation, import substitution and export development. These envisaged:

- The development of 10-20 innovative drugs by 2020
- The commercialization of 20-30 innovative products by 2025.

More recently, the government's 14th Five-Year Plan for the sector called for annual increases of 10% in industry R&D spending through the period to December 2025 (see Industry Structure). Significantly, it also sought a reduction in levels of duplicative R&D investment, which have been a feature of the local industry in recent years as multiple competitors target particularly promising areas of research.

The policy environment for innovators has improved considerably since 2015, notably where drug review and approval procedures are concerned. Multinationals have been the main beneficiaries to date, but the local industry's growing commitment to innovative research and development has also begun to bear fruit. Domestic companies have increased their R&D spending significantly in recent years, driven in part by active government support for companies investing in R&D.

At provincial and local level, authorities are keen to attract high-tech businesses and the well-paid jobs they create. Shanghai, which is at the forefront of developments in the sphere, has established a government-backed consortium targeting the discovery and development of first-in-class drugs.

Domestic R&D has traditionally been focused on the development of generics, while innovation has largely been incremental. The situation has begun to change, however, and the number of domestic companies engaged actively in the development of new compounds is growing fast, with a particular

focus on biotechnology. Until 2022, venture capital funding for the Chinese biotech industry was rising rapidly, bankrolling a steady stream of initial public offerings (IPOs) and secondary financing rounds by companies in the sector. Financing has been harder to come by since then, however. A funding crunch has forced local research-based companies to trim their pipelines and license out rights to a growing number of development-stage assets.

Multinational investment in local R&D has been encouraged by progressive improvements in regulatory frameworks governing both clinical trials and the review and approval of new drugs. Potential new challenges to foreign investment in the sector have emerged in the shape of a deteriorating geopolitical climate and the introduction of new laws governing data protection and handling of human genetic data. Most leading multinationals continue to operate research centers in China, however, while several have either expanded their existing capabilities or announced expansion plans.

Generics and Biosimilars

Generics

Generics and copy products account for more than 90% of sales by volume generated in IQVIA-audited channels, and for some 78% of the audited retail/hospital market by value (at ex-manufacturer prices). Branded generics and copies dominate, with unbranded generics claiming around 23% of the audited retail and hospital market in volume terms and 17% by value (see Appendix A: Tables).

Overcrowding and unfavorable pricing policies limit the appeal of the market for most of the world's leading international generics houses, and the sector is dominated by domestic companies marketing branded generics.

Keen to improve the quality of drugs circulating in the Chinese market, the government launched a generic quality-consistency evaluation (GQCE) program in 2015, under which bioequivalence data for generics are being reviewed. While original deadlines for compliance with GQCE certification have been extended, the number and range of generics with what amount to bioequivalence certificates have risen gradually (see Regulatory Environment: Quality Standards and Drug Safety).

Policymakers have pursued or announced a number of other initiatives designed to encourage increased use of locally manufactured generics. These were outlined in a 2018 State Council circular. A work plan for implementation of those policies was published in December 2018, calling on the NHC and the NMPA to draw up a list of 'encouraged generics' eligible for priority review and fast-track approval.

The first list of encouraged generics was published in June 2019 (34 active ingredients), a second list followed in December 2020 (17 active ingredients), and a third list was published in December 2023 (39 active ingredients). The lists prioritized drugs that were either already off patent or were approaching patent expiry, and for which no request for generic marketing authorization had been submitted to regulators, or which were in short supply.

Policy Proposals to Encourage the Domestic Generics Industry

Proposal	Objective
Substitution	Establishment of an equivalent to the US FDA's Orange Book, listing approved drugs that have undergone therapeutic equivalence evaluation, to guide substitution.
Reimbursement	Pursuit of a more dynamic approach to including new generics on the NRDL.
Prescribing	Strict enforcement of prescribing by generic name and closer scrutiny of individual prescribing behavior.
Dispensing	Handing pharmacists a greater role in reviewing prescriptions.
Procurement	Compilation of centralized drug procurement lists by generic name.
Usage	Incentives for providers to use generics.

Source: *Opinions Regarding Reforming and Perfecting Policies in the Supply and Usage of Generic Drugs*, April 2018

Since 2018, national VBP for multisource drugs has seen generics capture major shares of the public-hospital supply market, at the expense of a growing number of off-patent originators (OPOs). Only GQCE-certified generics may bid for national VBP contracts, but fierce competition for the substantial volumes on offer has driven sharp reductions in the price of some generic molecules.

More recently, amendments to the Patent Law – approved in 2020 and implemented in June 2021 – included provisions designed to encourage early patent challenges by generic manufacturers. Revised rules outline specific procedures under which generic manufacturers may challenge originator patents, offering 12-month market exclusivity periods to successful challengers.

The government's 14th Five-Year Plan, covering the period to December 2025, targets further development of the local generics industry. It calls for regular updates to the list of encouraged generics, and improvements to existing policies designed to promote the development of generics urgently needed in clinical practice.

Biosimilars

With originators not filing or adequately protecting their IP assets in China during the 1990s, local manufacturers seized opportunities to register copies of biologics such as interferon, filgrastim, and epoetin. Other copy biologics have been brought to market since then. Few have been tested in head-to-head trials with originators, however, and most are best described as 'alternative biologics' rather than biosimilars.

The publication of a regulatory guideline for biosimilars in February 2015 clarified the route to market for biologics that demonstrate similarity with a reference brand (see Regulatory Environment: Drug Registration: Biosimilars). Follow-on products are reviewed under the new drug approval process, but technical reviews follow the principles and processes outlined in the 2015 guideline – notably, proof of biosimilarity with the reference product (usually the originator). Explicit regulatory guidance boosted biosimilar-development activity by domestic companies, with over 200 biosimilars cleared for clinical trials in China towards the end of the last decade, according to the NMPA.

Regulators approved China's first 'true' biosimilar – Henlius Biotech's rituximab product – in February 2019. By the end of 2023, biosimilars of adalimumab, bevacizumab, denosumab, etanercept, infliximab, rituximab, trastuzumab and tocilizumab had been launched. With multiple biosimilar

versions of most of those molecules now competing for shares of the market, regulators are expected to begin including biologics in national VBP rounds at a relatively early stage in the forecast period.

Distribution

The operating environment for pharmaceutical distribution in China differs significantly from western markets. Large distributors play a direct role in selling products, participate in provincial tendering and hospital listing, and assume the credit risks of lower-tier distributors. Most leading distributors have their own manufacturing operations, enabling them to supply generics through their own networks.

The challenging nature of drug distribution, which is characterized by increasingly slim margins, has prompted distributors to explore diversification opportunities, including investments in the hospital sector. Some leading wholesalers are part of vertically integrated conglomerates that also include drugstore or retail pharmacy chains.

Gross distributor margins – which depend on local circumstances and the working capital required to support individual company operations – vary widely, ranging from around 4% to 15-20%. Net margins, however, often average between 1% and 2%. Financing is a major contributor to costs, since up to 12 months of working capital may be needed to finance inventory and accounts receivable.

Structure of the Distribution Market

While the distribution market is still highly fragmented –NMPA data suggest that some 14,800 companies held distribution licenses in 2023 – the sector has witnessed a period of progressive consolidation. Over 70% of the market is now concentrated in the hands of the 100 leading distributors, while the top three players – Sinopharm (a joint venture involving Fosun Pharma and the China National Pharmaceutical Group), Shanghai Pharmaceutical, and China Resources Pharmaceutical Group (CRPG) – hold a combined share of more than 30%. While it is smaller than the 'big three', Jointown is also a significant force in the distribution market.

Further consolidation of the distribution market is targeted under the 14th Five-Year Plan for the period to December 2025. By then, policymakers want to see the 100 leading players accounting together for more than 98% of the pharmaceutical wholesale market. The plan envisages that between one and three distributors will post revenues of more than RMB500 billion in 2025, while a further five to 10 will report revenues in excess of RMB100 billion.

Traditionally, drugs often passed through the hands of several intermediaries before reaching the end-user, with many smaller distributors providing access to local markets or even just a few hospitals. With each of those players charging a mark-up of 7-10%, the complex, multi-layered nature of distribution inflated end-user drug costs. Efforts to shorten the distribution chain and regulate the sector more effectively, which have been pursued since 2015, have encouraged significant restructuring in the sector. The two key drivers of this process have been:

- The imposition of good supply practice (GSP) standards, which have driven up distributor operating costs.
- The introduction of a 'two-invoice' system, which has truncated the distribution chain, limiting commercial opportunities for smaller players.

'Two-Invoice' System

The 'two-invoice' system for drug distribution was launched in 2017. It allows a maximum of two invoices to be issued in the distribution chain: one from the manufacturer to the distributor; and one from the distributor to the hospital or pharmacy. By truncating a previously multi-layered chain, the system is designed to:

- Streamline drug distribution.
- Enable better regulation of the distribution sector.
- Improve price transparency.
- Reduce mark-ups, limiting end-user costs.

Like many other healthcare reforms, the two-invoice initiative was pioneered initially by Sanming city, Fujian province. Implementation in other provinces and cities piloting healthcare and public hospital reforms was mandated in 2017. The initiative was rolled out across the remainder of the country in 2018. The vast majority of drugs now pass through the hands of a single distributor, although involvement of sub-distributors is still permitted to guarantee the supply of drugs to remote hospitals, health centers and pharmacies.

Since a copy of the manufacturer's invoice to the distributor has to be submitted to the hospital alongside product deliveries, the 'two-invoice' system has increased price transparency. This has significant implications for manufacturers, as hospitals now have full visibility of ex-manufacturer prices and distributor margins.

Leading distributors have expanded significantly through mergers and acquisitions, increasing their geographical coverage and driving market consolidation. That trend quickened following the imposition of the two-invoice system. Two erstwhile foreign entrants divested or downsized their interests in the Chinese pharmaceutical distribution sector during 2018.

Major Foreign Arrivals and Departures in the Distribution Market

Company	Presence
Cardinal	Acquired Zuellig Pharma China in 2010 and emerged as a top-10 distributor following a string of subsequent acquisitions – but sold its Chinese operations to Shanghai Pharmaceutical in 2018. The deal saw Shanghai Pharma take over Cardinal's 17 distribution centers supplying hospitals, retail pharmacies and sub-distributors in 320 cities.
Allianz Boots	Entered the Chinese market in 2008 through a 50/50 joint venture with Guangzhou Baiyunshan Pharmaceutical (GBP) and became the second largest shareholder in Nanjing Pharmaceutical – a top-20 wholesaler operating primarily in east and south-east China – in 2014. Alliance Boots was acquired by US pharmacy giant Walgreens in 2014, and Walgreens Boots Alliance (WBA) reduced its stake in the joint venture with GBP from 50% to 20% in June 2018.

Source: IQVIA research

Retail Pharmacy

Reflecting the traditionally hospital-led nature of the healthcare system and the limited availability of prescription-drug reimbursement by BMI funds through retail pharmacies, most outpatient prescription drugs are dispensed in the hospital sector. Selected retail pharmacies are registered to accept medical

insurance cards under the urban BMI schemes in some cities, notably Shanghai. Generally, though, the availability of prescription drug reimbursement in retail pharmacies remains limited, rendering the channel a predominantly self-pay sector.

With hospitals reluctant to list and dispense non-reimbursed products on their premises, retail pharmacies are an important sales channel for non-reimbursed drugs. The rollout and expansion of national VBP has encouraged companies that fail to win VBP contracts to devote more resources to generating sales through retail pharmacies, especially for well-known brands.

NRDL listing shifts sales from the retail sector into hospitals, where reimbursement is available, although the process may be lengthy, and some hospitals may be reluctant to list costly drugs even after inclusion in the NRDL. Direct-to-patient (DTP) pharmacies have emerged as a sales channel for specialty products and have broadened access to costly drugs that may not be available at hospital pharmacies – either because they are not reimbursed, or because they have not been added to the hospital's formulary.

A 'dual-channel' system for the reimbursement of selected drugs on the NRDL, dispensed through designated retail pharmacies, was launched in 2021. It is being rolled out progressively (see Prescribing and Dispensing: Dispensing).

Retail Pharmacy Numbers and Market Shares

Retail pharmacy numbers, which have increased progressively in recent years, rose by a further 7.0% to nearly 667,000 in 2023. Pharmacy chains have spearheaded recent expansion. Chain-store numbers exceeded independent pharmacy outlet numbers for the first time in 2017. By 2023, almost 58% of retail pharmacies were part of chains or franchises.

The government would like to see significant consolidation of the sector, with the Ministry of Commerce's *Guiding Opinions on Promoting the High-quality Development of the Pharmaceutical Distribution Industry during the '14th Five-Year Plan' period* envisaging five to ten specialized and diversified pharmaceutical retail chains operating across China by 2025, each with turnover of more than RMB50 billion. Targets also include the top 100 retail pharmacy chains generating more than 65% of retail market value by the end of 2025 (up from 52% in 2021), with chains accounting for close to 70% of all retail pharmacies.

Retail Pharmacy Outlets 2021-2023

Category	2021	2022	2023	Change 2023 (%)
Independent outlets	252,300	263,276	281,366	6.9
Chain outlets	337,400	360,023	385,594	7.1
Total	589,700	623,299	666,960	7.0

Source: NMPA

While many chains are pursuing in-house growth, competition within the sector is also driving M&A activity, concentrating more substantial market shares in the hands of fewer, larger players, which target major urban areas where footfall is heaviest. Only a small number of chains have achieved cross-regional coverage, and most smaller multiples are focused on a single city or region. The fragmented nature of the sector means even large players command relatively modest market shares.

Leading chains have begun to embrace franchising more widely, inflating the overall number of outlets they control. Excluding franchise activity, all of the five biggest chains now operate directly managed store networks numbering more than 8,000 outlets. Those numbers are increasing rapidly, reflecting the impact of M&A deals and in-house investments.

Leading Chains by Directly Managed Store Count

Chain	Store Numbers 2021-2022	Store Numbers 2022-2023	Store Numbers 2023-2024	% Change 2023-2024
Yifeng	6,877	8,306	10,264	23.6
Yixintang	8,560	9,206	10,255	11.4
Dashenlin	7,258	8,038	9,909	23.3
Laobaixing	6,120	7,903	9,180	16.2
Sinopharm GuoDa	7,252	7,730	8,528	10.3

Source: Menet.com

The presence of foreign drugstore chains in China remains limited to date. Watsons has a modest presence, while Walgreens Boots Alliance gained an interest in the Chinese retail pharmacy sector through its acquisition of a stake in Nanjing Pharmaceutical. WBA acquired a 40% stake in Sinopharm GuoDa in June 2018.

Online Pharmacy

While online sales of prescription drugs were prohibited in 2004, e-commerce companies circumvented the ban by requiring payment for medicines on delivery, enabling them to classify such transactions as 'offline' purchases. E-pharmacy sales values have risen dramatically since 2015. The COVID-19 crisis prompted a further sharp rise in online drug sales, reflecting more widespread e-prescribing and patient reluctance to visit healthcare facilities. Explicit rules governing the online sale of prescription drugs, which were published in 2021, came into force in December 2022, cementing prospects for further growth.

OTC Market

The OTC market is characterized by multiple channels, different product segments, and the inclusion of reimbursed medicines. Rules governing the sale of non-prescription bound products reflect perceived levels of risk associated with the use of individual medicines, which are split into two categories:

- Class A products – identified by a red symbol on the label. The sale of red-label products, which account for around three-quarters of all non-prescription medicines, is restricted to hospitals and retail pharmacies.
- Class B products – identified by a green symbol on the label. Green-label products, which are deemed safer for use by self-medicating patients, may be sold in supermarkets and other mass-market outlets, but must be stocked in separate drugstore units within such premises.

Non-prescription bound products for conditions not widely self-medicated tend to be prescribed initially by a hospital doctor, after which patients may obtain refills directly from a retail pharmacy. Hospitals

and clinics generate around 40% of OTC sales, with retail pharmacies accounting for approximately 60%.

OTCs are reimbursable if they are included in the EDL or the NRDL. Since reimbursable OTCs sold through the hospital channel can be good business for manufacturers, this has dissuaded companies from delisting their products. Patients insured under the UEBMI scheme can pay for OTC medicines using their personal medical accounts at dedicated retail pharmacy stores.

The availability of OTCs listed on the EDL or NRDL free of charge to patients covered by BMI schemes in hospitals and CHCs limits retail pharmacy sales of those products. TCM formulations, which dominate the OTC market, have spearheaded overall growth in the sector.

OTC products are subject to the same registration rules as prescription drugs. This adds to costs and contributes to delays in market access for new OTC products. Requirements for prescription drugs switching to OTC status were outlined in guidance published in 2012. The switch process is slow and lacks transparency, while regulators have traditionally adopted a cautious approach to deregulation.

OTC advertising regulations were tightened in 2015, in response to the prevalence of false or misleading claims. While direct-to-consumer advertising of OTC products is permitted, advertisements require prior approval by provincial regulatory authorities, and efficacy or safety claims are not allowed. Generating sales of a particular brand can be challenging, since the choice of product sold will usually depend on the profit margin earned by the pharmacist.

Sales and Promotion

Pharmaceutical sales and marketing practices have evolved significantly in the face of government crackdowns on corruption and, since 2018, the emergence and expansion of volume-based purchasing. The COVID-19 crisis and the introduction of new rules governing promotional activity in 2020 has driven further change in the sector, with dramatic shifts in the promotional mix. IQVIA Channel Dynamics data show that face-to-face contacts, which accounted for 91% of interactions with healthcare professionals in 2019, were responsible for 60% of the total in 2024.

Regulation of Promotional Activity

The implementation of 'nine prohibitions' in 2013 (ruling out some traditional forms of interaction between manufacturers and health professionals, including commercial bribery), and a 'blacklist' system for drug and medical device companies found to have engaged in bribery, changed the way manufacturers engaged with customers. Face-to-face detailing remains a key part of promotional strategies, however, and leading companies still employ large sales forces.

Amendments to the Anti-Unfair Competition Law (AUCL), which took effect on 1 January 2018, tightened rules on misleading promotional activity, and increased fines for breaches of those rules. The revised law also redefined bribery and holds companies responsible for acts of bribery committed by their employees. Fines and other penalties for bribery were increased, and egregious breaches expose companies to potential loss of business licenses.

Rules implemented in December 2020 require compliance by a range of stakeholders with codes of conduct and impose additional restrictions on sales representative activity. Key provisions include:

- MAHs or their designated agents must register and manage all representatives acting on their behalf (including contract sales forces as well as in-house staff).

- Sales representatives must meet specific education and training requirements and will be subject to registration on a database managed by the China Pharmaceutical Association (CPA).
- Medical reps are prohibited from undertaking 'drug sales tasks' or engaging in sales-related activities, and are explicitly prohibited from providing sponsorships, donations or financial aid to physicians, medical institutions or individuals in charge of procurement.
- As well as sales reps, physicians, pharmacists and hospital managers violating the new rules will be subject to punitive action outlined in the new Drug Administration Law.

A new drive to root out corruption was launched in July 2023, when the National Health Commission, together with nine other government agencies, announced a year-long campaign to address corruption in the healthcare and pharmaceutical sectors. While the drive also targets insurance fraud and provider behavior, scrutiny of industry and its interactions with other stakeholders has been stepped up, affecting the entire pharmaceutical industry value chain, from clinical research through registration, manufacture, distribution, procurement, to promotion and prescribing.

Impact of Regulatory Developments on Promotional Strategies

In the wake of these developments, access to hospital doctors has become more difficult in parts of the country, forcing manufacturers to alter their promotional strategies. Reps are meeting with doctors outside hospital premises and engaging with them more frequently by telephone or via electronic means. Face-to-face meetings with individual physicians remain important, but group meetings and seminars have begun to feature more prominently in the promotional mix.

While hospital physicians are still the main promotional target, separate sales forces are employed to target doctors working in CHCs – primarily to ensure that the use of brands prescribed in hospitals continues in the primary care sector. Activity at this level has traditionally been at portfolio rather than product level, but the importance of promoting to physicians in low-tier hospitals and CHCs is increasing. A more significant promotional presence in the retail channel is also gaining importance for manufacturers with brands that fail to secure VBP contracts for supply of the public hospital sector.

Advertising

Rules governing advertising claims and the ban on direct-to-consumer (DTC) promotion of prescription drugs remain tightly enforced, with tough penalties for companies that fail to comply. Advertising materials for medicines must be reviewed and approved by provincial regulatory authorities prior to dissemination, while health-related TV programs must be reviewed by the media regulator. Penalties for serious breaches of advertising standards – such as making false claims for a drug – can include imprisonment in addition to fines. Penalties for misleading promotional activity were increased by the 2018 amendments to the AUCL.

APPENDIX

Published: March 2025

APPENDIX A: TABLES

Sales of Leading Corporations (2024)

Audited Retail and Hospital Sector at Ex-Manufacturer Prices

Rank		Company	RMB (million)	Market Share (%)	Annual Growth (%)
2023	2024				
1	1	AstraZeneca	23301	2.8	10.4
2	2	Jiangsu Hengrui	18065	2.2	6.5
6	3	Novartis	17238	2.1	17.8
5	4	Roche	15778	1.9	6.1
4	5	Qilu Pharmaceutical	15620	1.9	4.5
7	6	CSPC Ouyi Pharmaceutical	14236	1.7	8.9
3	7	Pfizer	13856	1.7	-12.6
9	8	Sanofi	13557	1.6	15.9
10	9	Bayer	12991	1.6	14.4
8	10	Sino Biopharmaceutical	12263	1.5	-3.7

Source: IQVIA

Sales of Leading Products (2024)

Audited Retail and Hospital Sector at Ex-Manufacturer Prices

Rank		Product	Molecule	Manufacturer	RMB (million)	Market Share (%)	Annual Growth (%)
2023	2024						
2	1	En Bi Pu	butyl phthalate + sodium	CSPC-NBP Pharma	10597	1.2	14.4
3	2	Entresto	sacubitril + valsartan	Novartis	9066	1.0	39.4
1	3	Sulperazon	sulbactam + cefoperazone	Pfizer-Dalian	6236	1.0	-20.7
6	4	Forxiga	dapagliflozin	AstraZeneca	4040	0.6	23.0
8	5	Tpiao	thrombopoietin	SY Sunshine Pharma	3908	0.5	17.9
-	6	Ozempic	semaglutide	Novo-Nordisk	3831	0.5	-
7	7	An Ke Da	bevacizumab	Qilu Pharma	3702	0.5	3.5
5	8	Behring	albumin	CSL Behring AG	3636	0.5	-3.4
10	9	Lipitor	atorvastatin	Viatis	3223	0.4	6.4
11	10	Tagrisso	osimertinib	AstraZeneca	3200	0.4	4.2

Source: IQVIA

Value Share (%) by Licensing Status – Combined Retail and Hospital Sector

Ex-Manufacturer Prices

Category	2023	2024
Original Brands	13.3	14.5
Licensed Brands	6.8	7.2
Unbranded Generics*	18.3	17.3
Other Products**	61.6	61.0

* Includes products marketed under the generic name of their active ingredient(s)

** Includes branded generics, copy products, products where there is no evidence of a licensing agreement, products for which a licensing category has not been identified, and non-patentable products.

Source: IQVIA

Volume Share (%) by Licensing Status – Combined Retail and Hospital Sector

Standard Units

Category	2023	2024
Original Brands	4.2	4.1
Licensed Brands	2.9	3.0
Unbranded Generics*	22.9	23.2
Other Products**	70.0	69.7

* Includes products marketed under the generic name of their active ingredient(s)

** Includes branded generics, copy products, products where there is no evidence of a licensing agreement, products for which a licensing category has not been identified, and non-patentable products.

Source: IQVIA

APPENDIX B: GLOSSARY

Glossary

AIDS	Acquired Immune Deficiency Syndrome
API	Active Pharmaceutical Ingredient
BMI	Basic Medical Insurance
CAGR	Compound Annual Growth Rate
CCP	Chinese Communist Party
CFDA	China Food and Drug Administration
CHC	Community Health Centre
CHPA	China Hospital Pharmaceutical Audit
DRG	Diagnosis-Related Group
DRR	Drug Registration Regulations
EDL	Essential Drug List
EIU	Economist Intelligence Unit
EU	European Union
FDA	US Food and Drug Administration
GCP	Good Clinical Practice
GDP	Gross Domestic Product
GMP	Good Manufacturing Practice
GP	General Practitioner
GSP	Good Supply Practice
M&A	Merger and Acquisition
MAT	Moving Annual Total
MoH	Ministry of Health (former)
MoHRSS	Ministry of Human Resources and Social Security
MRTC	Multi-Regional Clinical Trial

NHC	National Health Commission
NHSA	National Healthcare Security Administration
NMPA	National Medicinal Products Administration
NPC	National People's Congress
NRCMS	New Rural Co-operative Medical Insurance Scheme
NRDL	National Reimbursement Drug List
OTC	Over-the-Counter
PBC	People's Bank of China
PhRMA	Pharmaceutical Research and Manufacturers of America
R&D	Research and Development
RMB	Renminbi (national currency)
SAMR	State Administration for Market Regulation
TCM	Traditional Chinese Medicine
USTR	United States Trade Representative
WHO	World Health Organization
WTO	World Trade Organization

APPENDIX C: SOURCES AND METHODS

For details on Market Prognosis sources and methods, please refer to the link below:

<http://syndicatedanalyticslibrary.imshealth.com/files/MarketPrognosisSourcesandMethods.pdf>